

Anatomy of Respiratory System

Major module contents

□ *Major contents of this module includes:*

- ✓ Introduction to the respiratory system
- ✓ Thoracic wall
- ✓ Gross structures and functions of the upper respiratory tract
- ✓ *Gross structures and functions of the lower respiratory tract*
- ✓ *Posterior mediastinum*
- ✓ Development of respiratory system

Introduction to Respiratory System

- Humans can live without water for days and without food for weeks, but they cannot live without oxygen for even a few minutes.
- **Breathing is our most urgent need.**
- From our first breath at birth, the rate and depth of our breathing are unconsciously matched to our activities.
- Breathing is so characteristic of life that, **along with the pulse**, it is one of the **first things health professionals check to determine if an unconscious person is alive**

Introduction to Respiratory System,,

- Studying, sleeping, talking, eating, and exercising all involve breathing.
- **The trillions of cells in the body need a continuous supply of oxygen to produce the energy needed to carry out their vital functions**
- As the cells use oxygen, they produce carbon dioxide, a waste product the body must eliminate.
- **The primary role of the respiratory system is to make oxygen available to cells for cellular respiration and to remove carbon dioxide, the main by-product of that metabolism.**

Introduction to Respiratory System,,,

- **The respiratory and cardiovascular systems** collaborate to deliver oxygen to tissues throughout the body and to transport carbon dioxide to the lungs for elimination.
- Also these two systems **have a close spatial relationship** in the thoracic cavity, but they also have such a **close functional relationship** (cardiopulmonary system).
- A disorder that affects the **lungs has direct effects on the heart, and vice versa**.
- **The term respiration can mean** ventilation of the lungs (**breathing**) or the use of oxygen in cellular metabolism.

Introduction to Respiratory System,,

- *The entire process of respiration encompasses 5 unique & sequential processes:*
 1. Breathing (pulmonary ventilation)-the movement of air into & out of the lungs.
 2. **Alveolar gas exchange** —the exchange of oxygen and carbon dioxide between the **air in alveoli and the blood in alveolar capillaries**.
 3. Gas transport —transport of oxygen and carbon dioxide between the lungs and tissues, accomplished by the cardiovascular system.
 4. **Systemic gas exchange** —the exchange of oxygen and carbon dioxide between the blood in systemic capillaries and the tissue cells.
 5. Cellular respiration—the use of oxygen and production of carbon dioxide during ATP production.

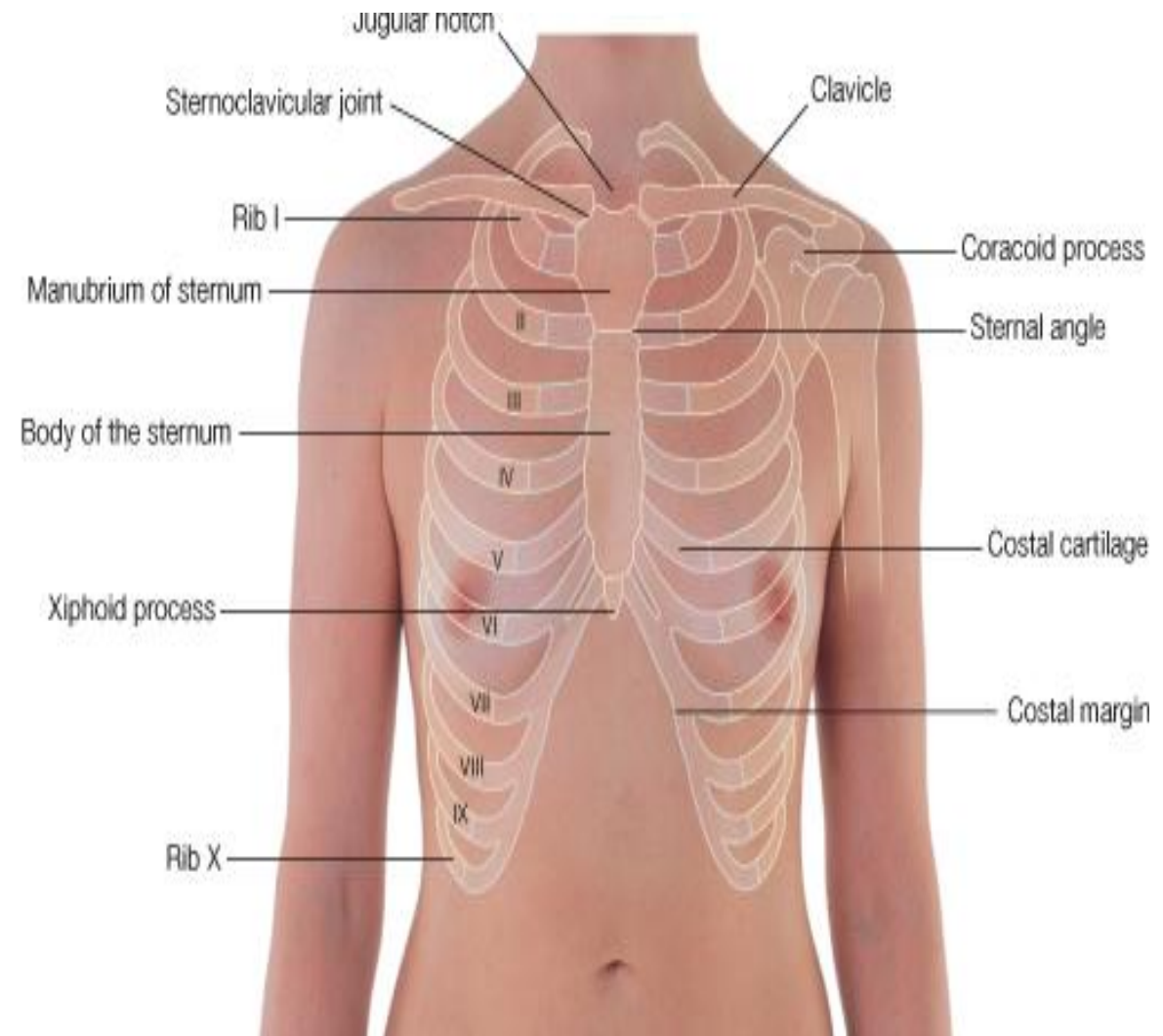
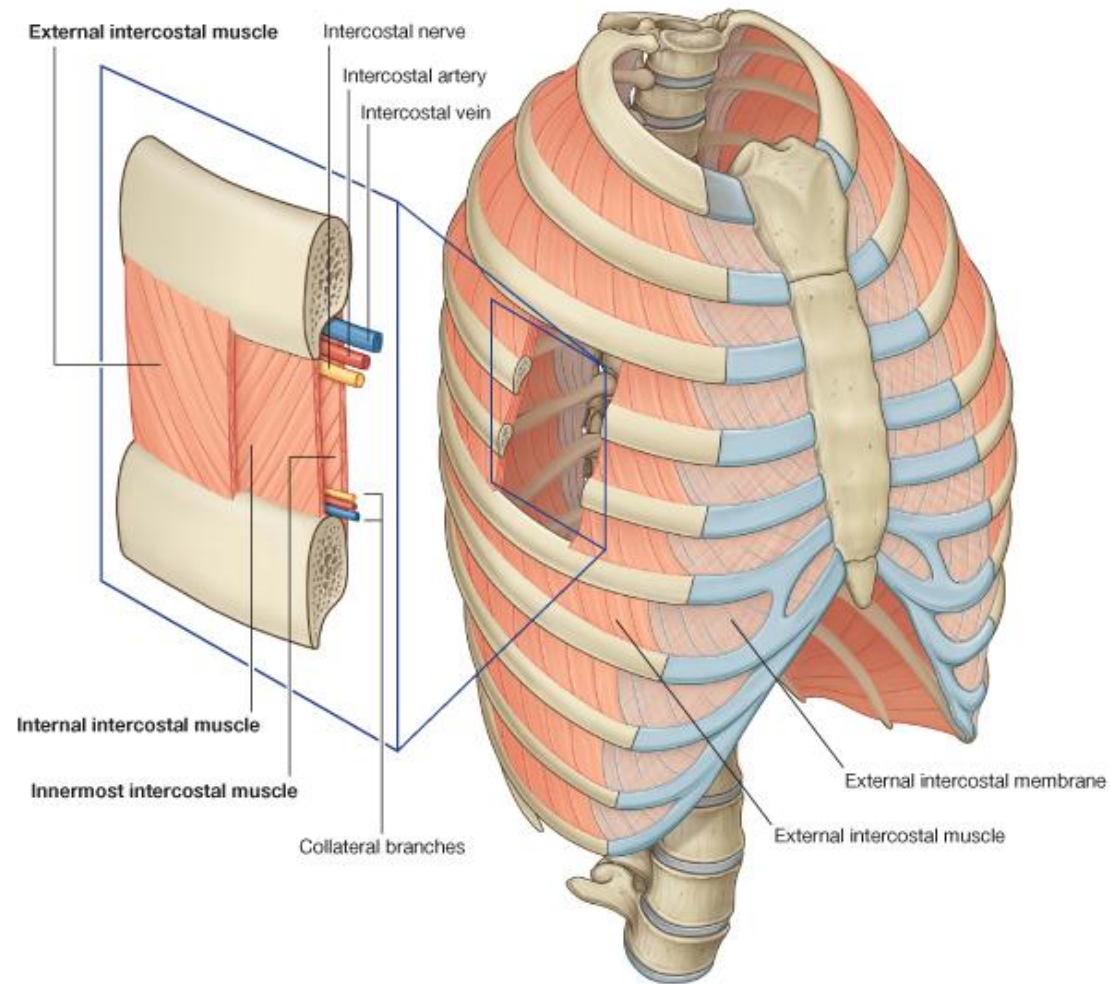
Structures of the Respiratory System

- ❖ **The organs of the respiratory system** include the nose, nasal cavity, and paranasal sinuses; the pharynx; the larynx; the trachea; the bronchi and their smaller branches; and the **lungs, which contain the terminal air sacs, or alveoli**
- The structures of the respiratory system are involved directly in **only two** of these processes: **breathing and alveolar gas exchange**, which is collectively referred to **as external respiration.**

Structures of the Respiratory System,,

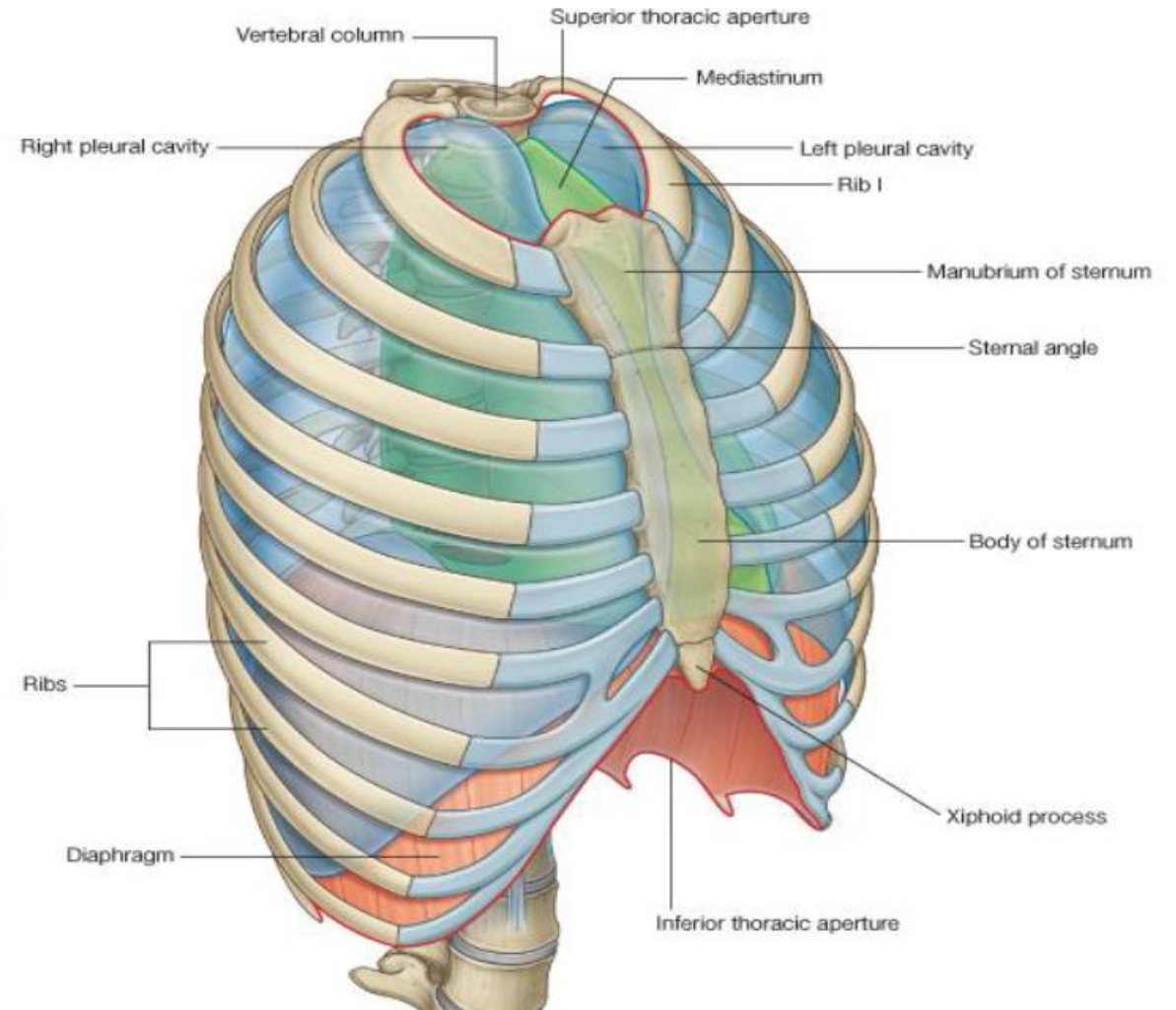
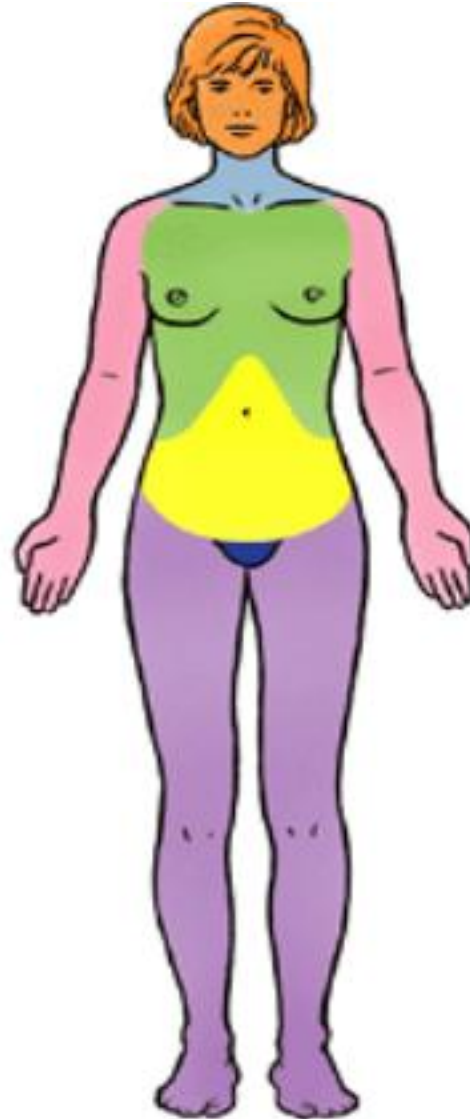
- **Systemic gas exchange and cellular respiration** together are referred to as **internal respiration**.
- **The respiratory system does more than just exchange respiratory gases.**
- ❖ **It also helps to detect odors, produce sounds, regulate blood pH, trap and defend the body from airborne pathogens, and assist in the movement of venous blood and lymph.**

Thoracic wall



Thoracic wall,,,

- **The thorax** is the region between the **neck and the abdomen**.
- It has the shape of a truncated cone.
- Is **narrow** superiorly but **circumference increase** inferiorly.
- It is flattened in **front & behind** but **rounded at the sides**.



Skeleton of Thoracic Wall

❖ **The thoracic skeleton forms osteocartilaginous which includes;**

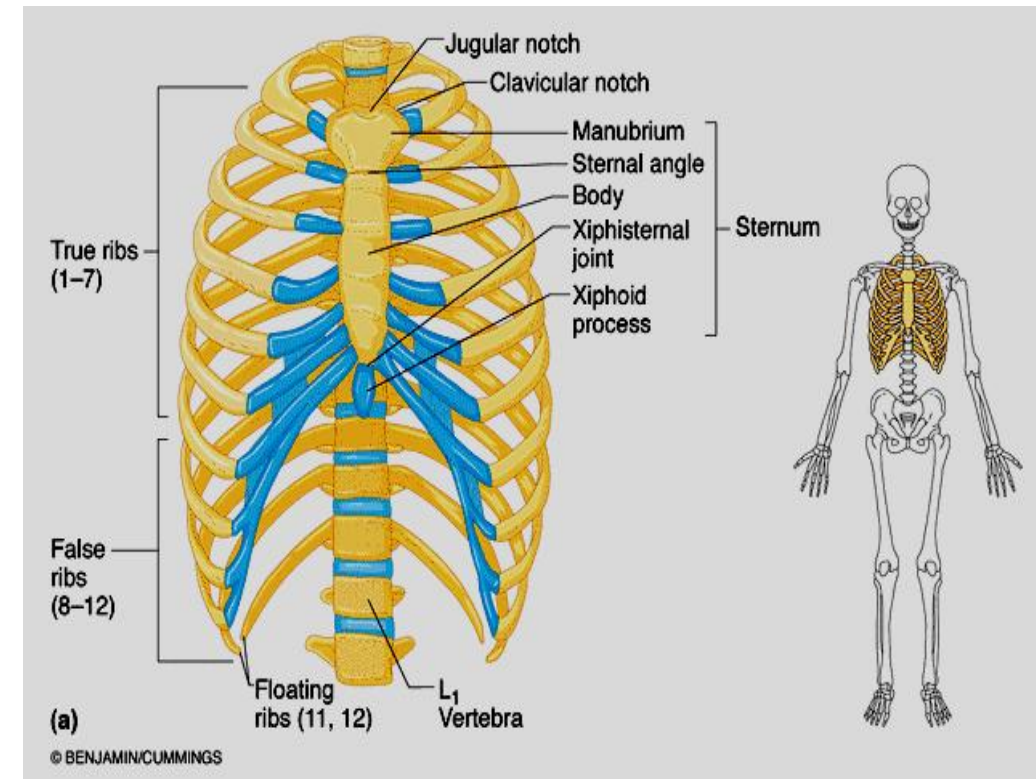
A. Skeletal elements(thoracic cage)

✗ **Anteriorly-** sternum (manubrium, body, & xiphoid process)

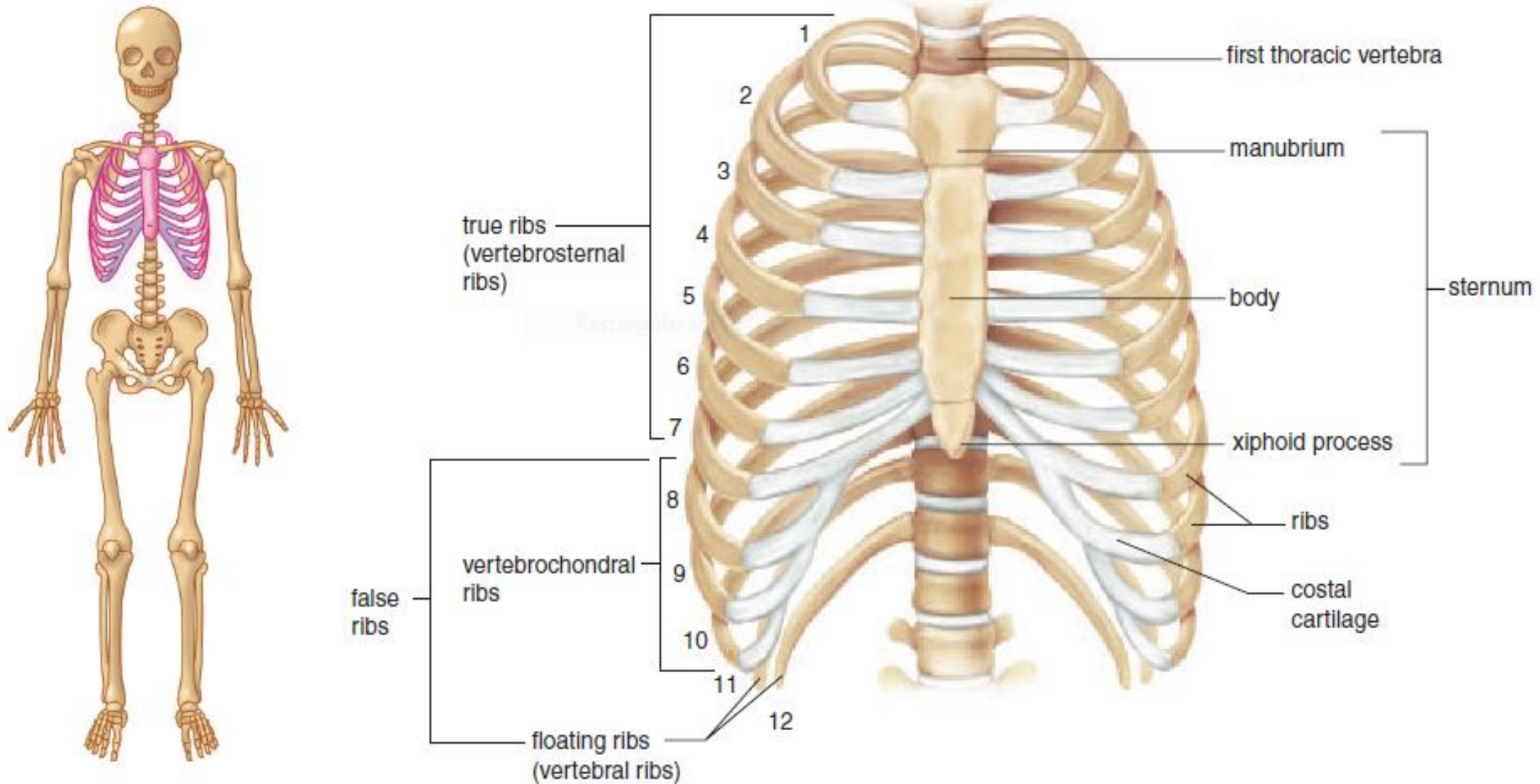
✗ **Posteriorly-**all thoracic vertebrae & their intervening IVDs

✗ **Laterally** -12 pairs ribs & their cartilages

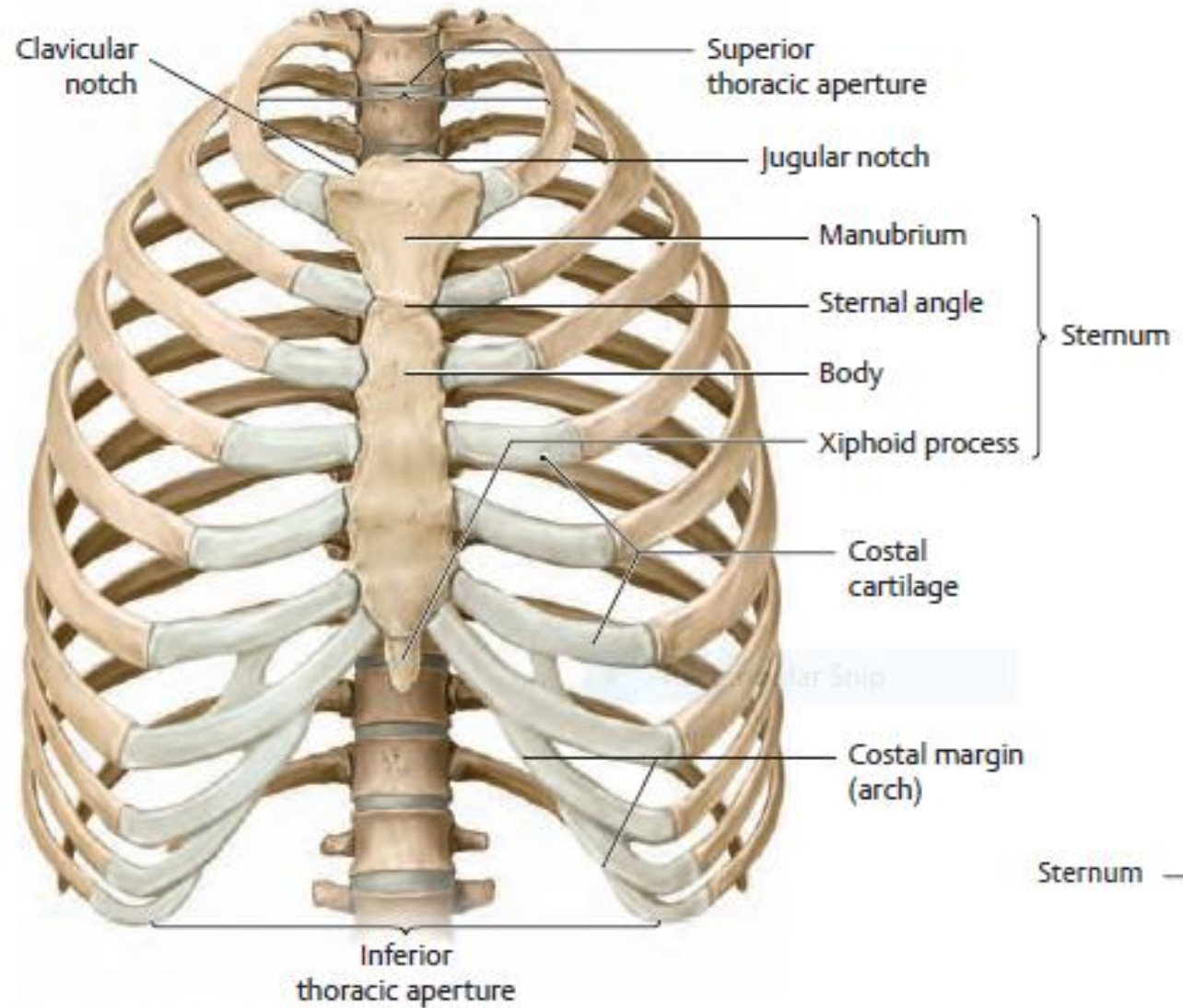
B. Skin: Mammary glands of breasts lie within the subcutaneous tissue of thoracic wall.



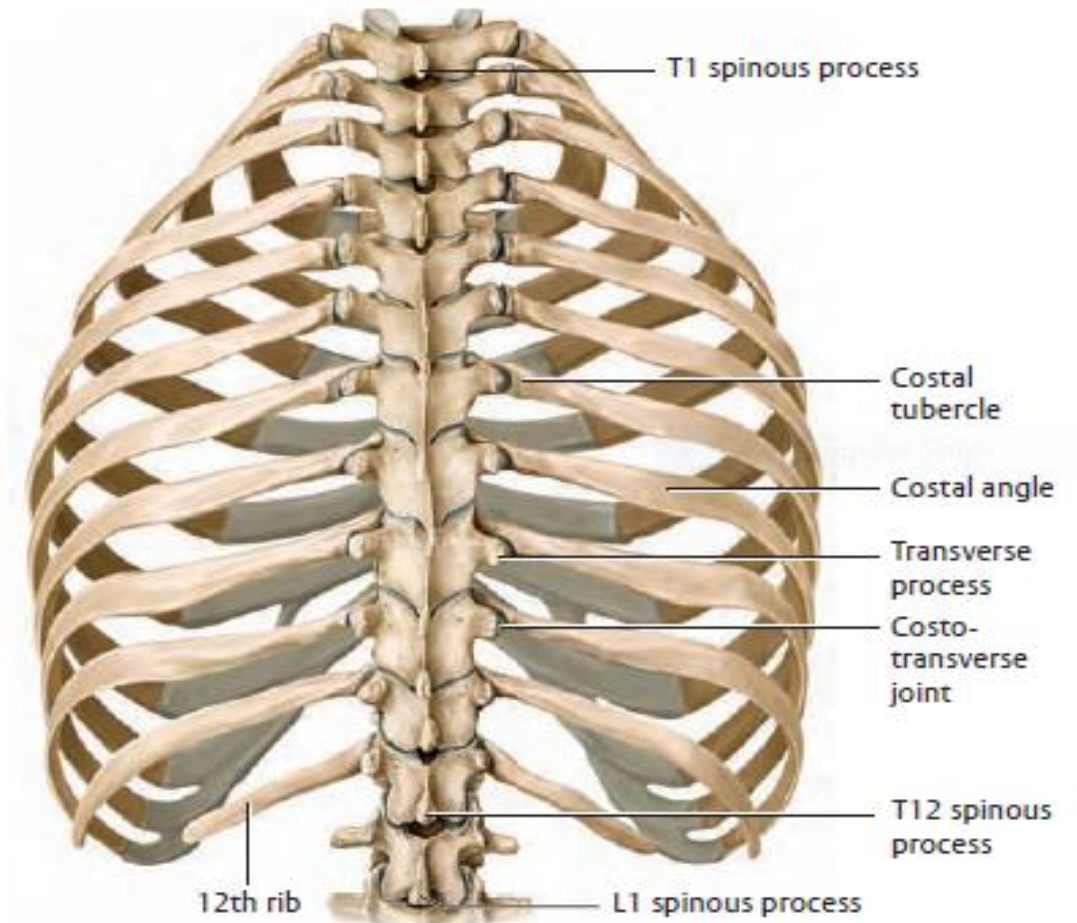
Skeleton of Thoracic Wall,,



Skeleton of Thoracic Wall,,,



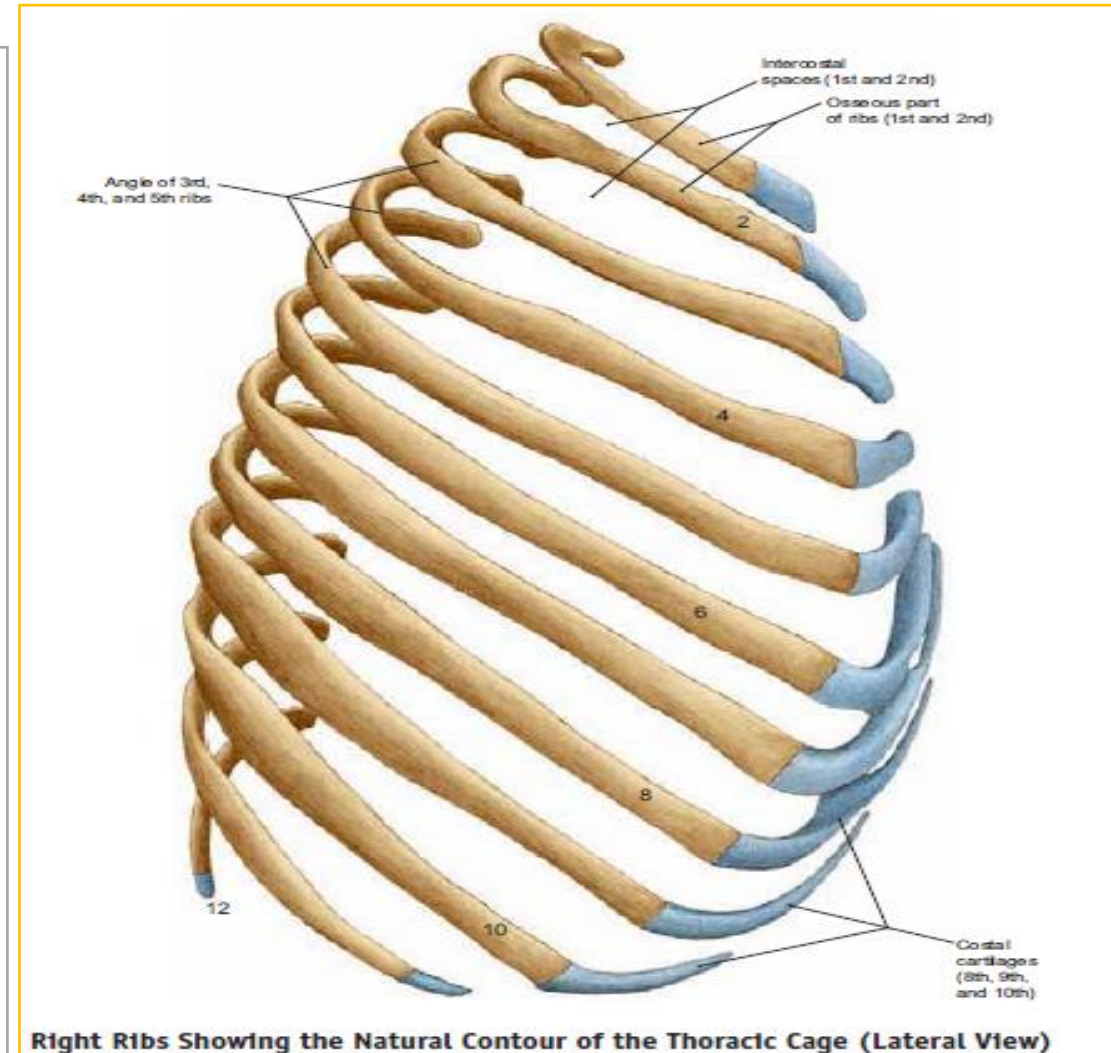
A Anterior view.



C Posterior view.

Skeleton of Thoracic Wall,,,

- **Ribs (L. *costae*)** are curved, flat bones that form most of the thoracic cage.
- There are twelve pairs of ribs, each terminating anteriorly in a costal cartilage
- They are **remarkably light in weight**
- Each rib has a spongy interior containing bone marrow (hematopoietic tissue).



Skeleton of Thoracic Wall,,,

□ There are *three types of ribs* that can be classified as typical or atypical:

1. True (vertebrosternal) ribs (1st–7th ribs):

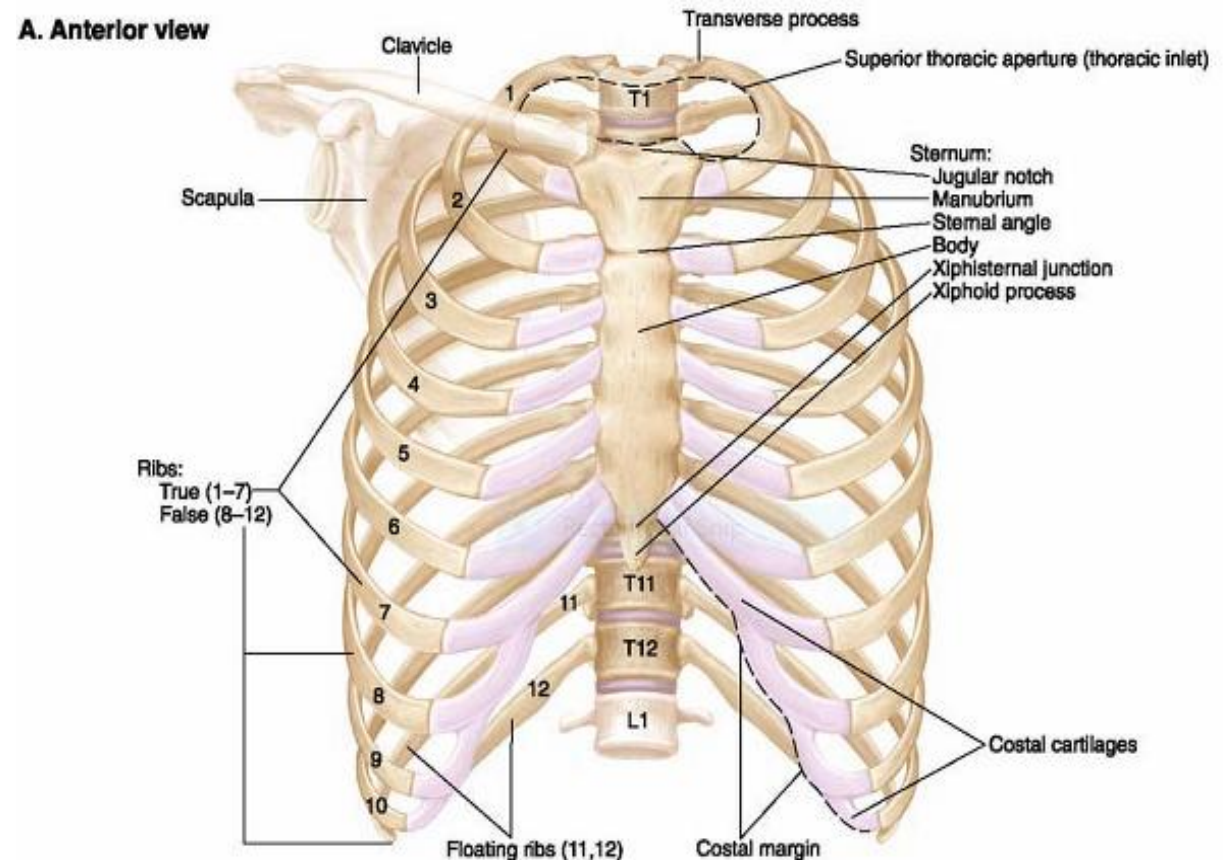
- They attach **directly to the sternum through their own costal cartilages.**

2. False (vertebrochondral) ribs (8th, 9th, and usually 10th ribs).

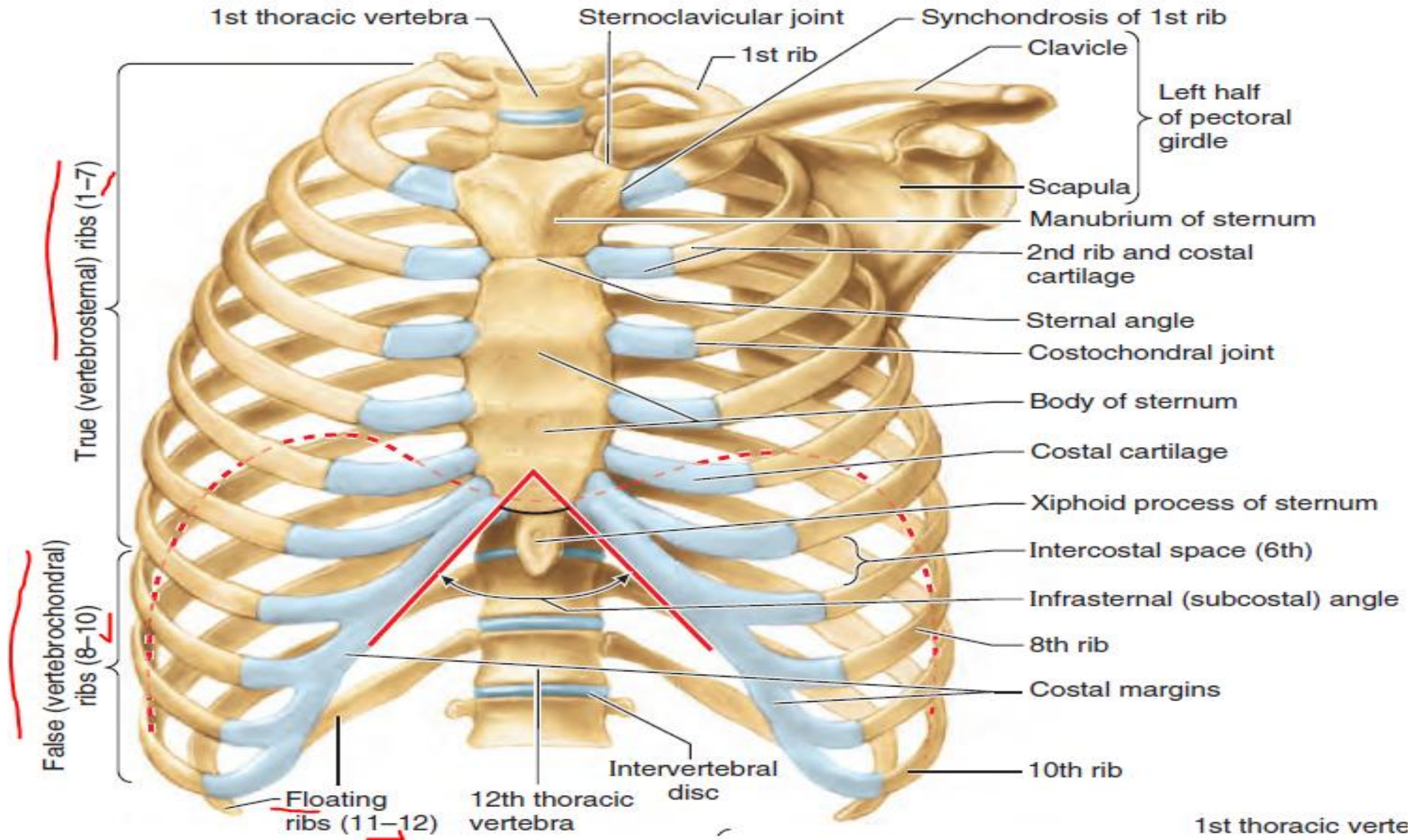
- Their cartilages are connected to the cartilage of the rib above them; thus **their connection with the sternum is indirect.**

3. Floating (vertebral, free) ribs (11th, 12th, and sometimes 10th ribs):

- The rudimentary cartilages of these ribs do **not connect even indirectly with the sternum.**



Skeleton of Thoracic Wall,,



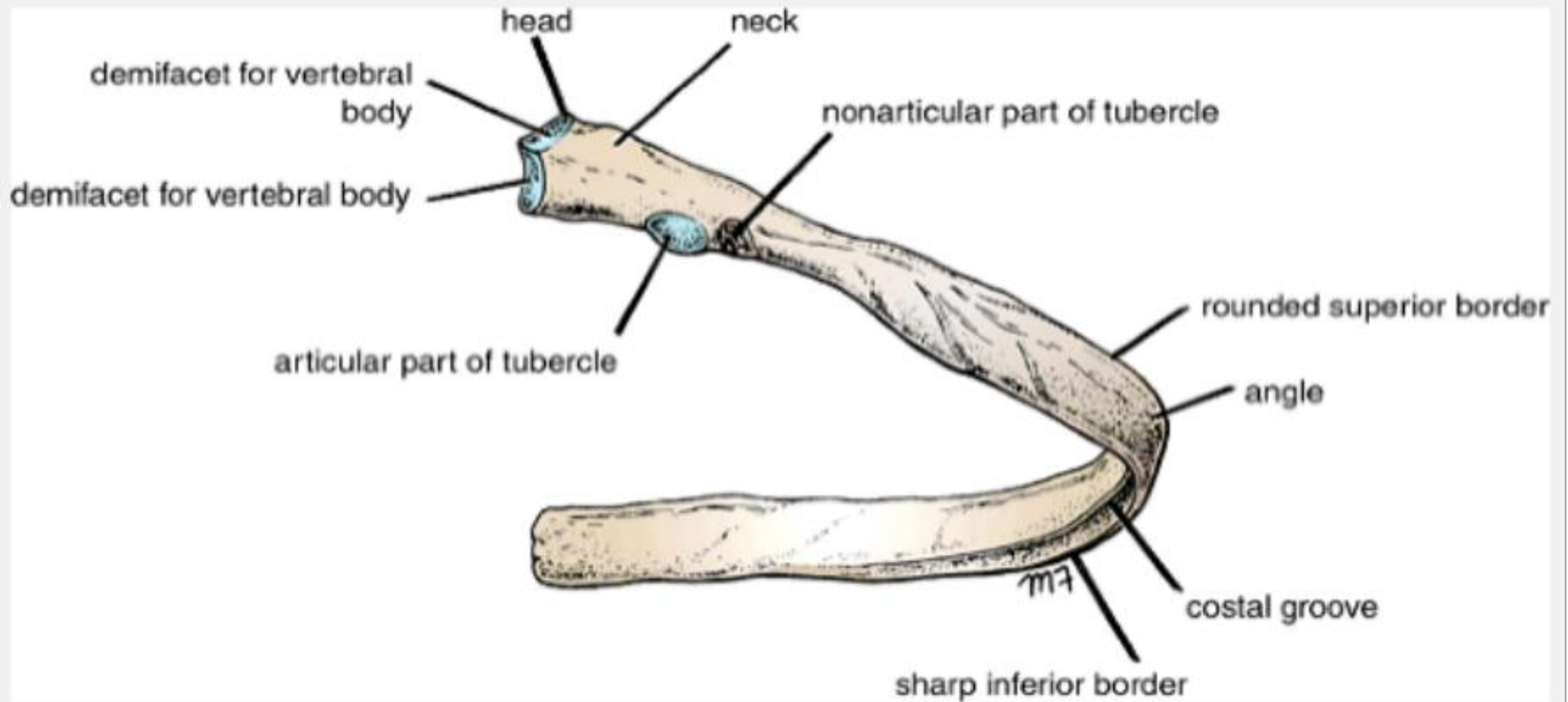
Classification of ribs as:

Typical VS atypical

Typical ribs (3rd - 9th)

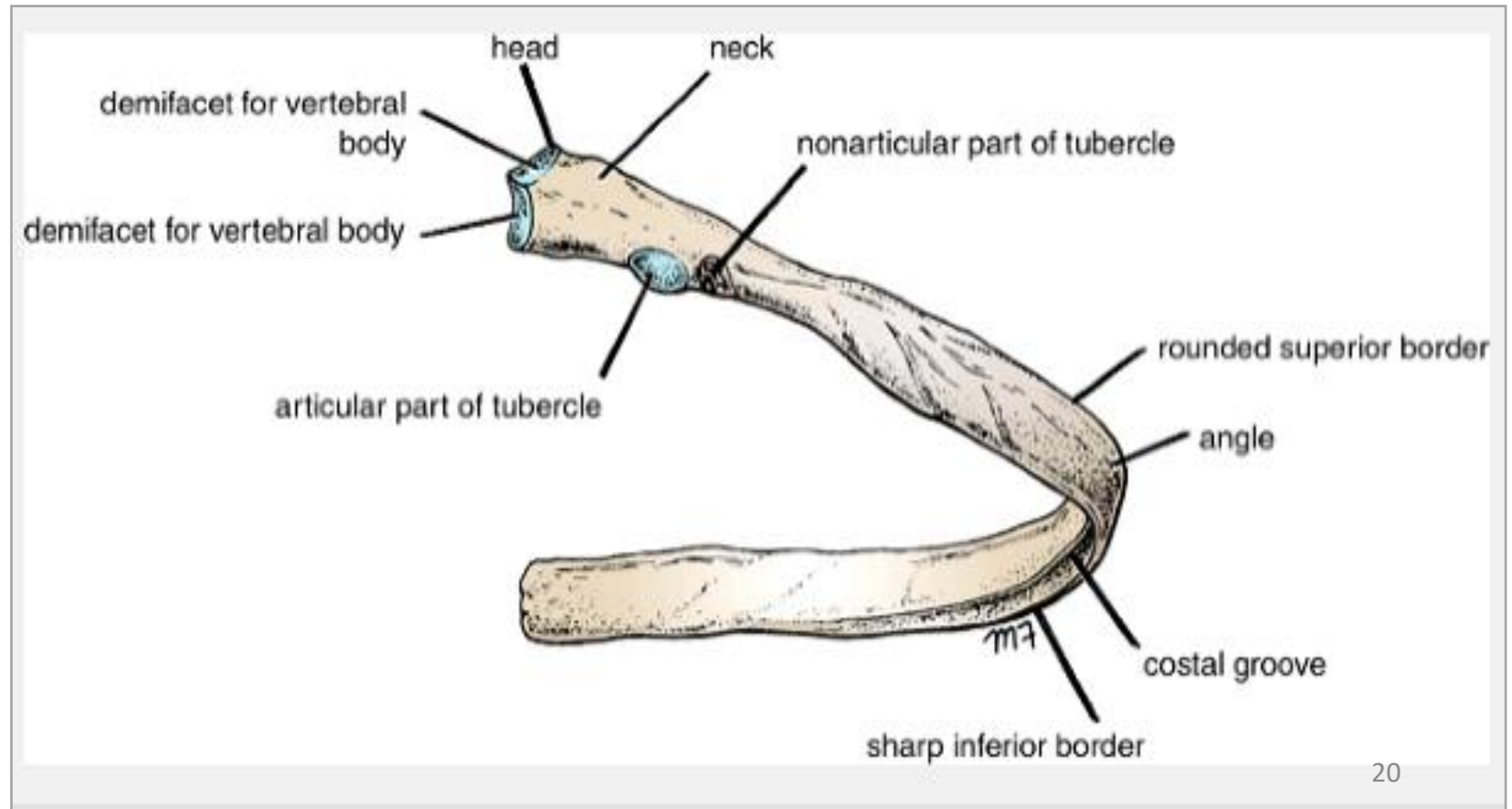
- A typical rib is a long, twisted, flat bone having a **rounded, smooth superior border** and a **sharp, thin inferior border**.
- **The inferior border, the costal groove accommodates** the **intercostal nva.**
- The anterior end of each rib is attached to the corresponding **costal cartilage.**
- The posterior end articulates with the vertebral column and is characterized by a **head, neck, and tubercle.**

Typical ribs (3rd - 9th)

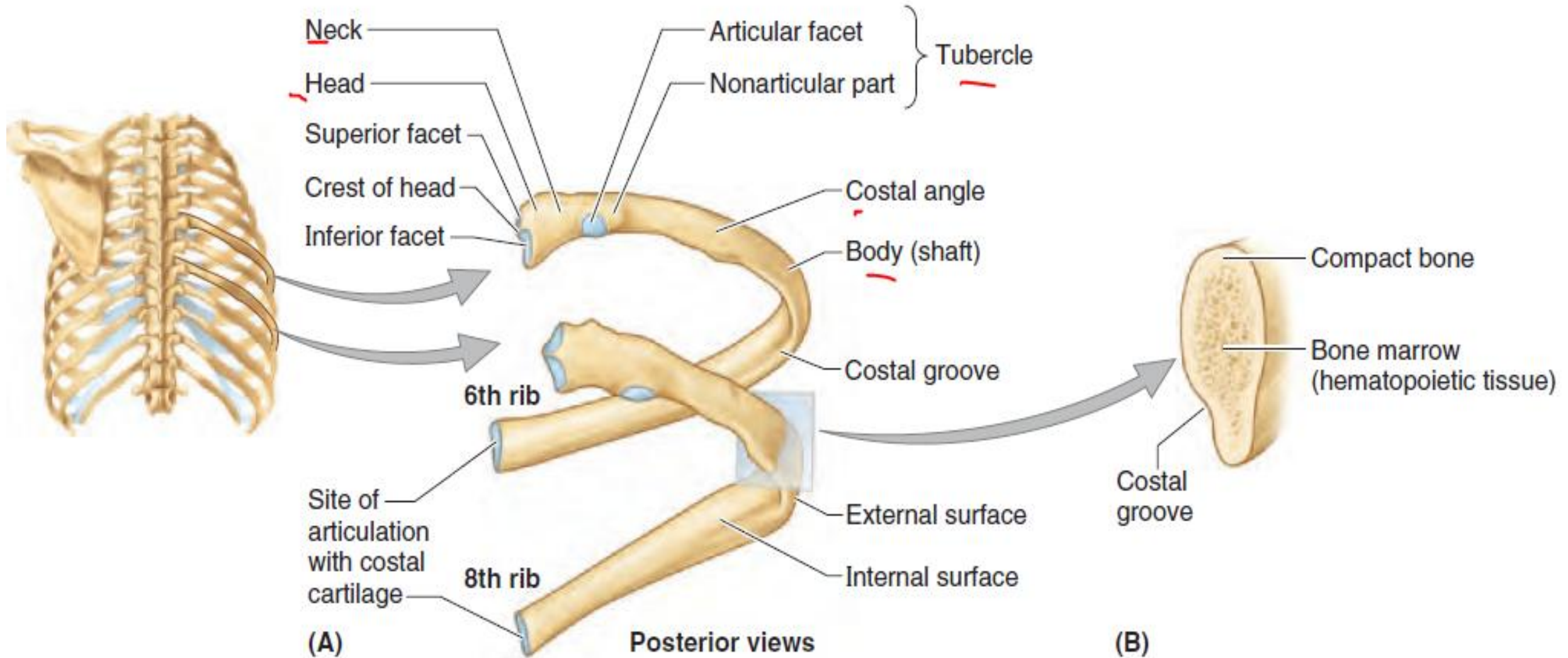


Parts of typical ribs

- **Head:** wedge-shaped and has **two facets**, separated by the **crest of the head**. one facet for **articulation with the numerically** corresponding vertebra and **one facet for the vertebra superior to it.**
- **Neck:** connects the head the of rib with the body at the level of the tubercle.

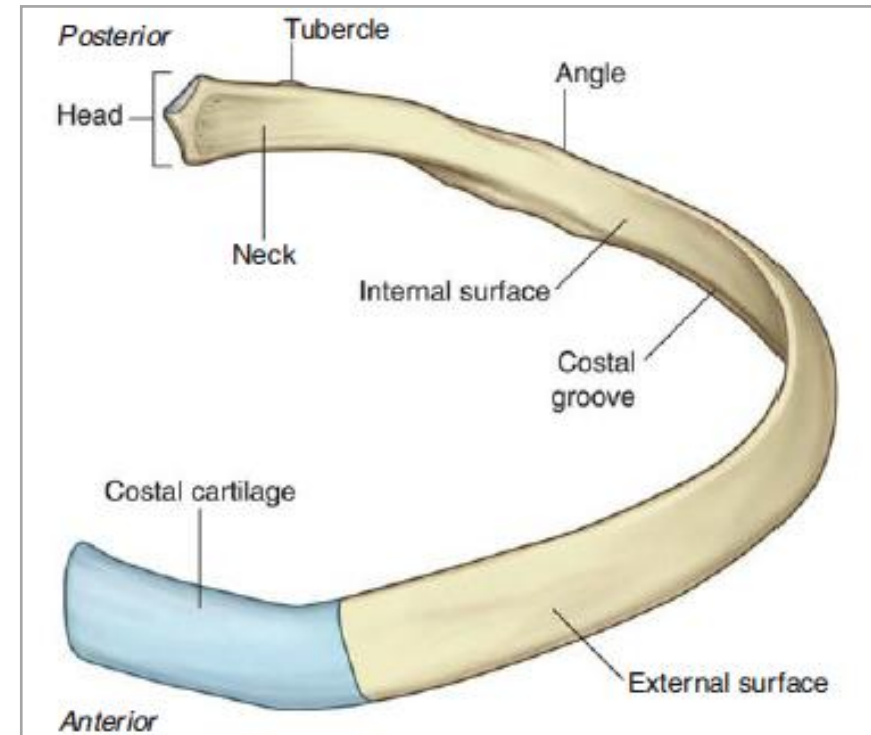


Parts of typical ribs,,,

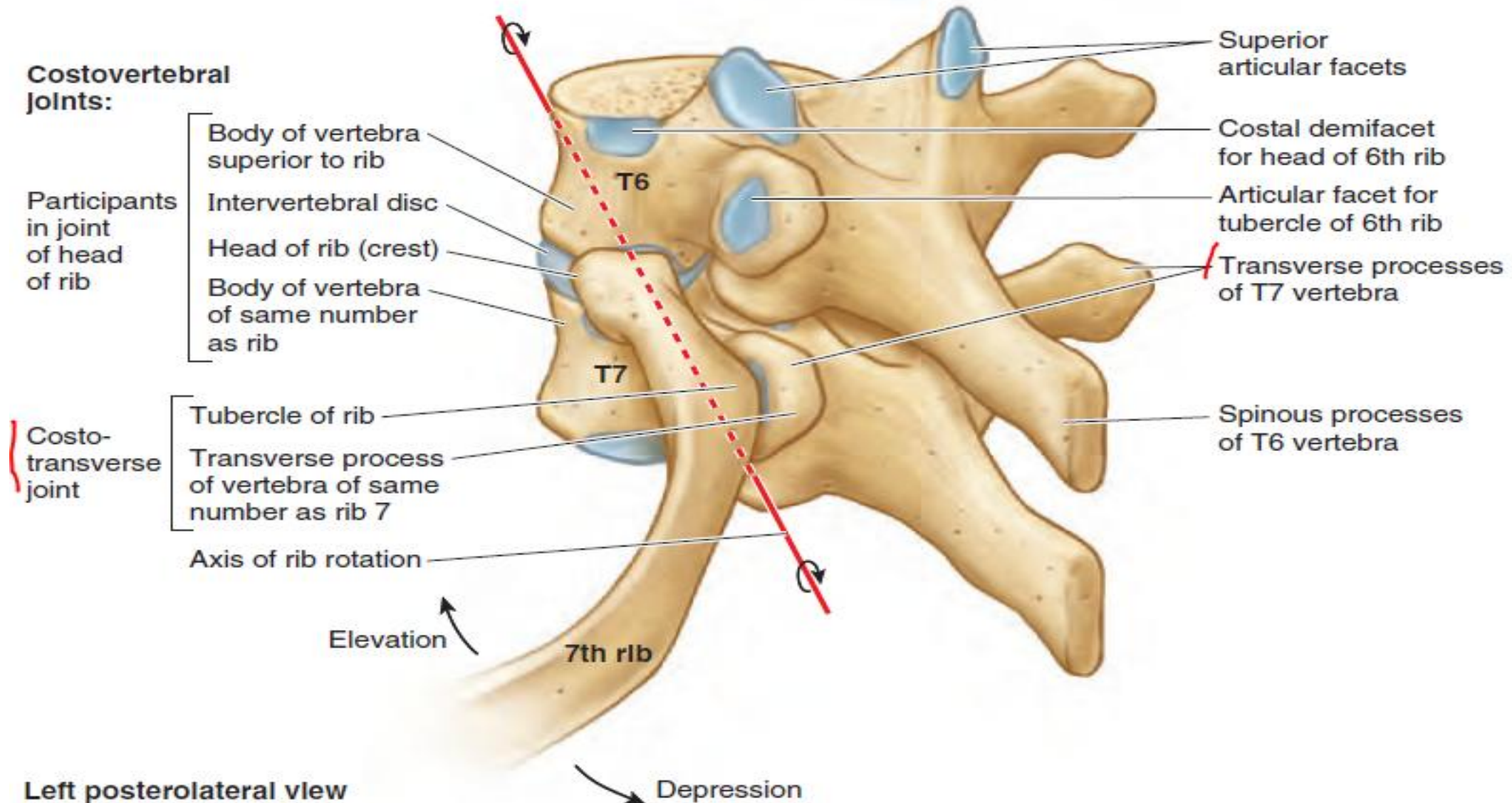


Parts of typical ribs,,,

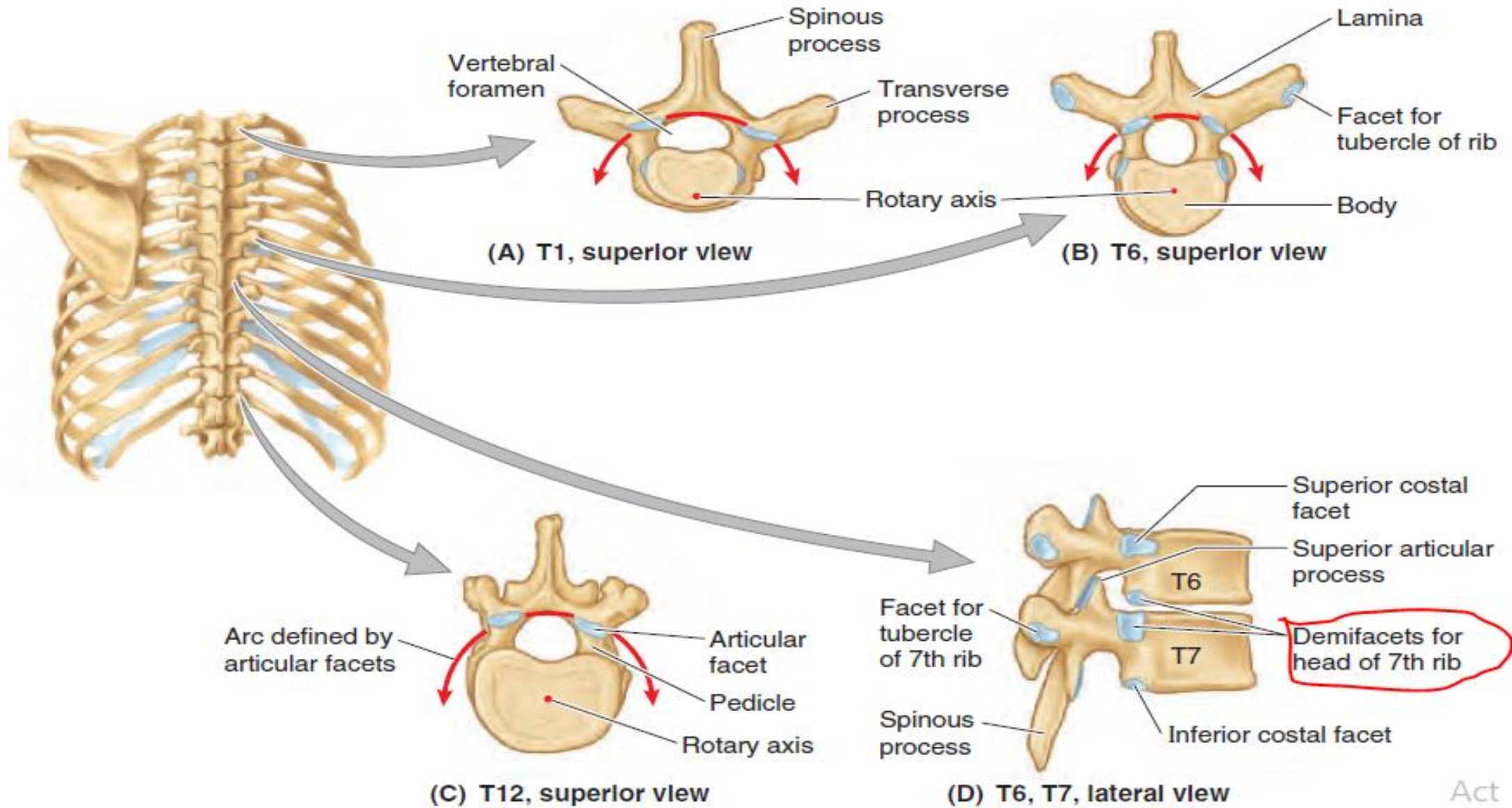
- **Tubercle:** located at the junction of the **neck** and **body**; a smooth *articular part* articulates with the corresponding **transverse process** of the vertebra, and a rough **nonarticular part** provides **attachment for the costotransverse ligament**.
- **Body (shaft):** thin, flat, and curved.
- Is concave on the internal surface but convex on their external.



Parts of typical ribs,,,



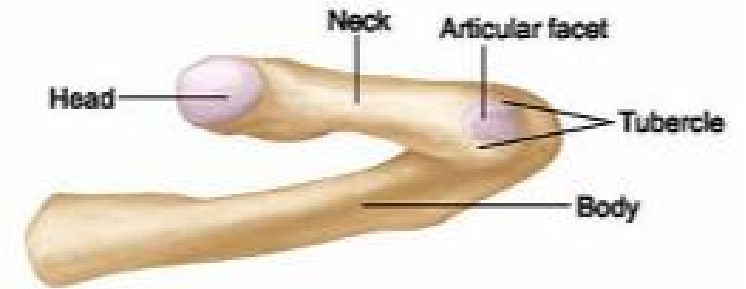
Parts of typical ribs,,,



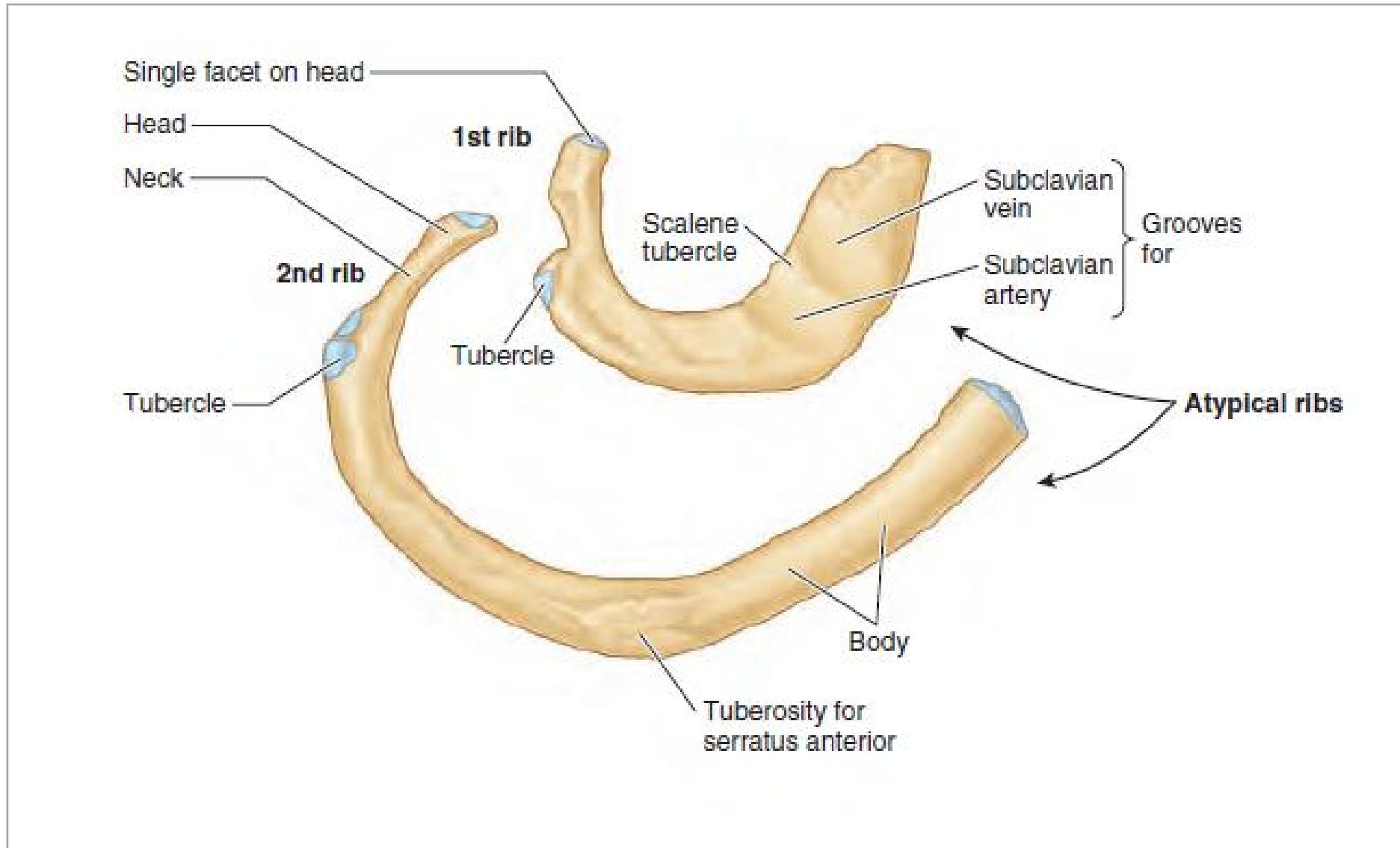
Atypical (1st, 2nd, and 10th–12th)

- These ribs are dissimilar each other.
- The 1st rib: is the broadest, shortest, and most sharply curved of the seven true ribs.
- It has *a single facet* on its head for articulation with the T1 vertebra only.
- Has two transversely directed *grooves* crossing its superior surface for the subclavian vessels; the grooves are separated by *a scalene tubercle* and *ridge*, to which the **anterior scalene** muscle is attached.

First rib, posterior view

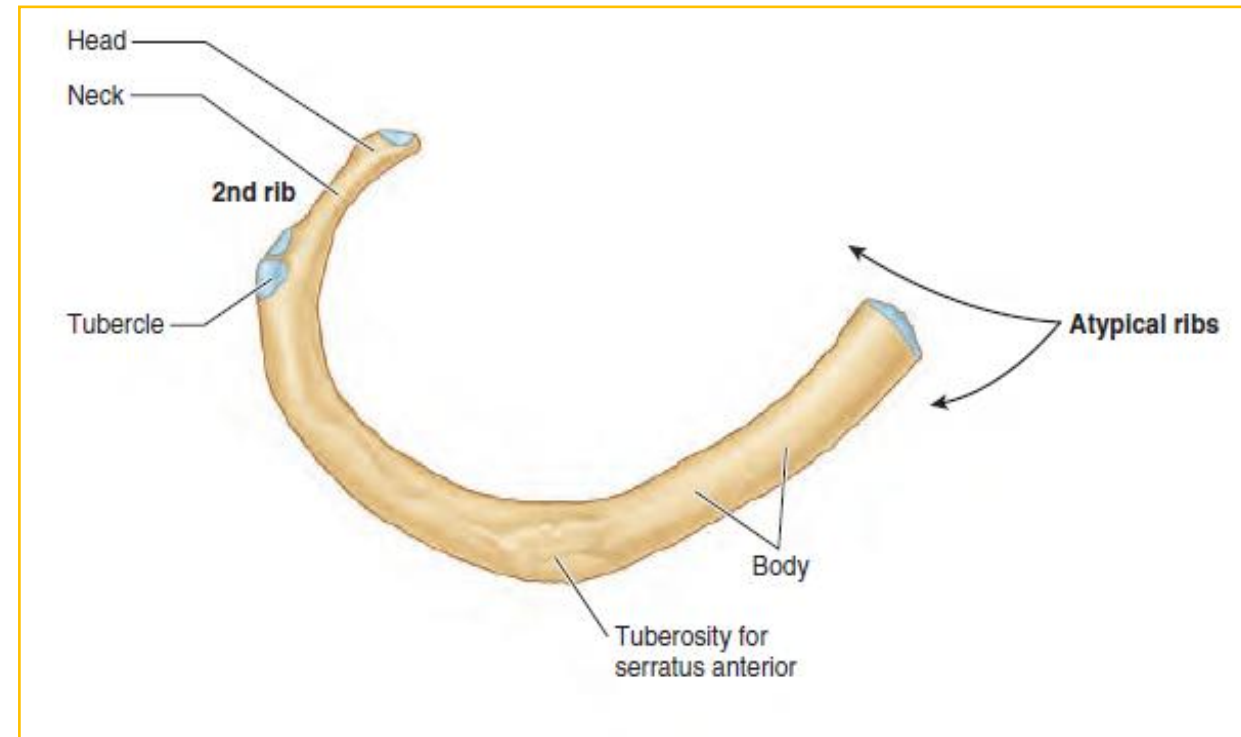


Atypical ribs (1st, 2nd, and 10th–12th),,,



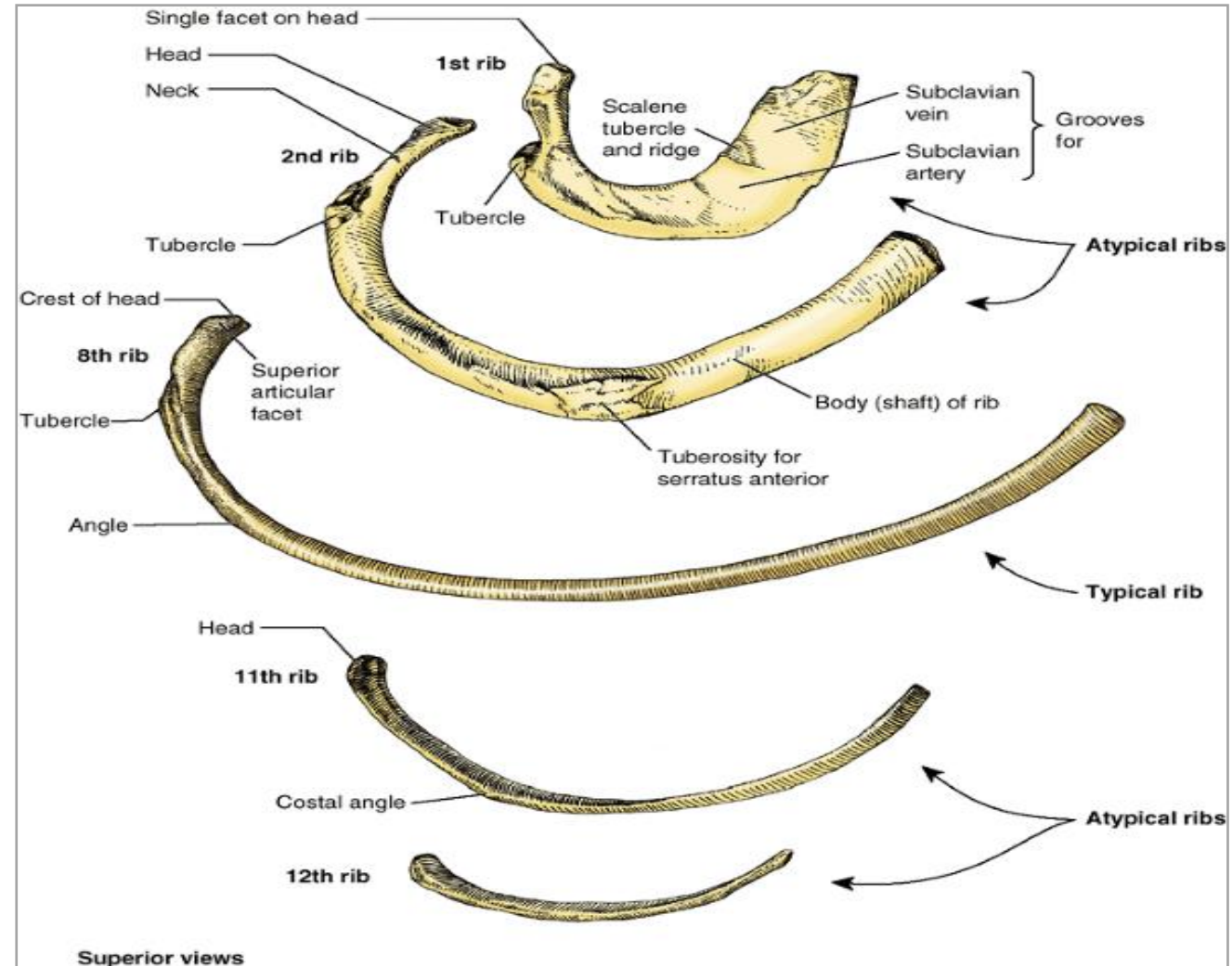
Atypical ribs (1st, 2nd, and 10th–12th),,,

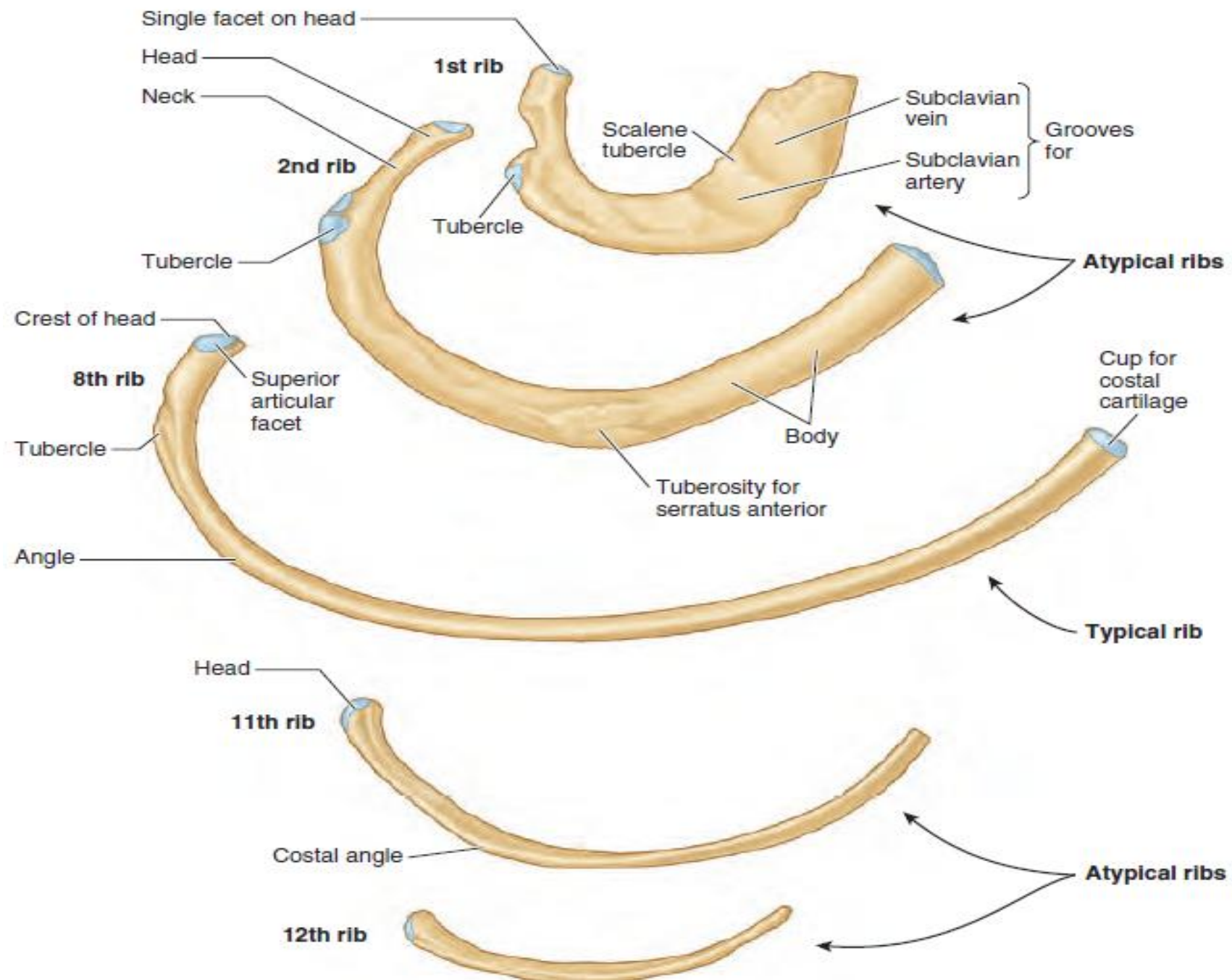
- **The 2nd rib:** Has a thinner, less curved body and is longer than the 1st rib.
- ✓ Its head has two facets for articulation with the bodies of the T1 and T2 vertebrae
- ✓ Its main atypical feature is a rough area on its upper surface, *the tuberosity for serratus anterior*, from which part of that muscle originates.



Atypical ribs (1st, 2nd, and 10th–12th),,,

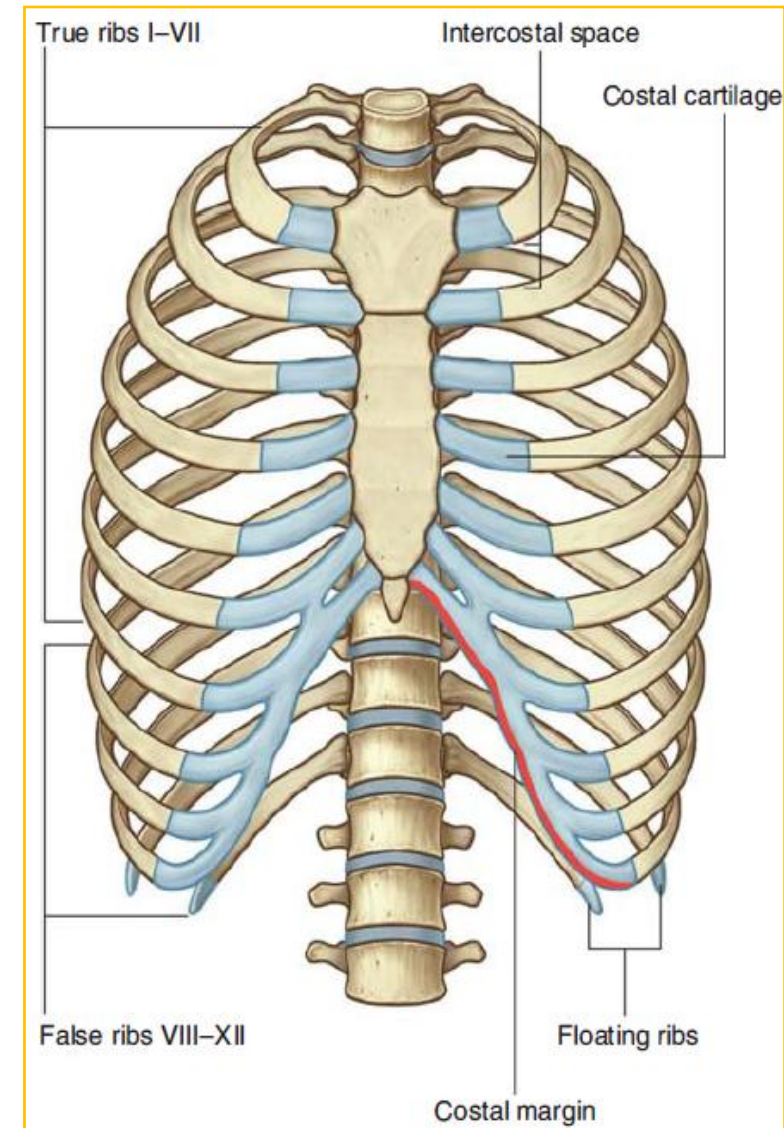
- The 10th - 12th ribs: like the 1st rib, have only one facet on their heads and articulate with a single vertebra
- The 11th and 12th ribs: are short and have no neck or tubercle.





Costal cartilages

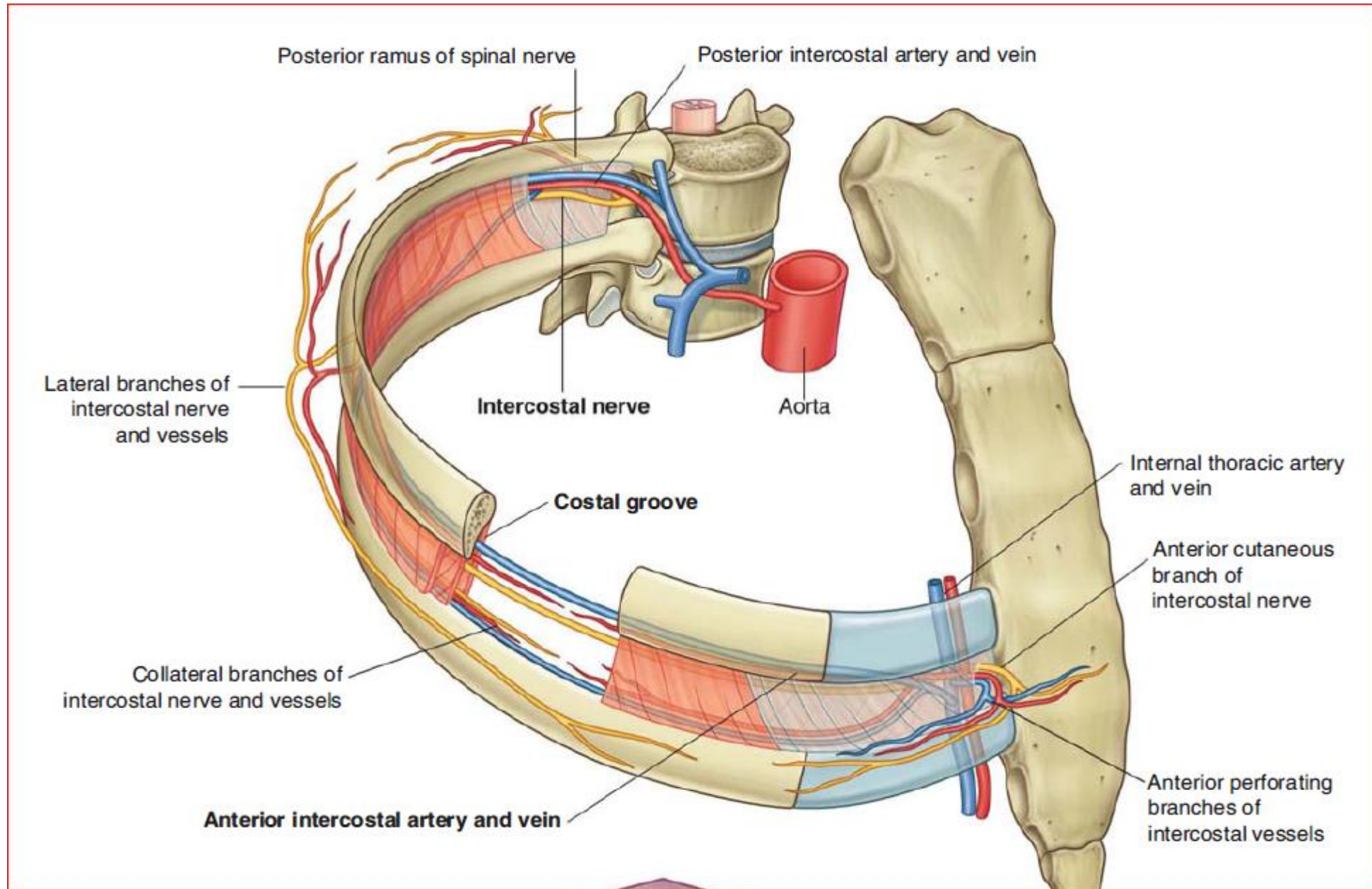
- Prolong the ribs anteriorly and contribute significantly to the **elasticity** and **mobility** of the thoracic walls.
- The 8th, 9th, and 10th articulate with the costal cartilages just superior to them, forming a **costal margin**.
- The 11th and 12th costal cartilages **do not reach or attach to any other bone or cartilage**.
- **The costal cartilages** of ribs 1–10 clearly anchor the anterior end of the rib to the sternum, limiting its overall movement.



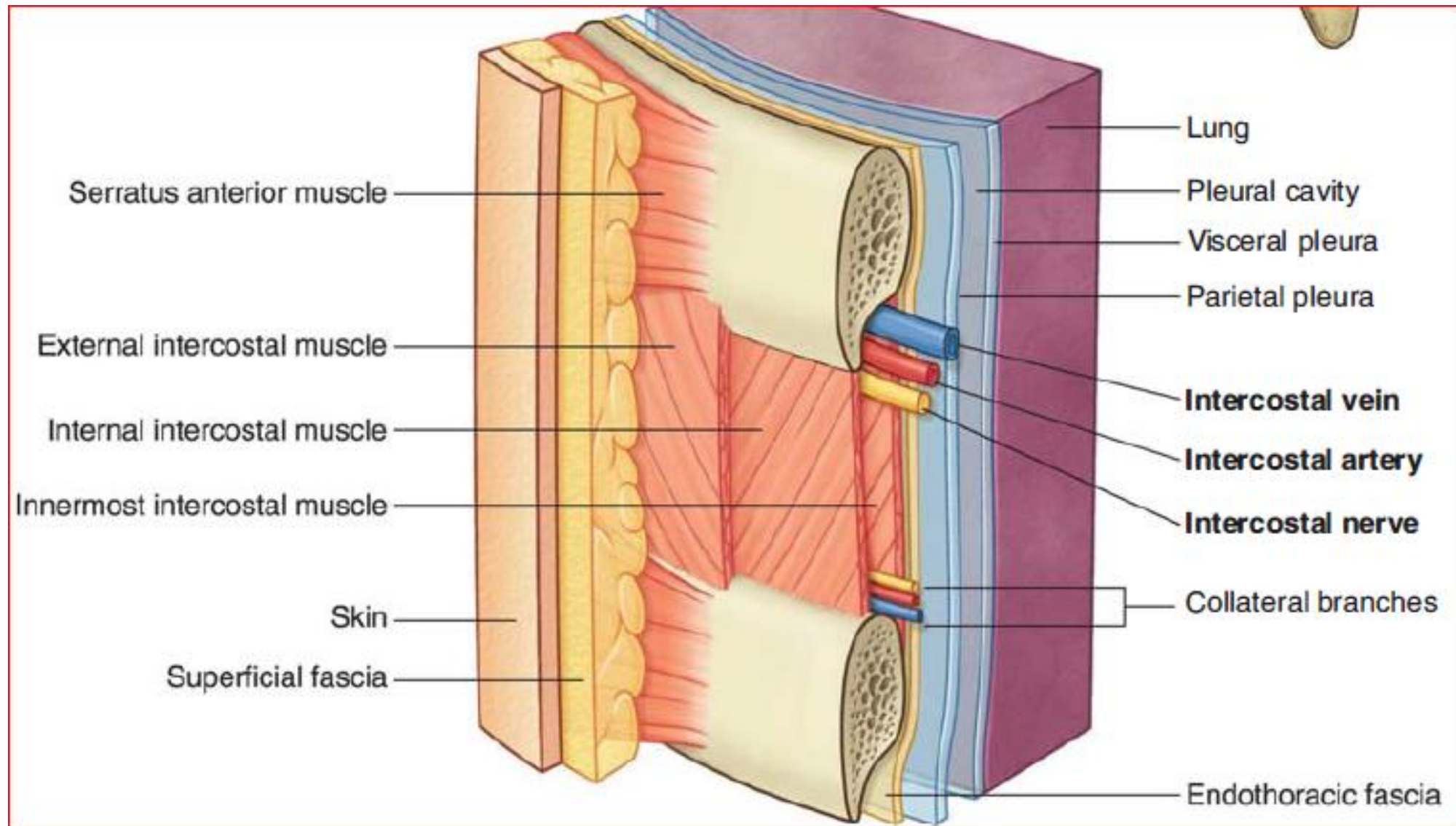
Intercostal spaces

- **Intercostal spaces** separate the ribs and their costal cartilages from one another.
- The spaces are named according to the rib forming its superior border.
- There are 11 intercostal spaces.
- The space below the 12th rib does not lie b/n ribs and is referred to as the **subcostal space**
- They are occupied by **intercostal muscles & membranes** & two sets (main & collateral) of intercostal blood vessels and nerves,

Intercostal spaces,,

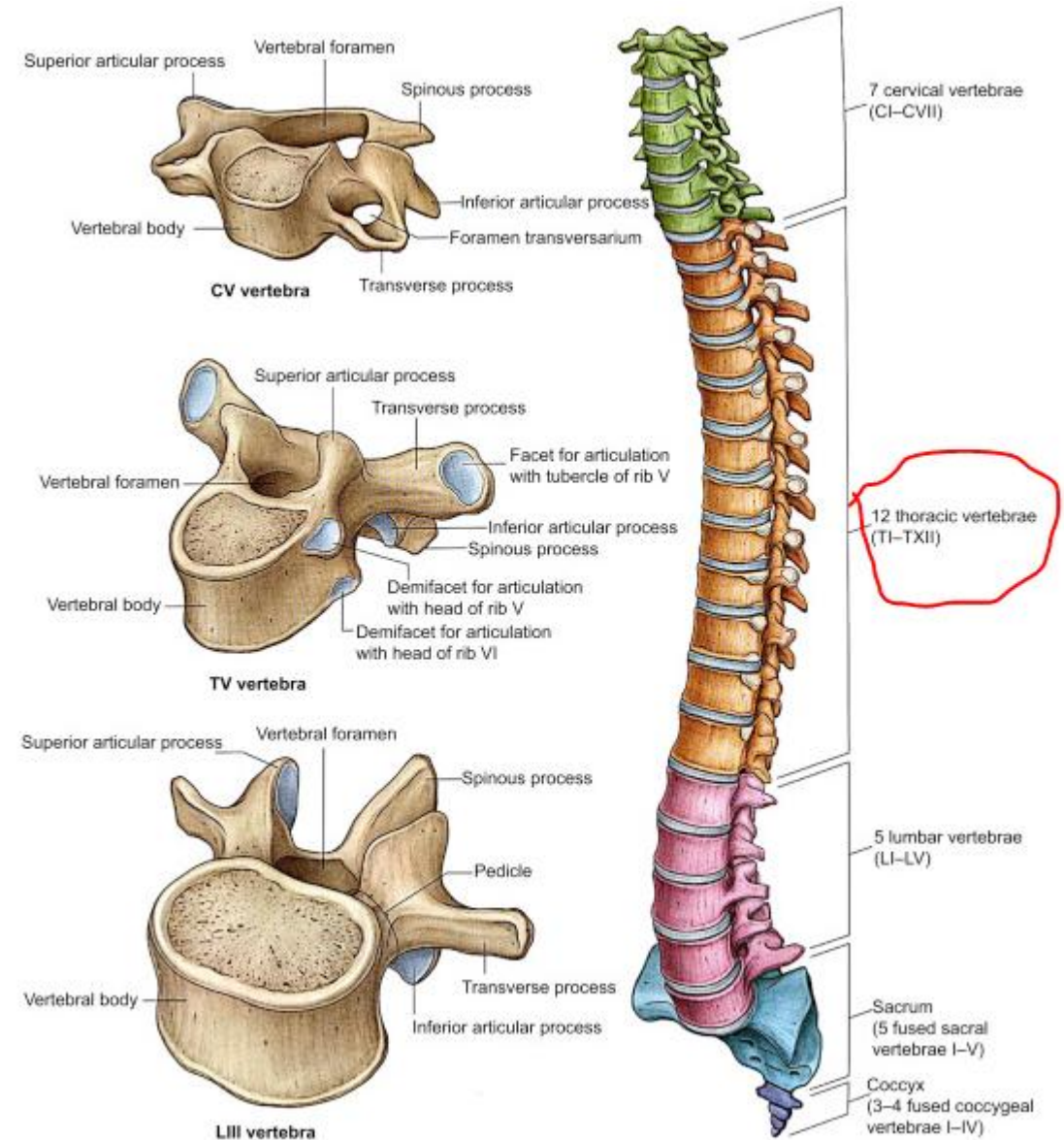


Intercostal spaces,,,



Thoracic vertebrae

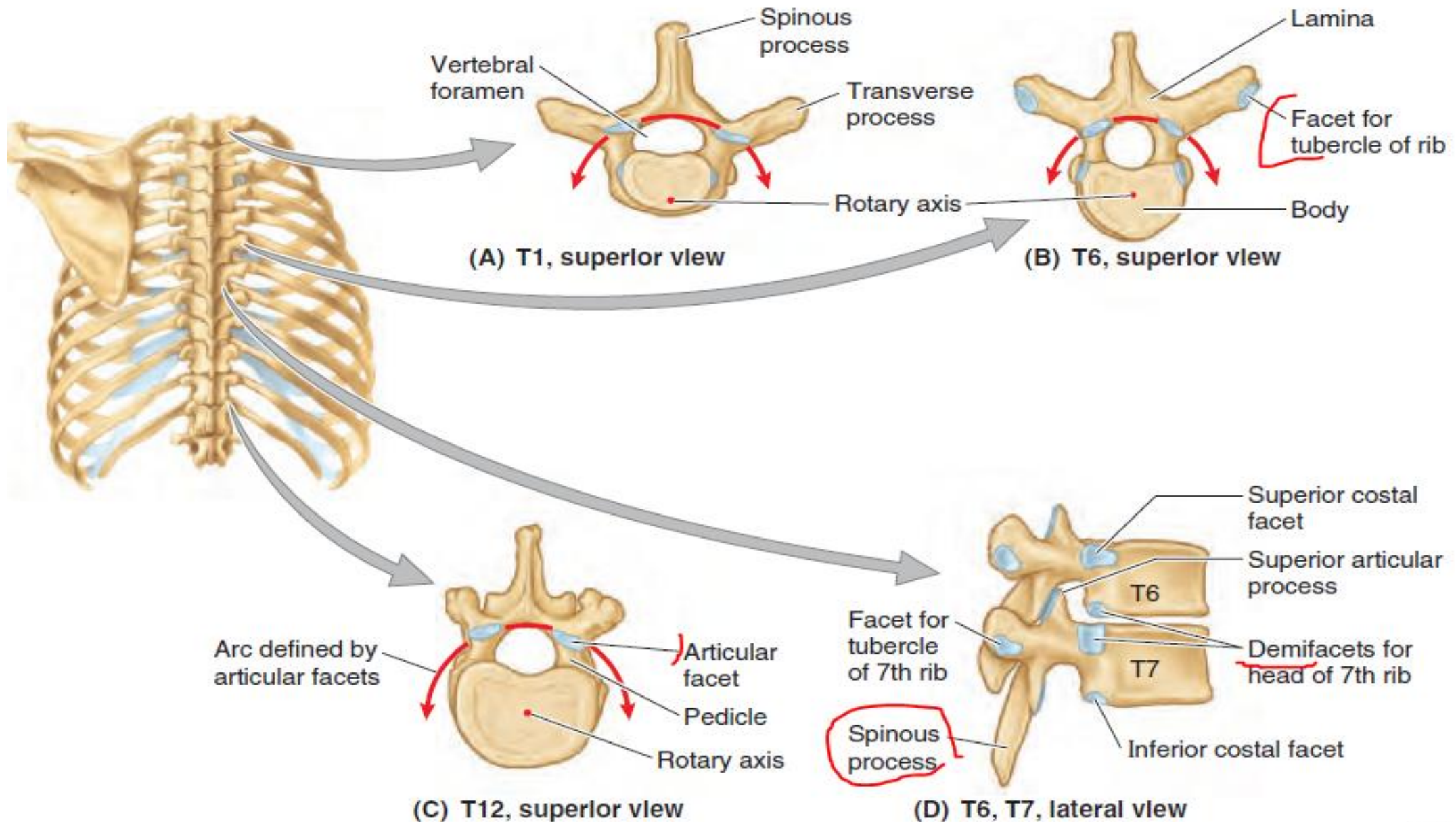
- The bony framework thoracic wall is held together by twelve thoracic vertebrae
- ✓ There are 12 thoracic vertebrae.
- ✓ Thoracic vertebrae are **typical vertebrae** in that they are independent and **have bodies, vertebral arches, and seven processes for muscular and articular connections.**



Characteristic features of typical thoracic vertebrae

- ✓ A typical thoracic vertebra has a **heart-shaped vertebral body**
- ✓ Roughly equal dimensions in the transverse and anteroposterior directions.
- ✓ **Bilateral costal facets (demifacets) on their bodies:** occurring in inferior and superior pairs, for articulation with the heads of ribs.
- ✓ **Costal facets on their transverse processes:** for articulation with the tubercles of ribs, except for the inferior two or three thoracic vertebrae.
- ✓ **Long, inferiorly slanting spinous processes.**

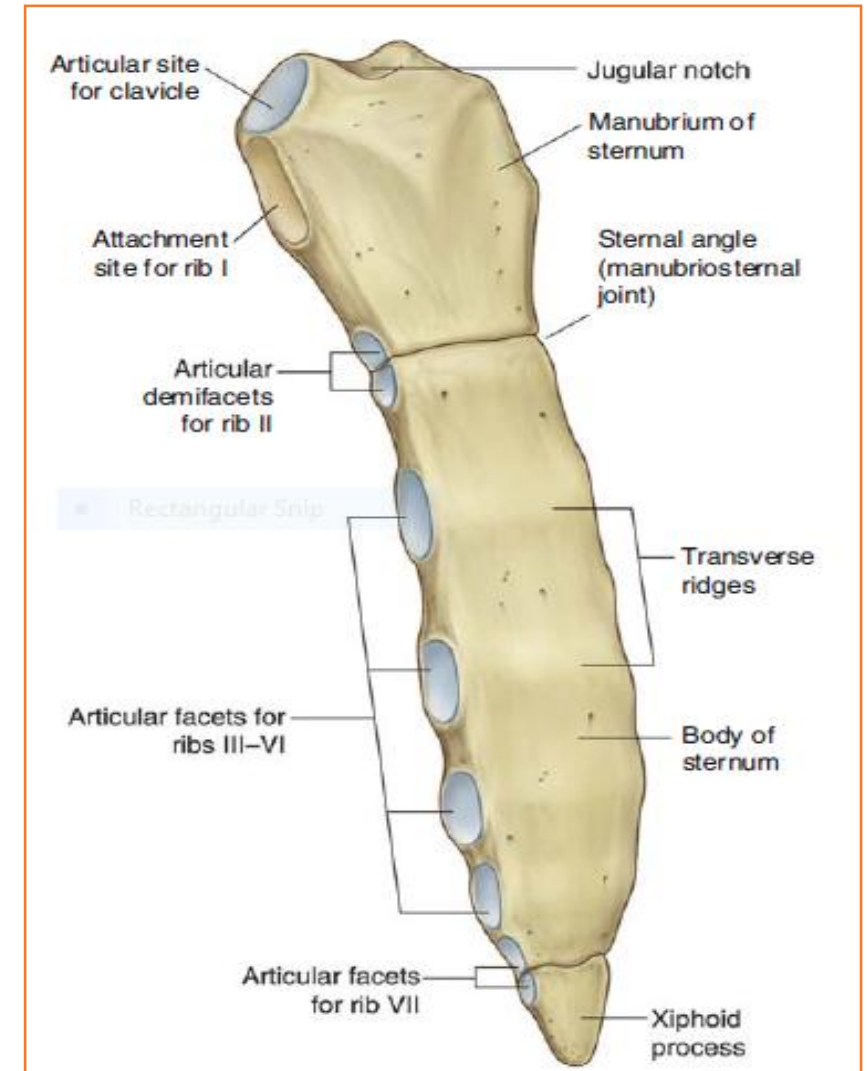
Characteristic features of typical thoracic verte,,



The sternum

- The **sternum** (Gr. *sternon*, chest) is the flat, elongated bone that forms the **middle** of the anterior part of the thoracic cage.
- lies subcutaneously in the **anterior median line** and is palpable through out its length.
- affords protection for mediastinal viscera in general and much of the heart in particular.

Sternum



The sternum,,,

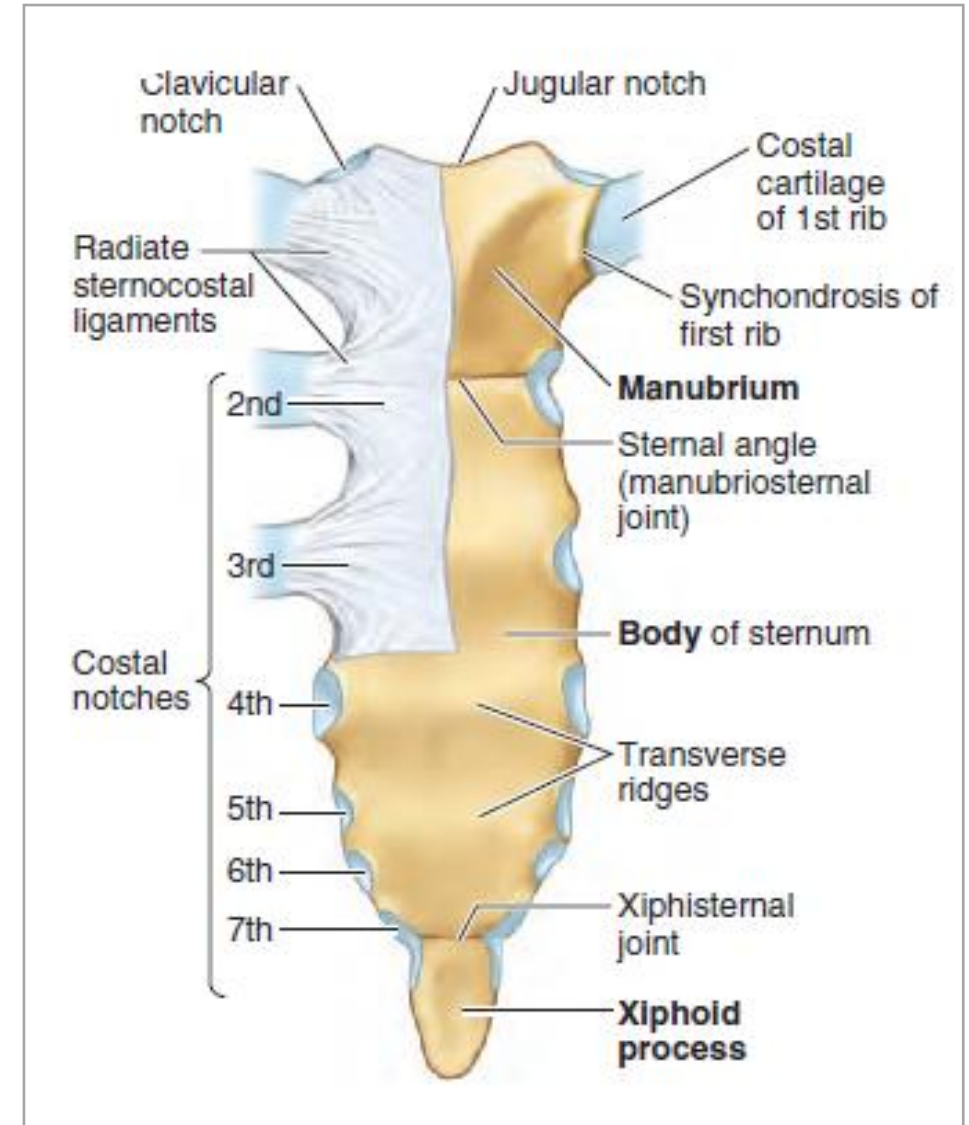
- The sternum consists of **three parts**.

- **manubrium**,

- **body**, and

- **xiphoid process**

❖ In adolescents and young adults, the 3 parts are connected by cartilaginous joints that ossify **during middle to late adulthood**.

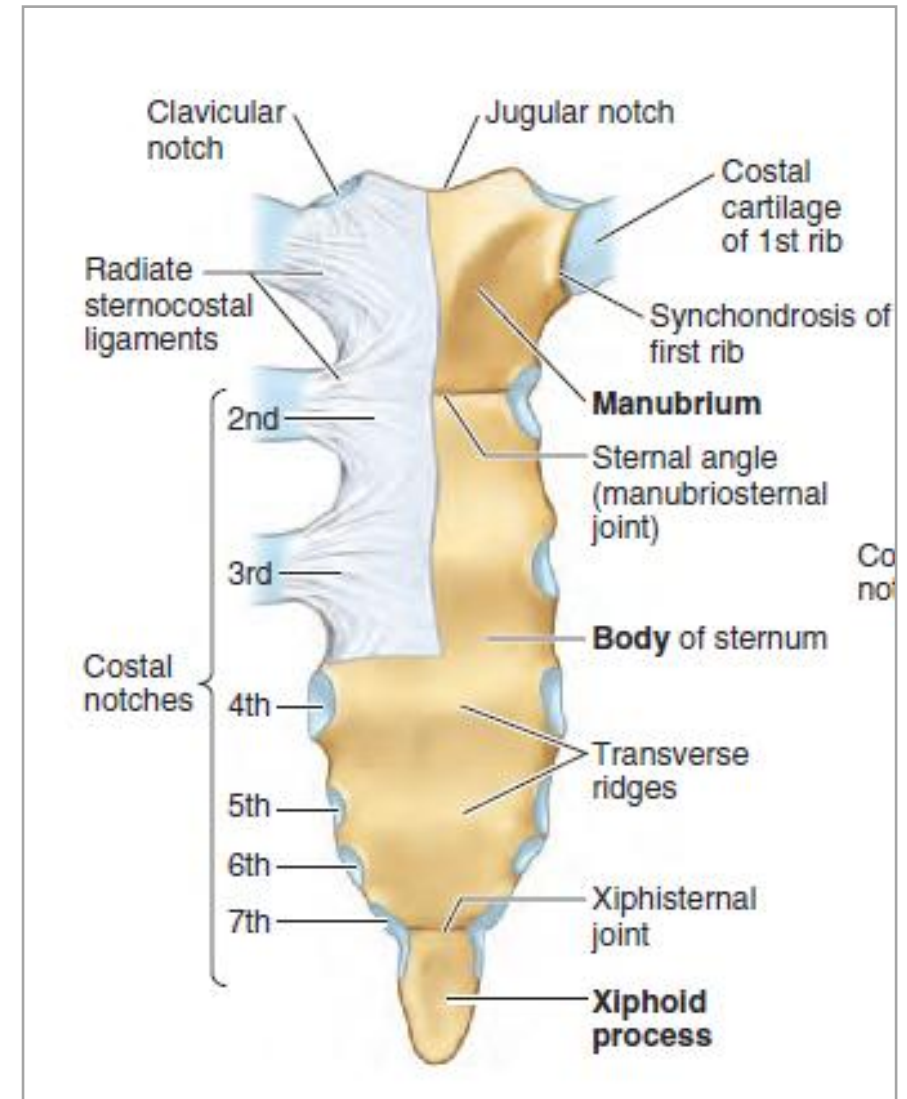


The manubrium

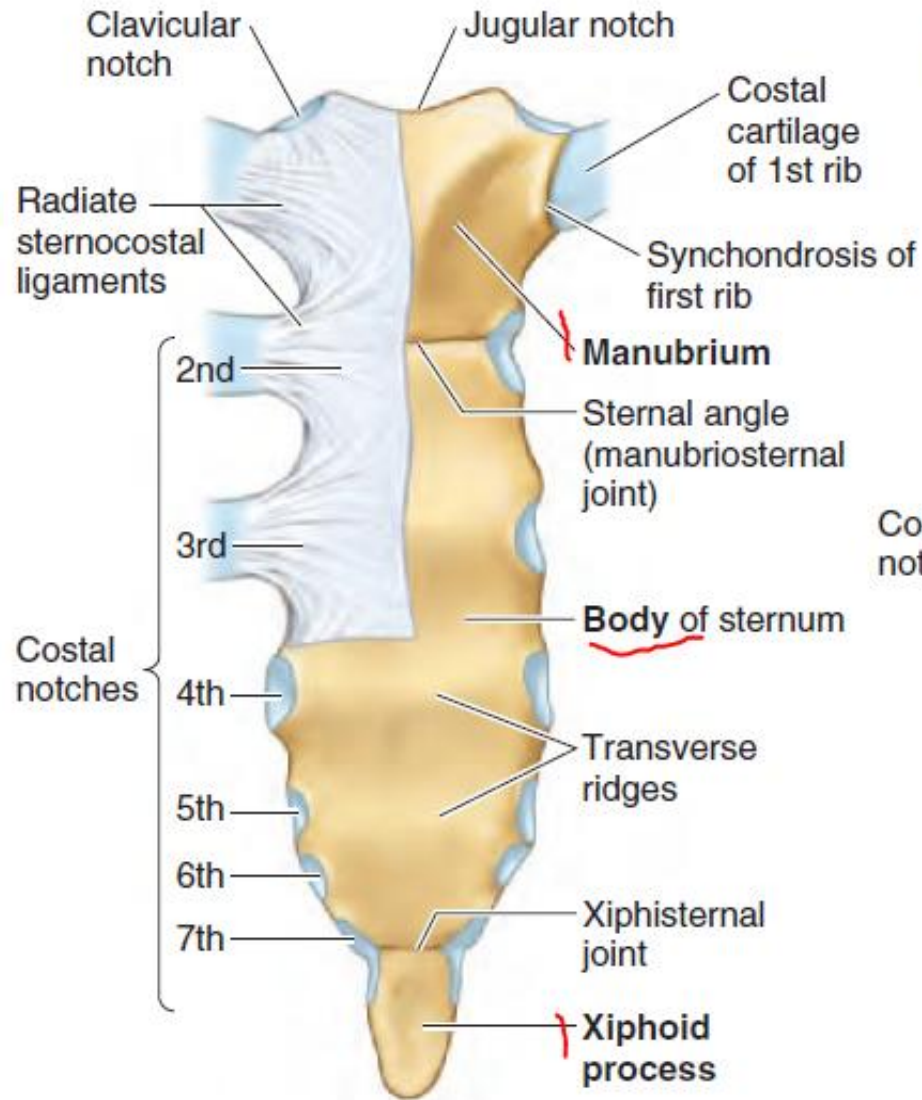
- The **manubrium** (L. handle) is a roughly trapezoidal bone.
- It is the widest & thickest of the 3 parts of the sternum.
- The concave center of the superior border of the manubrium is the **jugular notch** (**suprasternal notch**).
- The medial (sternal) ends of the clavicles forms *the sternoclavicular (SC) joints with suprasternal notch*.
- The manubrium & body lie in slightly different planes superior & inferior to their junction, the **manubriosternal** junction forms a projecting **sternal angle** (of Louis).

The body of the sternum

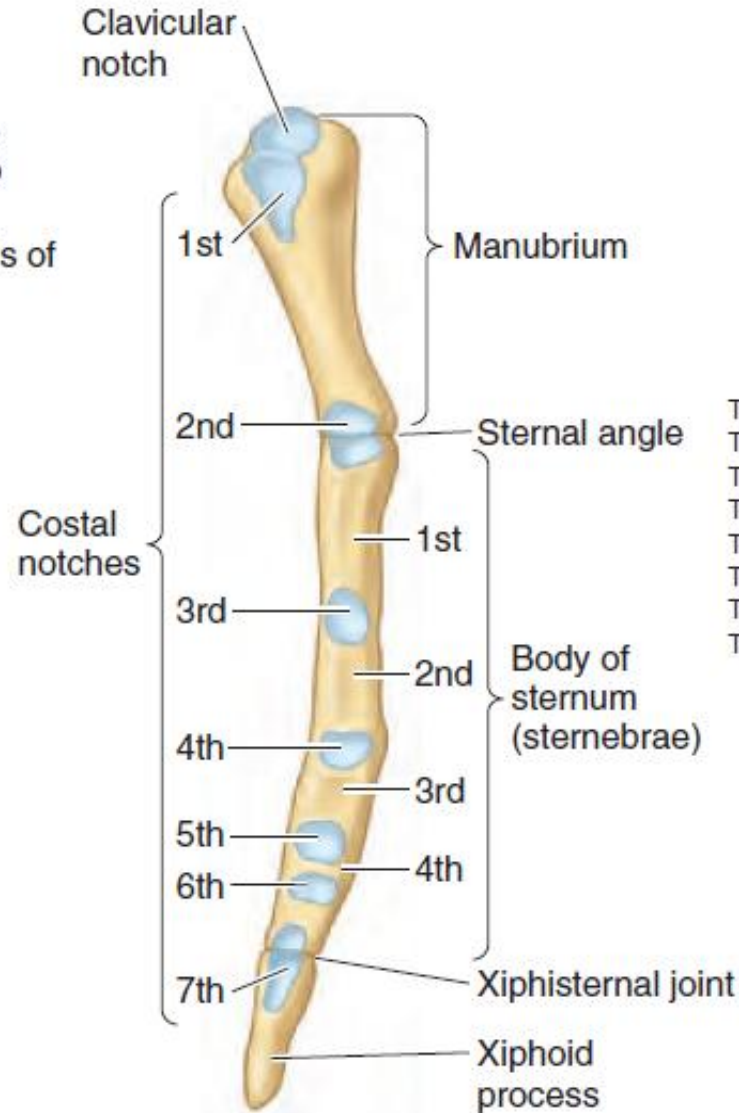
- The **body of the sternum**, is longer, narrower, and thinner than the manubrium, and is **located at the level of the T5–T9 vertebrae**.
- In young people, four *sternebrae* (primordial segments of the sternum) are obvious.



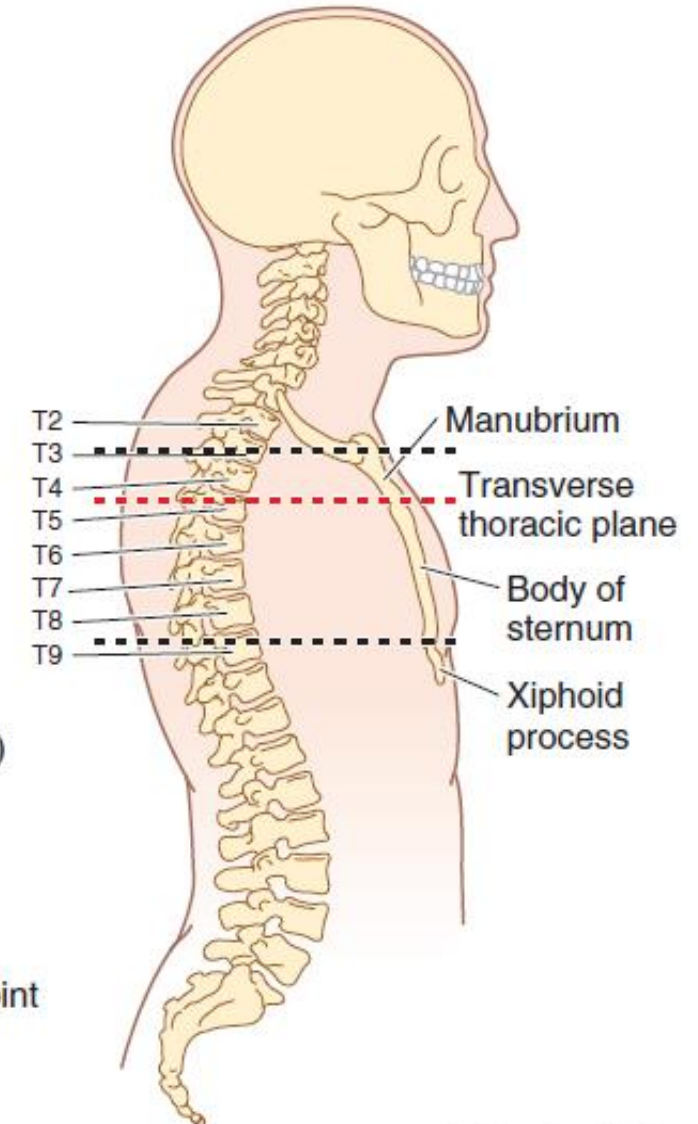
The body of the sternum...



(A) Anterior view



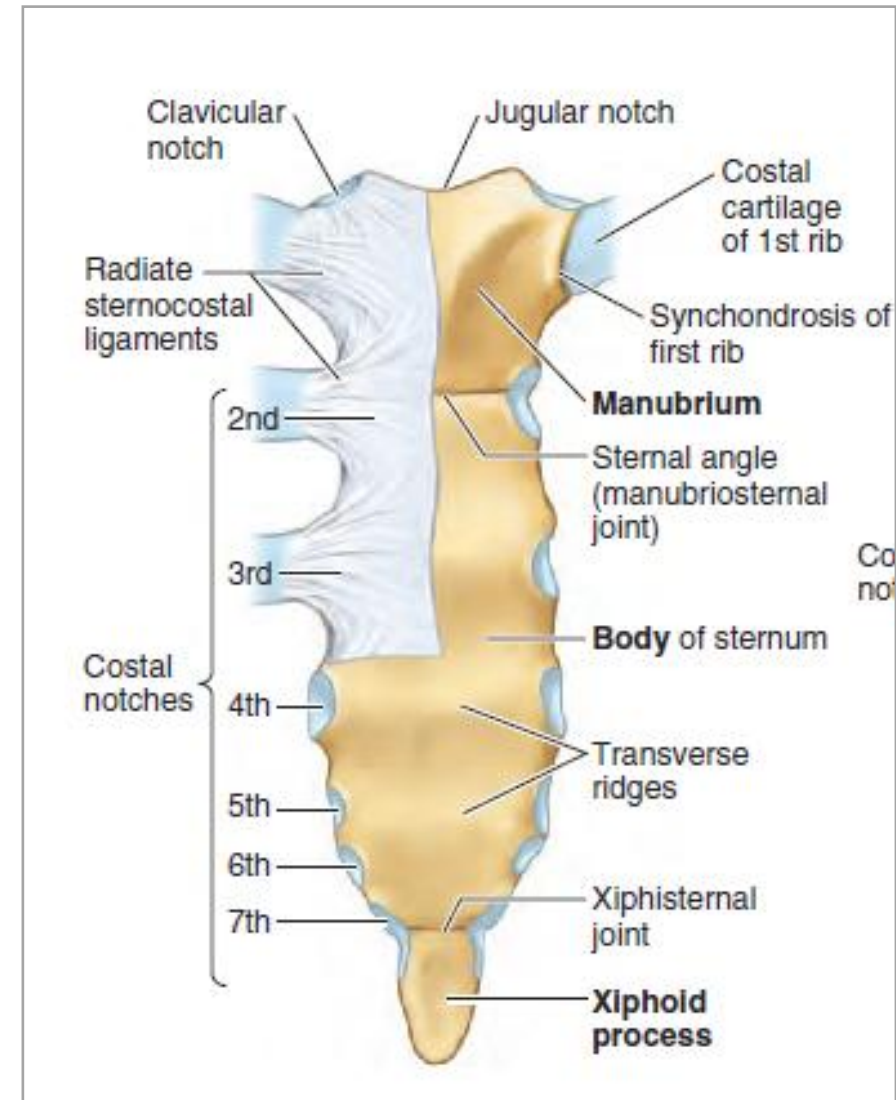
(B) Lateral view



(C) Lateral view

The xiphoid process

- Is the smallest and most variable part of the sternum.
- Is thin and elongated.
- Lies at the level of T10 vertebra.
- Although often **pointed**, the process may be **blunt**, **bifid**, curved, or deflected to one side or anteriorly.
- In elderly people, the xiphoid process may fuse with the sternal body.



The xiphoid process,,,

- **The xiphoid process is an important landmark in the median plane**
because: Its junction with the sternal body at the **xiphisternal joint** indicates
 - ✓ the inferior limit of the central part of the thoracic cavity
 - ✓ the site of the **infrasternal angle** formed by the **right and left** costal margins .
 - ✓ superior limit of **the liver,**
 - ✓ **the central tendon of the diaphragm, and**
 - ✓ **the inferior border of the heart.**

Joints of the Thoracic Wall

- Movements of the joints of the thoracic wall are frequent but the range of movement at the individual joints is relatively small.
 - **For example**, in association with normal respiration the range of movement at the individual joints is relatively small;
 - ✓ **Nonetheless**, any disturbance that reduces the mobility of these joints interferes with respiration.
 - ✓ **During deep breathing**, the excursions of the thoracic cage (anteriorly, superiorly, or laterally) are considerable.

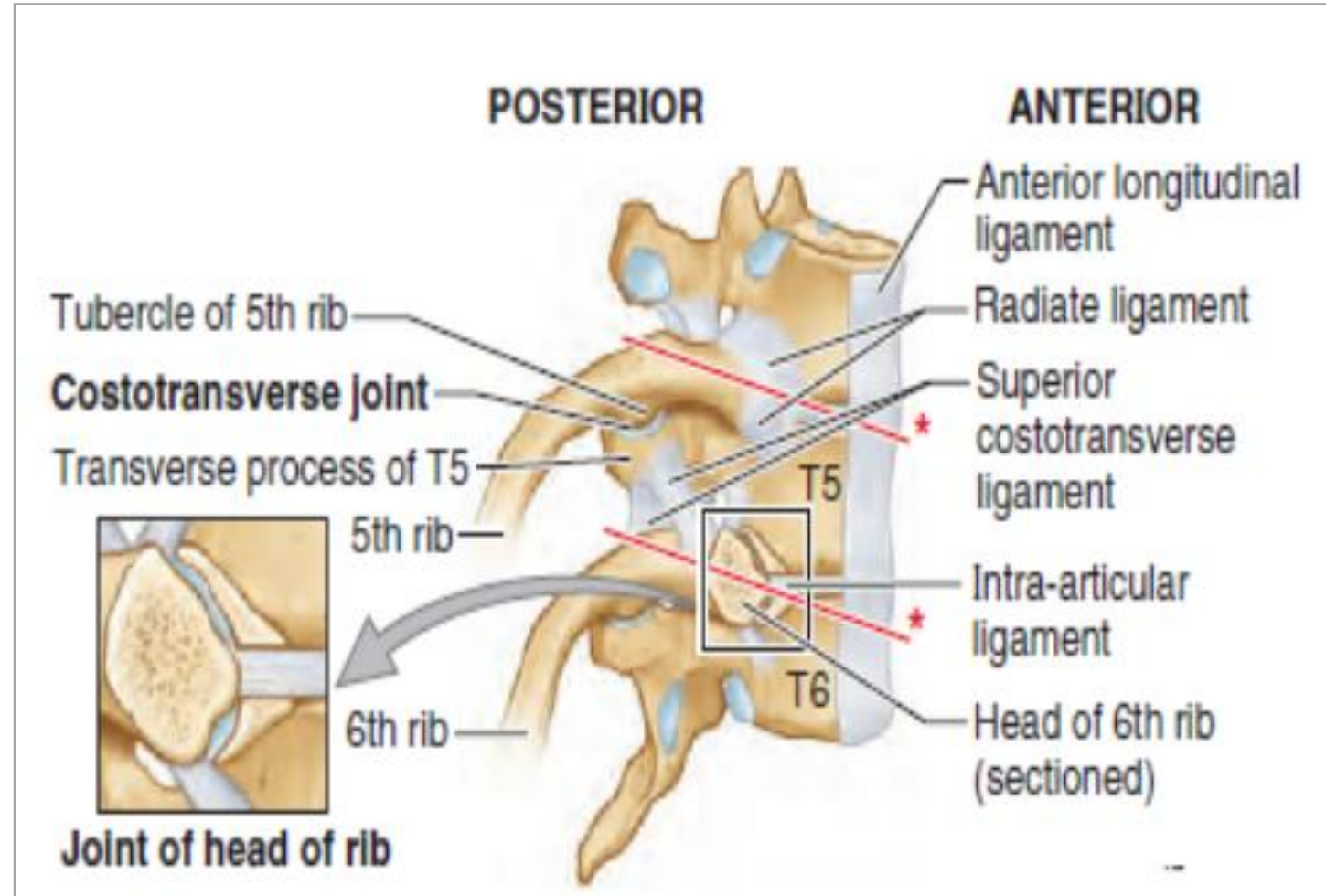
Joints of the Thoracic Wall,,

❖ **Joints of the thoracic wall occur between the:**

- Vertebrae (**intervertebral** joints)
- Ribs and vertebrae (**costovertebral** joints)
- Ribs and costal cartilages (**costochondral** joints)
- Between costal cartilages (**interchondral** joints)
- Sternum and costal cartilages (**sternocostal** joints)
- Sternum and clavicle (**sternoclavicular** joints)
- Parts of the sternum (**manubriosternal** and **xiphisternal** joints) in young people

Costovertebral Joints

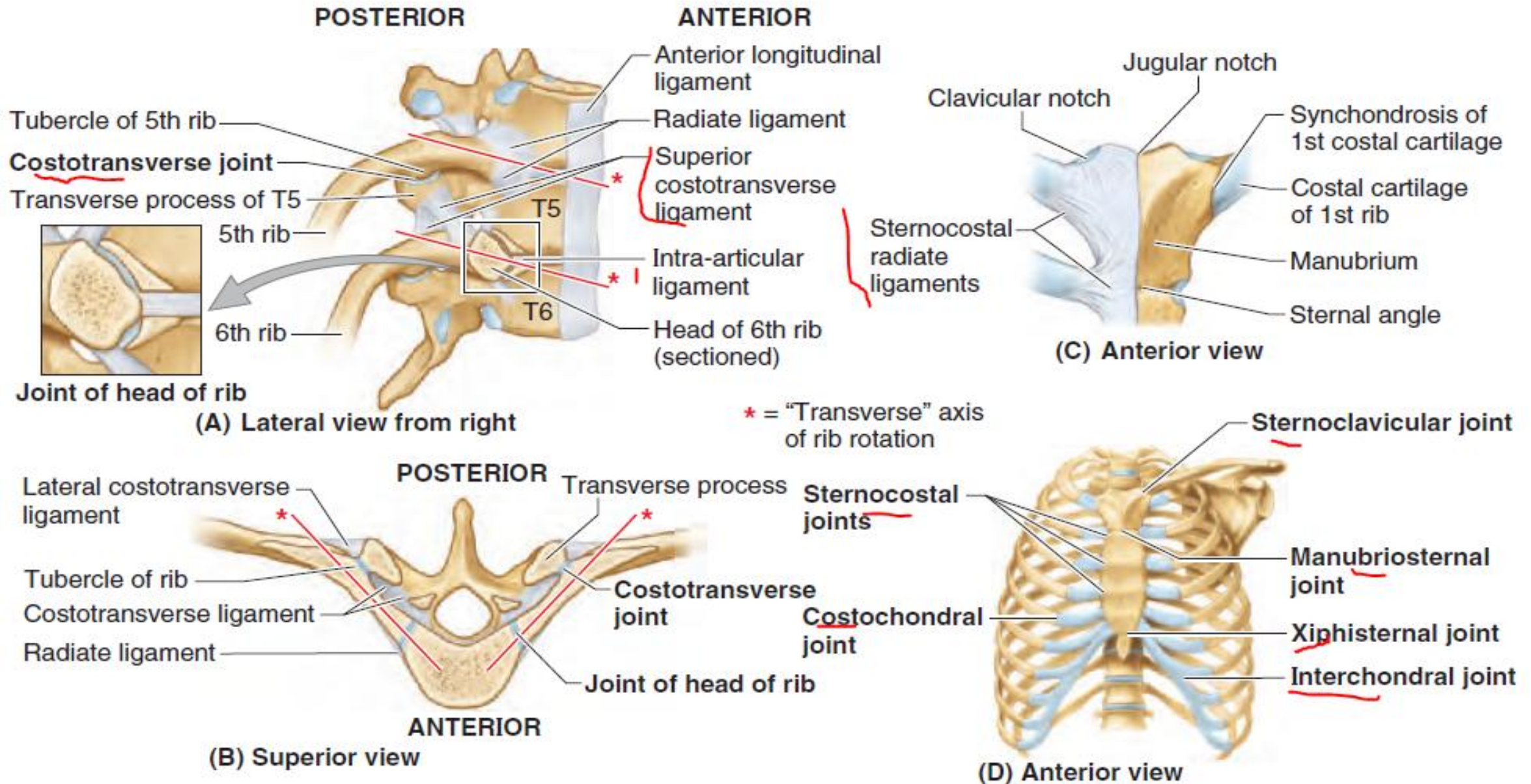
- A typical rib articulates with the vertebra at 2 joints.
- **A) b/n head of ribs & costal facet of vertebral body.**
- ✓ This is **synovial plane** type of joint.
- ✓ **Radiate and intraarticular** ligaments assist this joint.



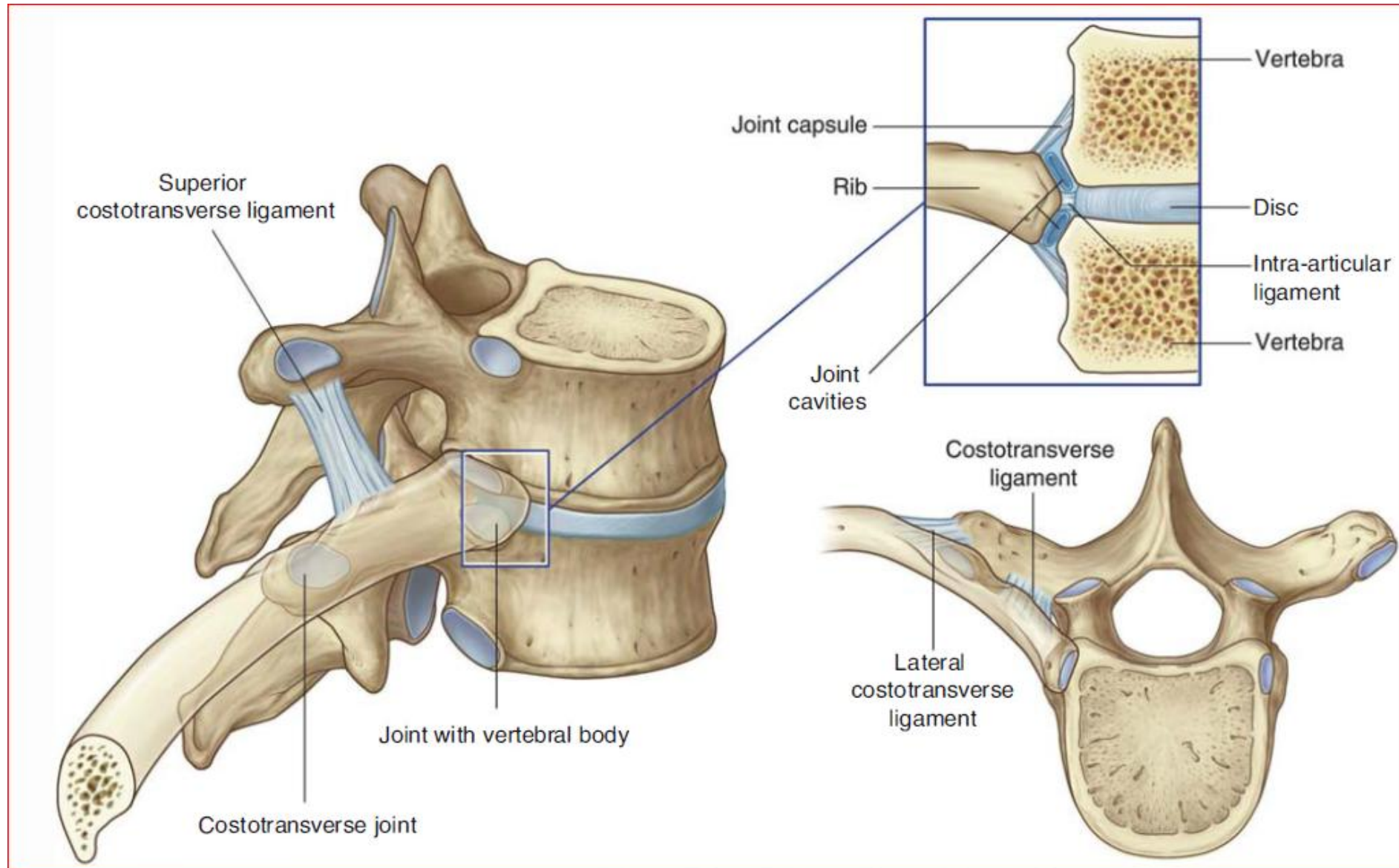
Costovertebral joints,,,

- **B) Costotransverse Joints.** are **synovial joints** b/n the **tubercle of a rib** and the **transverse process** of the related vertebra.
- 3 ligaments stabilize this joint.
 1. **Costotransverse ligament:** passing from the **neck of the rib** to the transverse process of rib of the same number.
 2. **Lateral costotransverse ligament:** passing from the **tubercle of the rib** to the **tip of the transverse process.**
 3. **Superior costotransverse ligament:** passing from neck of the rib to the **transverse process superior to it.**

Costovertebral joints,,



Costovertebral joints,,,



Sternocostal joints

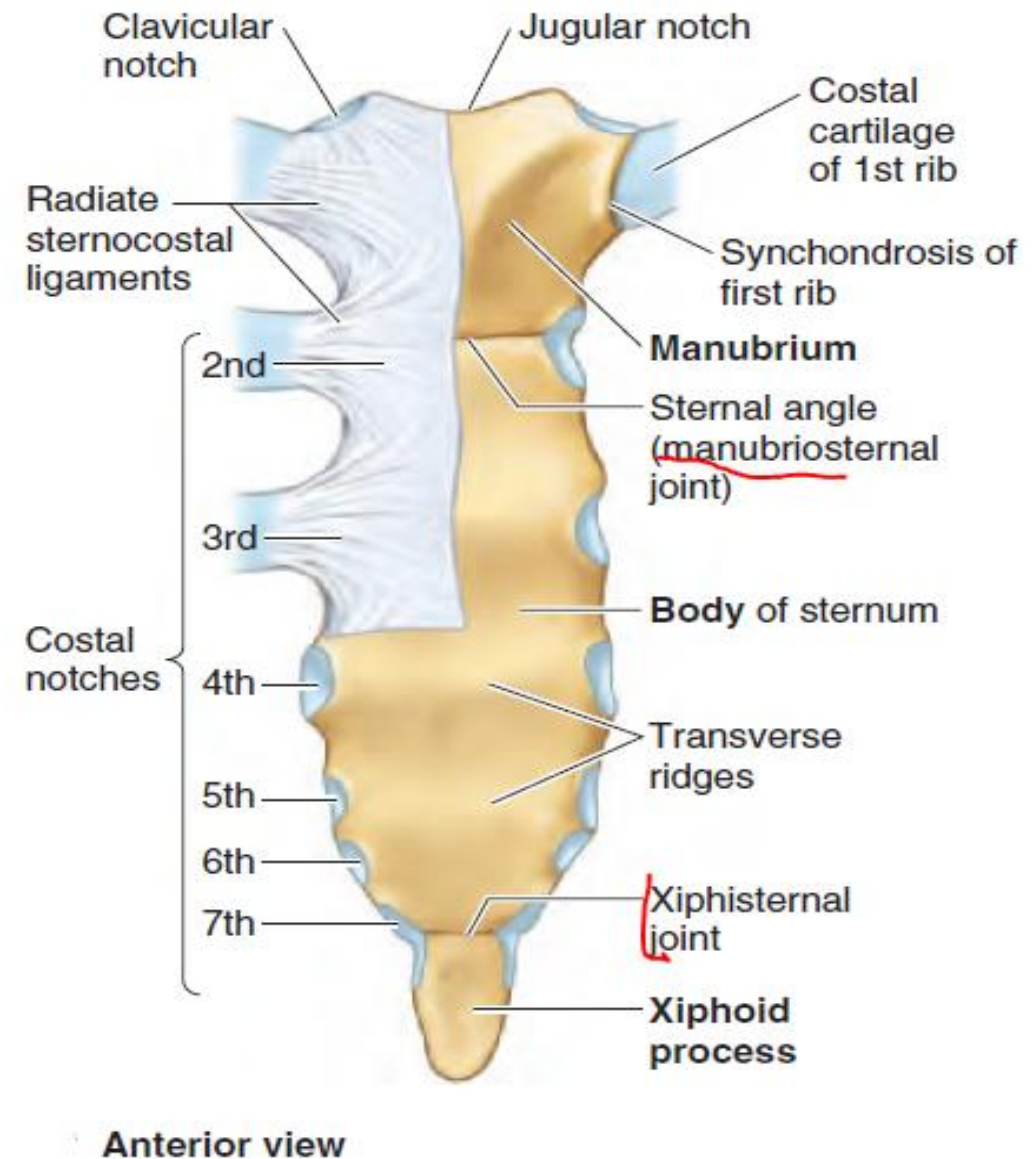
- are joints b/n the upper 7 costal cartilages and the sternum.
- ✓ The 1st costal cartilages articulates with the manubrium by means of a thin dense layer of tightly adherent **fibrocartilage** interposed b/n the cartilage & manbrium, the **synchondrosis of the 1st rib**.
- ✓ The 2nd–7th costal cartilages articulate with the sternum *at synovial joints* with fibrocartilaginous articular surfaces on both the chondral and sternal aspects, allowing movement during respiration.
- **Anterior and posterior radiate sternocostal and intra-articular ligaments support this joint.**

Interchondral joints

- **Occur** mainly b/n costal cartilages of ribs 6 to 10.
- They are **usually synovial plane joint**, and the **thin fibrous capsules** are reinforced by **interchondral ligaments**.

□ ***Manubriosternal and xiphisternal joints:*** The joints b/n the manubrium & body of sternum and b/n the body of sternum and the xiphoid process are usually **symphyses**.

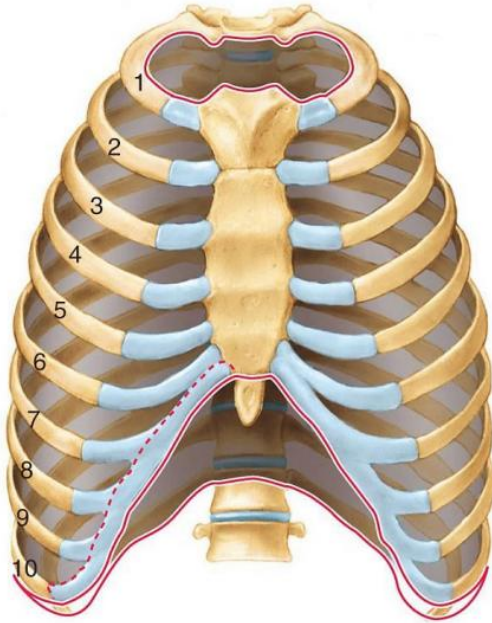
- **Only slight angular movements occur b/n** the manubrium & body of sternum during respiration.
- **The joint b/n body of sternum & the xiphoid process** often becomes **ossified with age**.



Interchondral joints,,,

- Sternal angle (angle of Louis) : Marks level of *2nd pair of costal cartilage* at **manubriosternal joint**. The *manubriosternal* (Sternal angle/angle of Louis) facilitate anatomical & clinical description of the body. For example:
 - ✓ *trachea bifurcates* into the main (primary) bronchi at this angle (or at level of *transverse thoracic plane*).
 - ✓ *arch of aorta* (begins posterior to **2nd right sternocostal joint**)
 - ✓ *thoracic aorta* begin at level of sternal angle.
- ❖ **Body of sternum:** -approximately 10 cm long; lies anterior to **right border of the heart & vertebrae T5 - T9.**

Interchondral joints,,,



Between ribs 1 – 10 & sternum

Sternocostal joints

Btw costal cartilages & sternum

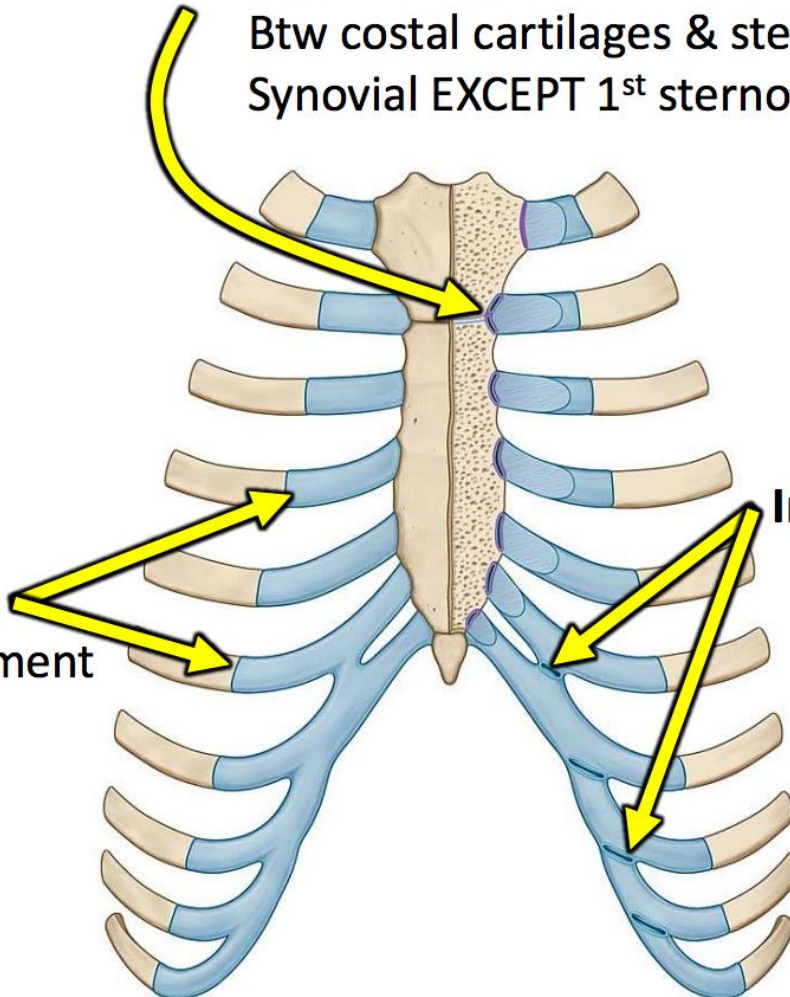
Synovial EXCEPT 1st sternocostal joint (synchondrosis)

Costochondral joints

Normally no movement

Interchondral joints

Synovial



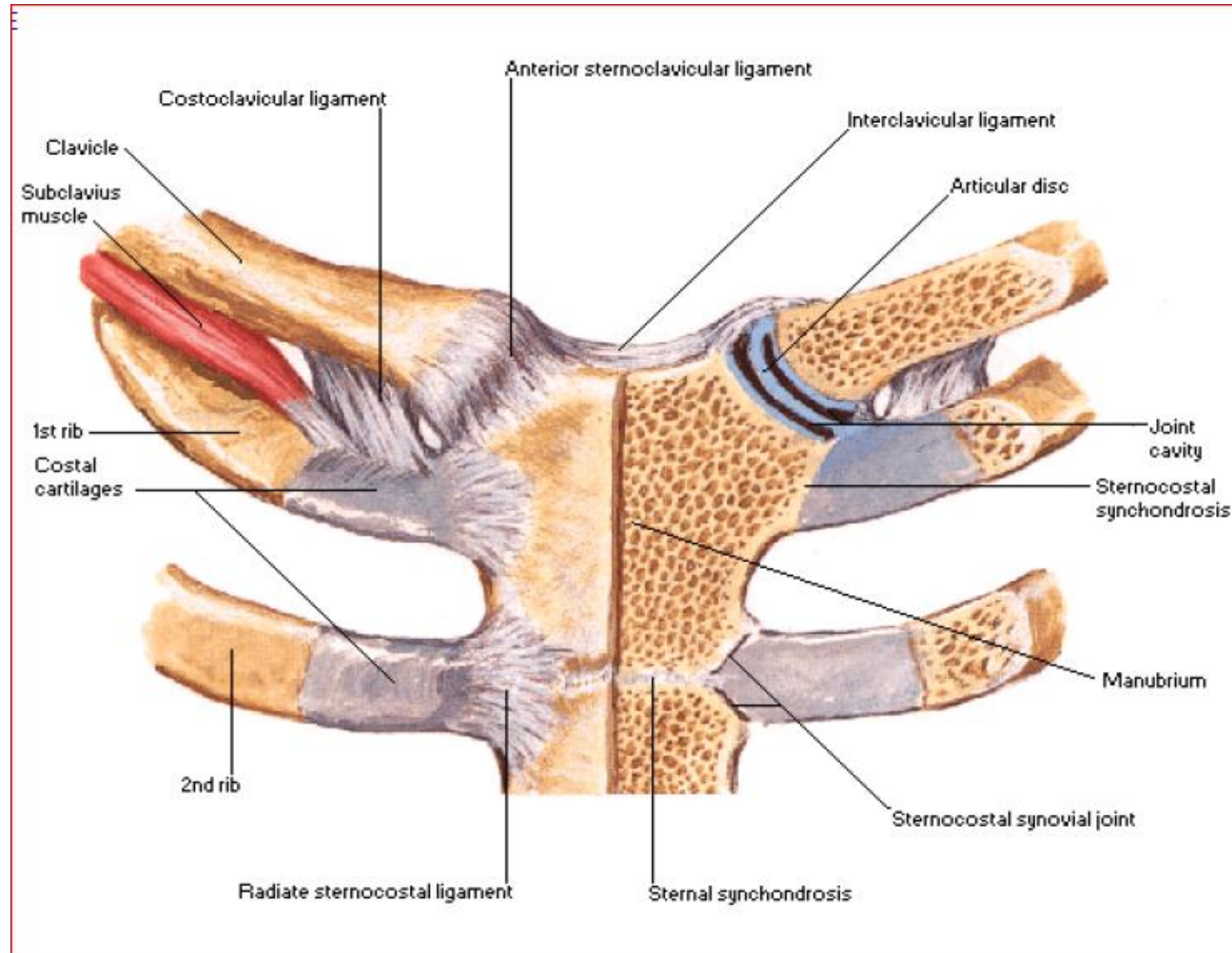
Costochondral joints

- The costochondral articulations are **hyaline cartilaginous joints**
- Each rib **has a cup-shaped depression** in its sternal end into which the costal cartilage fits.
- **The rib and its cartilage** are firmly bound together by the continuity of the **periosteum** of the rib with the **perichondrium of the cartilage**.
- No movement normally occurs at these joints.

Sternoclavicular joint

- **Is a synovial articulation** b/n the sternal end of the clavicle and the sternum
- The **sternoclavicular (SC) joint** is a **saddle type of synovial** joint but functions as a ball-and-socket joint.
- **Anterior and posterior SC ligaments** reinforce the joint capsule anteriorly and posteriorly.
- The **interclavicular ligament** strengthens the capsule superiorly.
- The **costoclavicular ligament** anchors the inferior surface of the sternal end of the clavicle to the 1st rib and its costal cartilage

Sternoclavicular joint



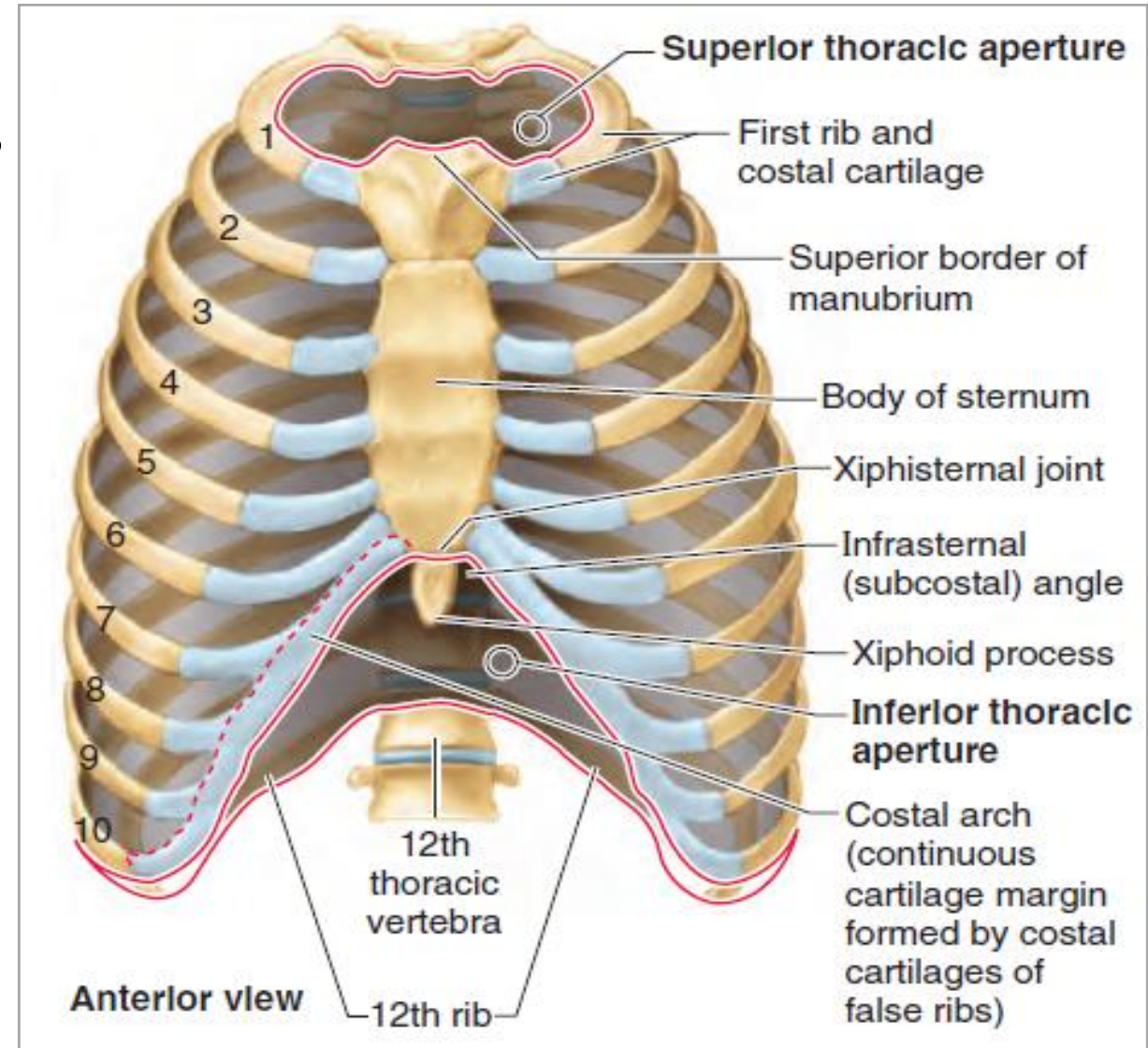
Thoracic apertures

- The thoracic cavity is open superiorly and inferiorly.
- The much smaller superior opening is a **passageway that allows communication with the neck and upper limb.**
- The larger inferior opening provides the ring-like origin of the **diaphragm**, which completely occludes the opening except diaphragmatic perforations.

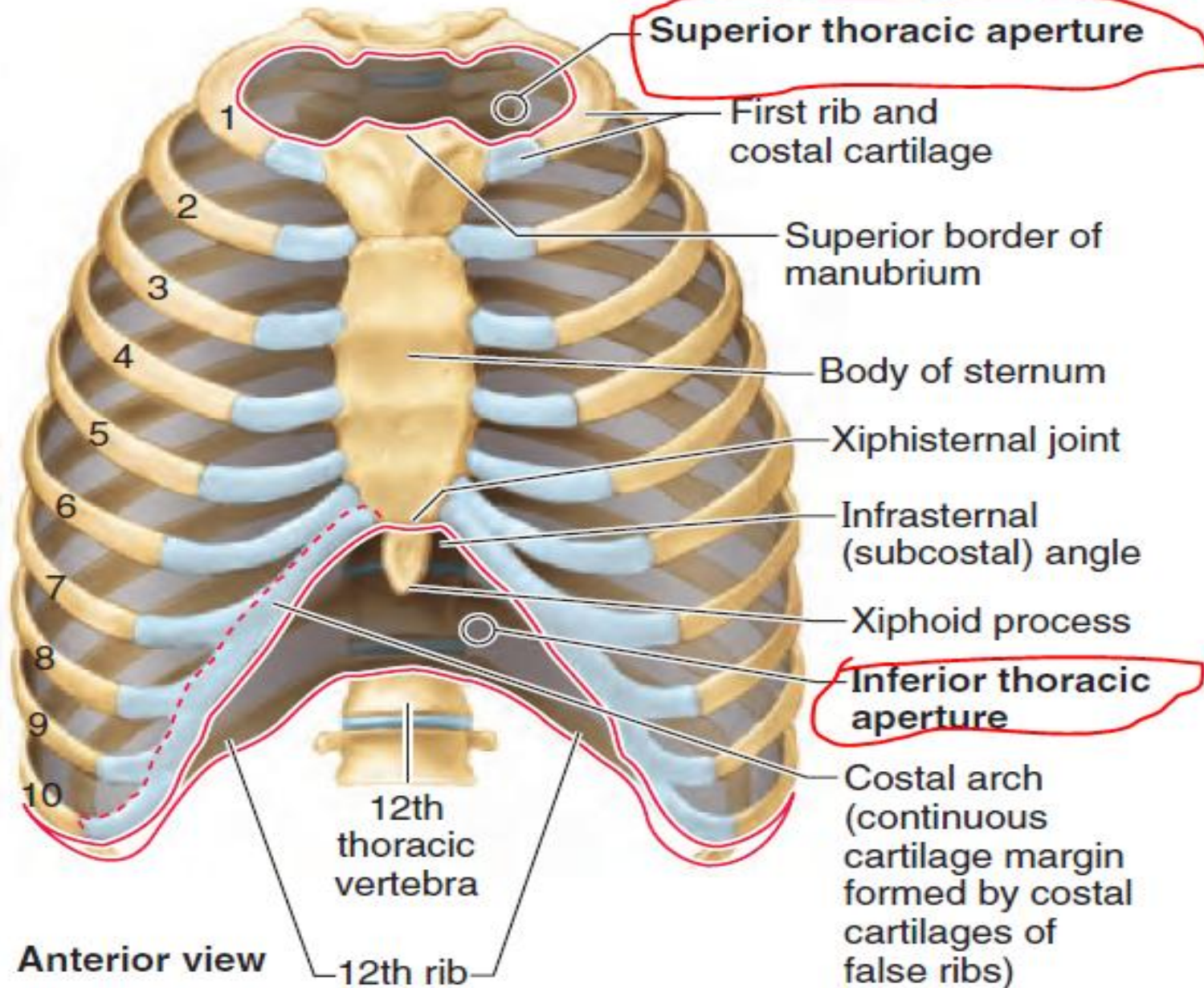
Superior thoracic apertures

✓ The superior thoracic aperture is bounded:

- Posteriorly, by vertebra T1(its body)
- Laterally, by the 1st pair of ribs
- Anteriorly, by the superior border of the manubrium.



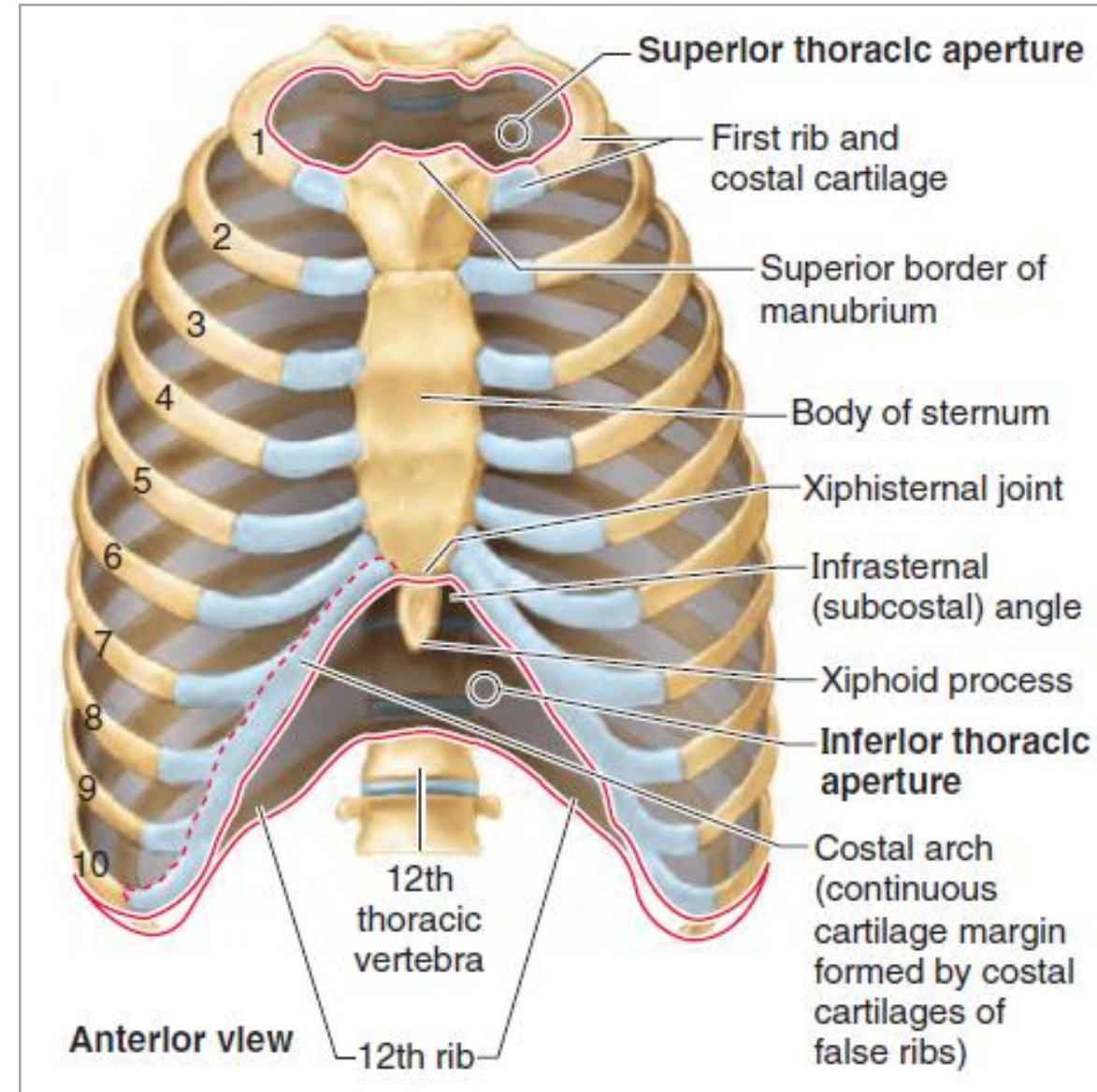
Superior thoracic apertures,,,



Inferior thoracic aperture

❖ It is bounded by:

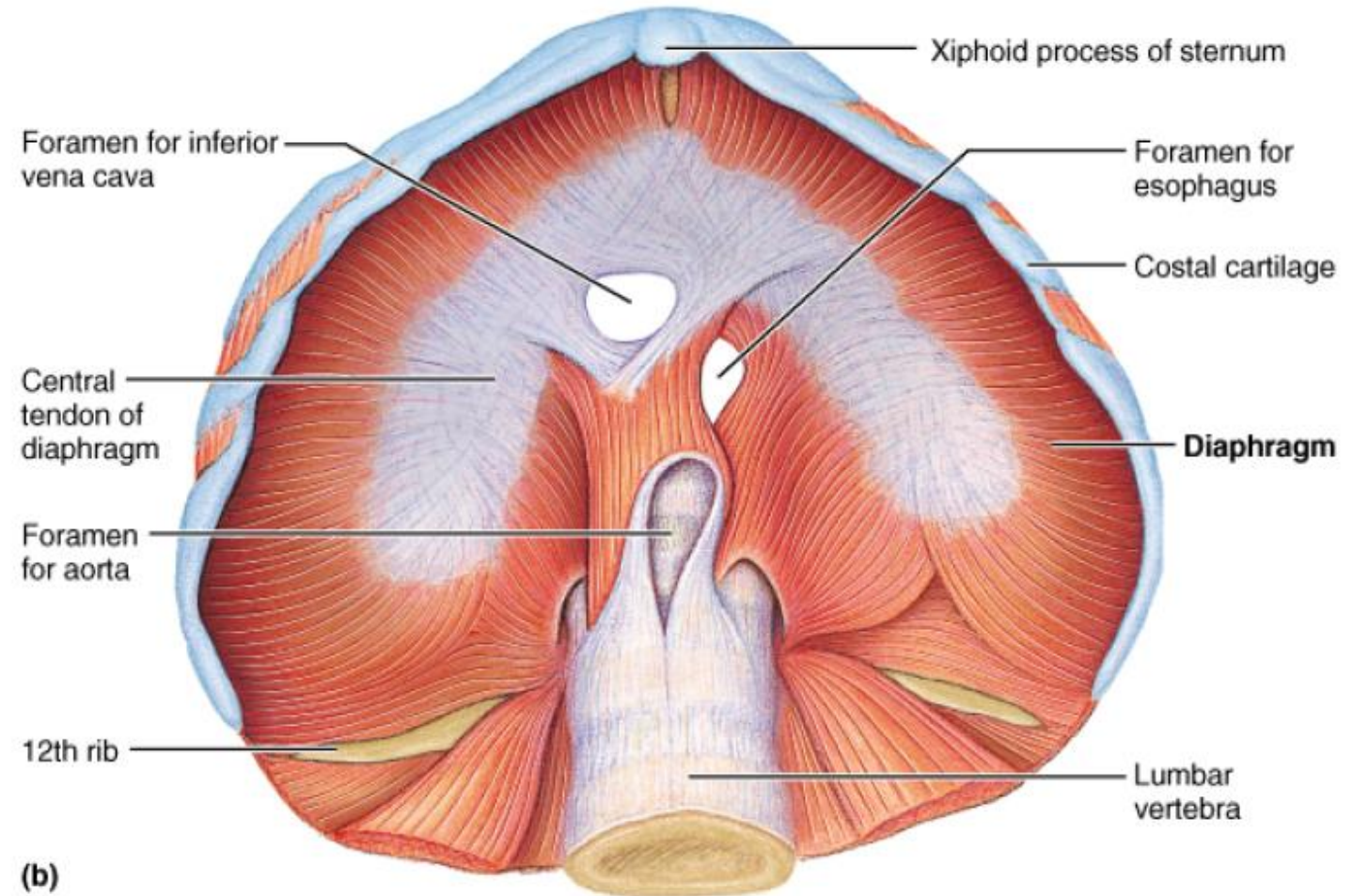
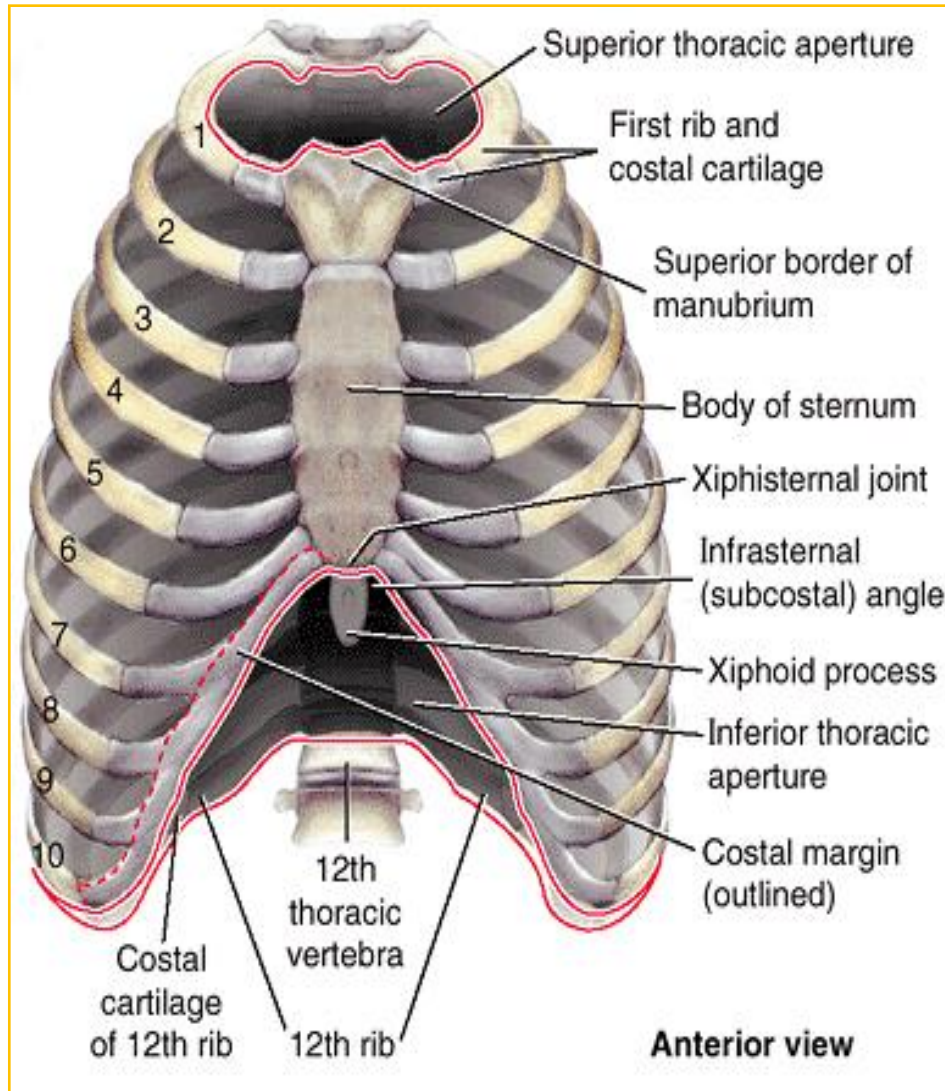
- ✓ Posteriorly by the 12T(its body).
- ✓ Posterolaterally, by 11th & 12th ribs.
- ✓ Anterolaterally, by the joined costal cartilages of ribs 7–10, forming the **costal margins**.
- ✓ Anteriorly, by the xiphisternal joint.



Inferior thoracic aperture,,,

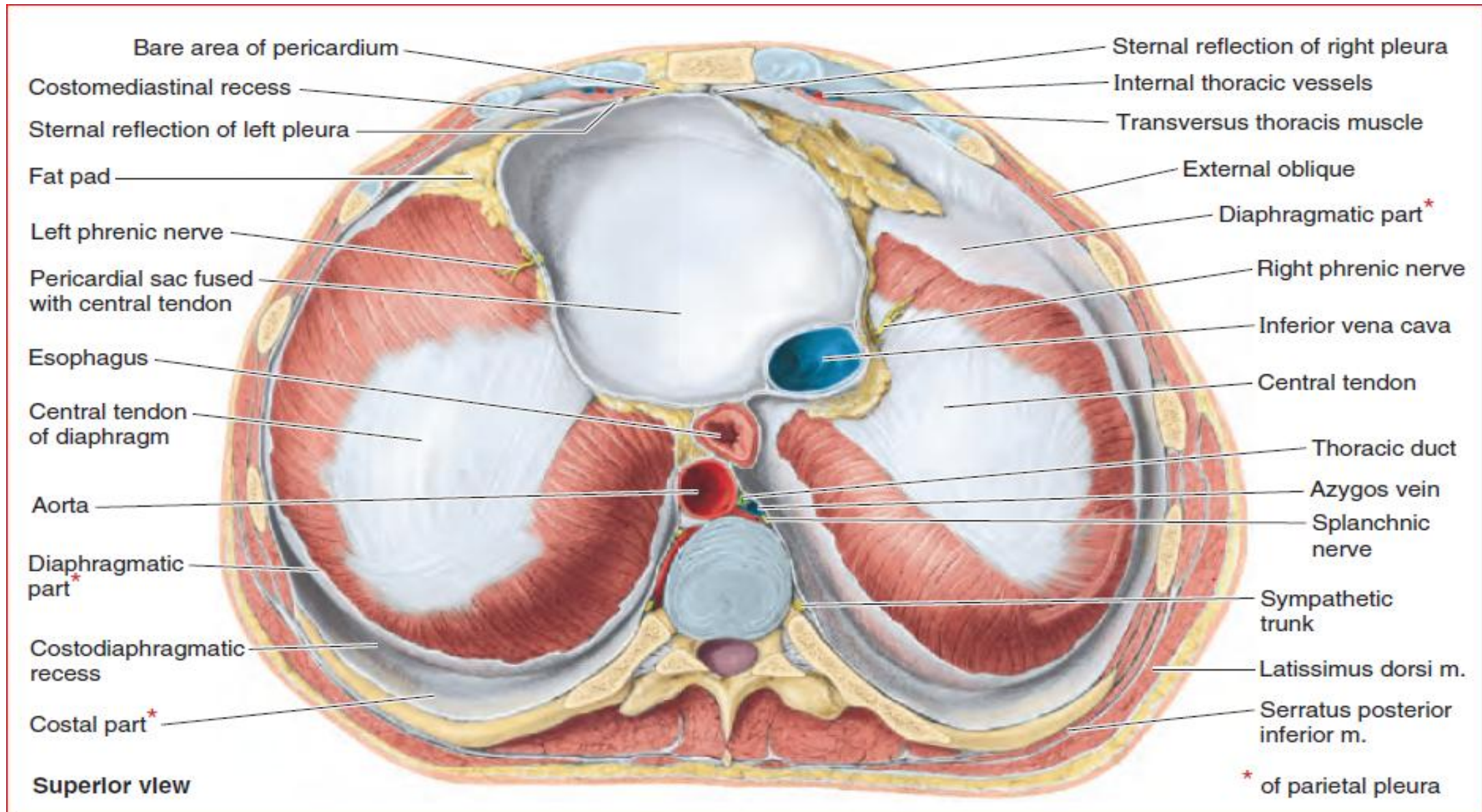
- The inferior thoracic aperture is **larger** than the superior thoracic aperture.
- It is oblique b/c the posterior thoracic wall is much longer than the anterior wall.
- The diaphragm separates the thoracic and abdominal cavities almost completely.
- Structures passing b/n the abdomen & thorax pass through **openings of the diaphragm, or pass posterior to it.**
- The domes of the diaphragm rise to the **level of the 4th intercostal space, and abdominal viscera**, including the liver, spleen, and stomach, lie superior to the plane of the **inferior thoracic aperture**, within the thoracic wall.

Inferior thoracic aperture,,



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Inferior thoracic aperture,,,



Muscles of the thoracic wall

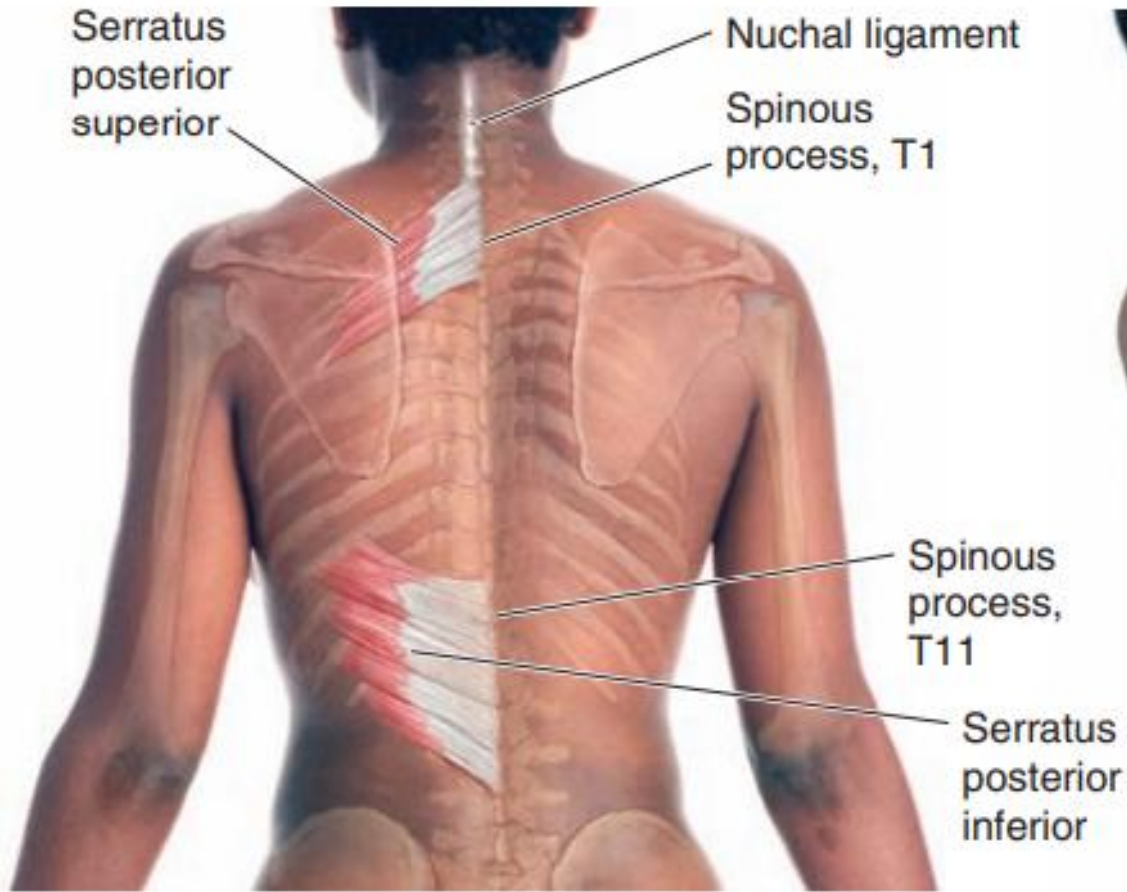
Several upper limb (thoracoappendicular) muscles attach to the thoracic wall – including

- ✓ Pectoralis major and pectoralis minor.
- ✓ Subclavius.
- ✓ Serratus anterior muscles anteriorly.
- ✓ Latissimus dorsi muscles posteriorly.
- ✓ **Anterolateral abdominal muscles and.**
- ✓ **Some back and neck muscles.**

Muscles of the thoracic wall,,,

- **The true muscles of the thoracic wall are** the serratus posterior, levatores costarum, intercostal, subcostal, and transversus thoracis.
- Muscles of the thoracic wall include those that fill and support the intercostal spaces, those that pass between the sternum and the ribs, and those that cross several ribs between costal attachments
- **The true muscles of the thoracic wall move the wall during breathing.**

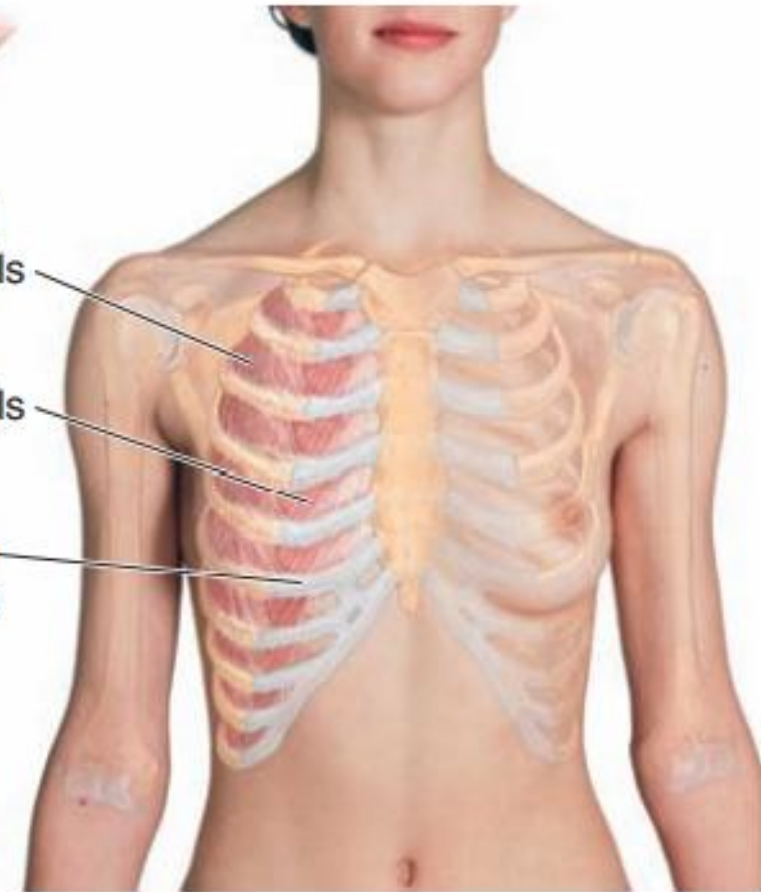
Muscles of the thoracic wall,,,



(A) Posterior view

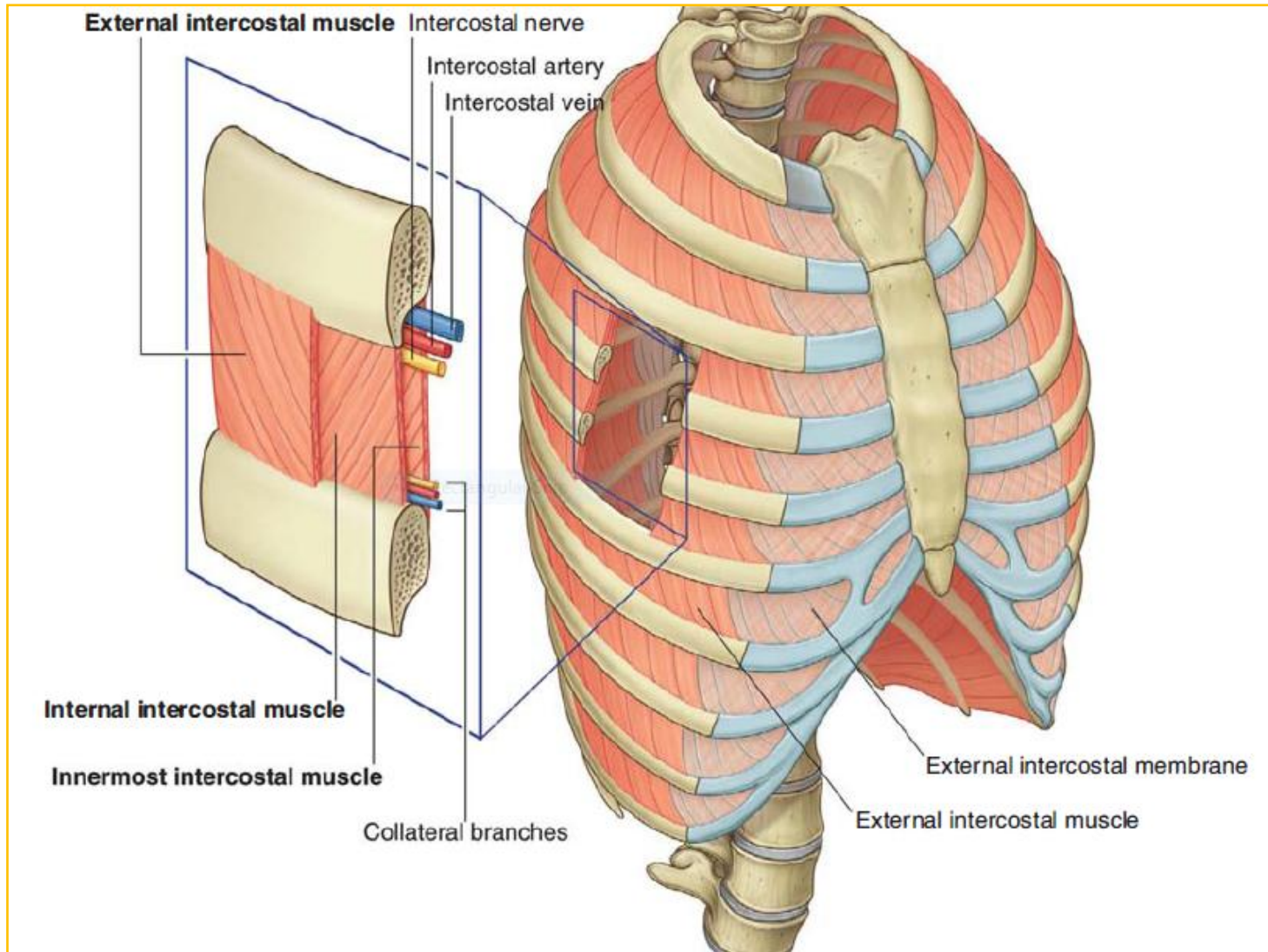


(B) Lateral view

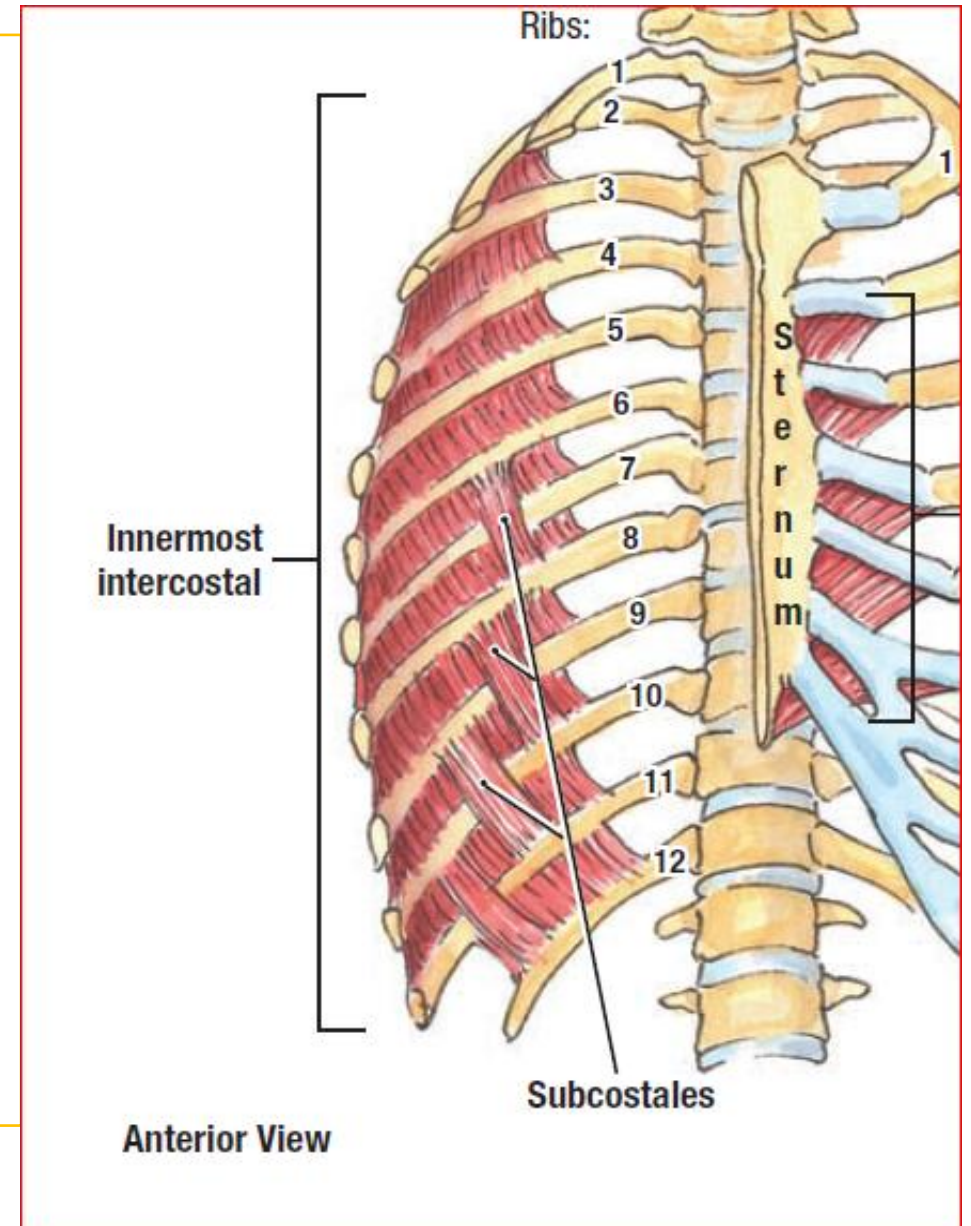


(C) Anterior view

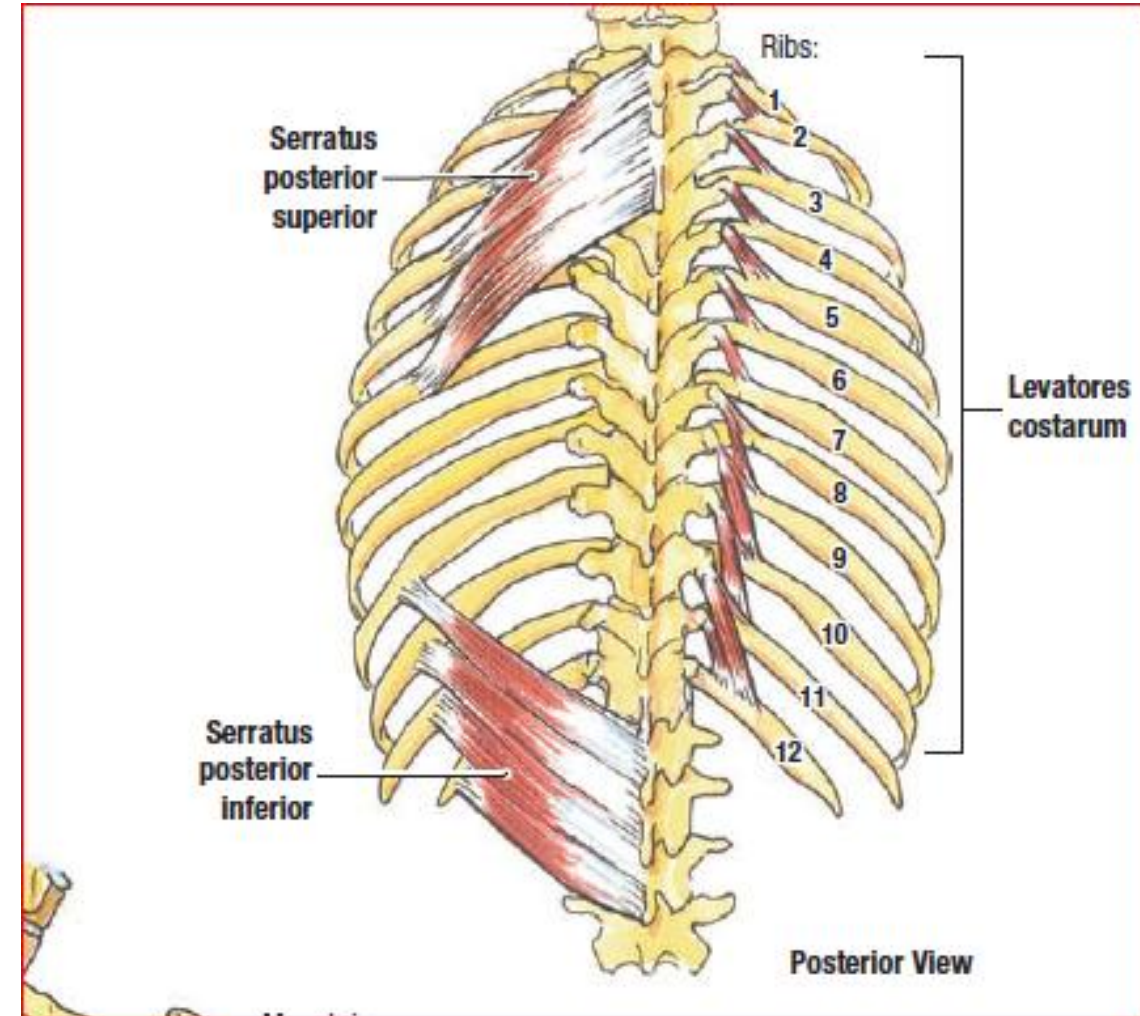
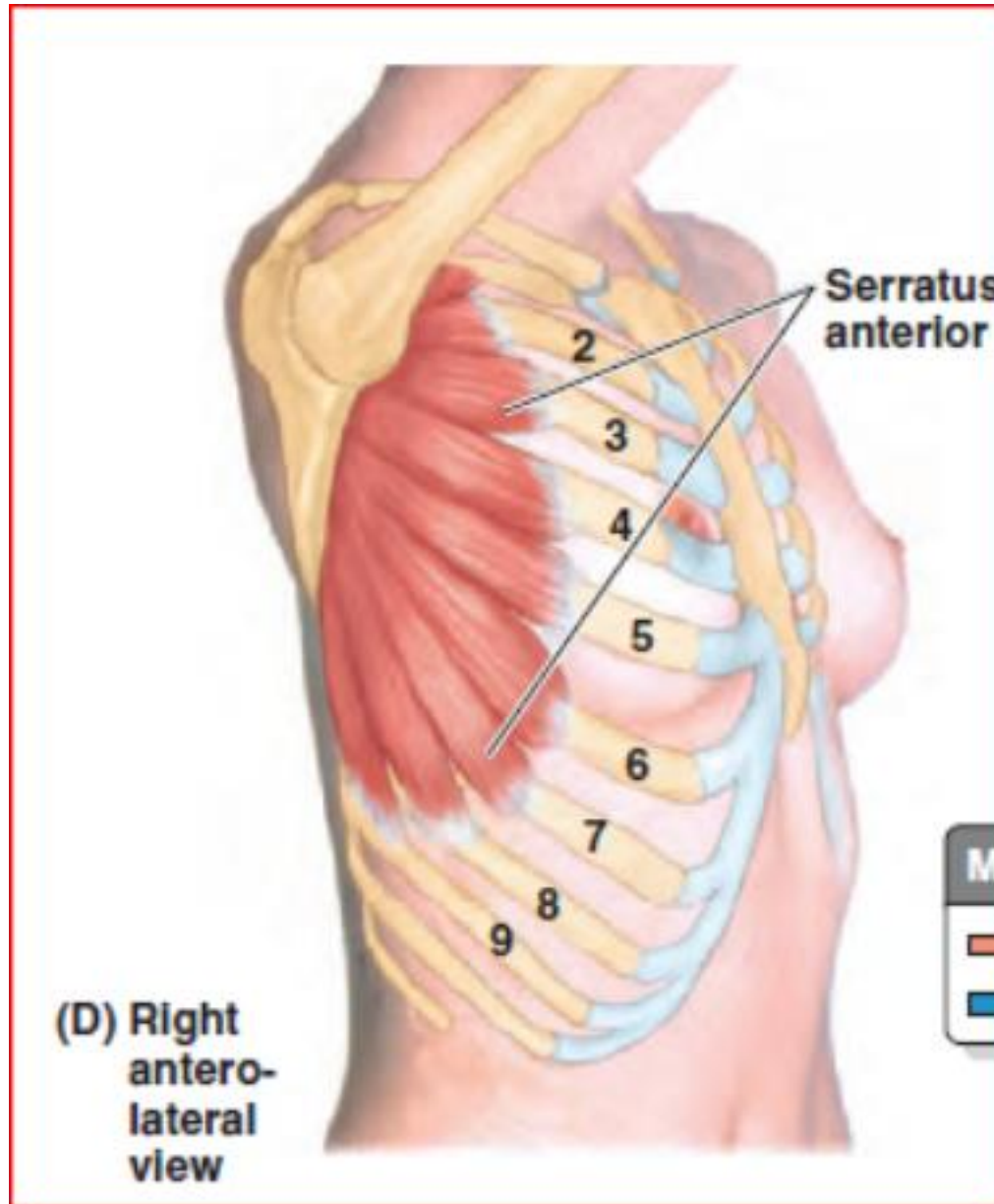
Muscles of the thoracic wall,,



Intercostal muscles

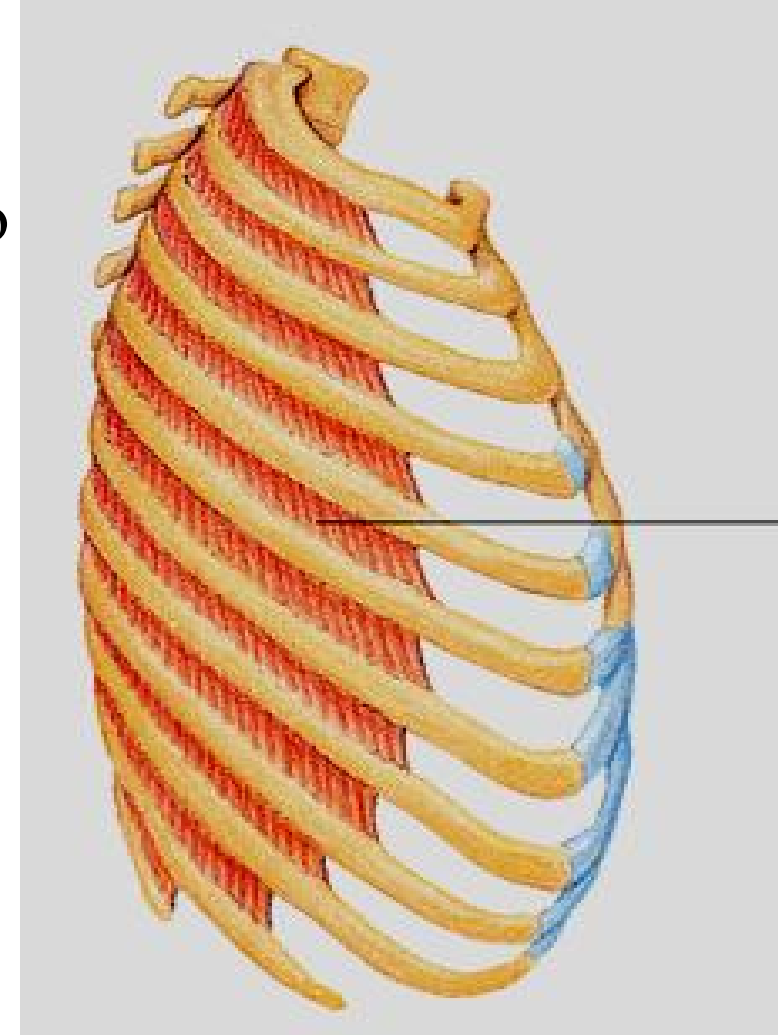


Muscles of the thoracic wall,,

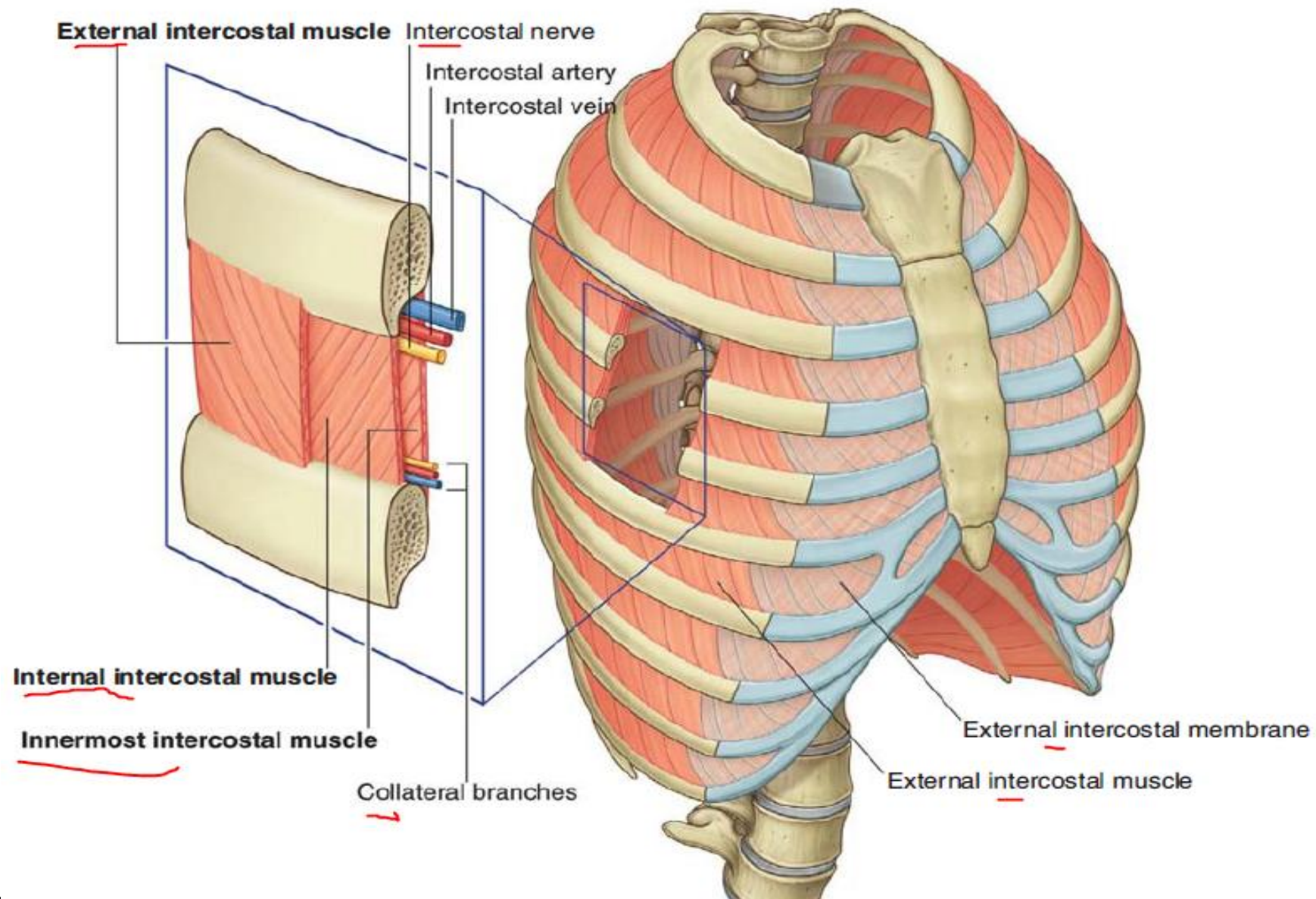


Intercostal muscles

- The *intercostal muscles* occupy the intercostal spaces.
- The **external intercostal muscles** (11 pairs) occupy the intercostal spaces from the tubercles of the ribs posteriorly to the **costochondral junctions**.
- Anteriorly, the muscle fibers are replaced by the **external intercostal membranes**.
- These muscles run **infero-anteriorly** from the rib above to the rib below.
- *The external intercostals are most active during inspiration.*



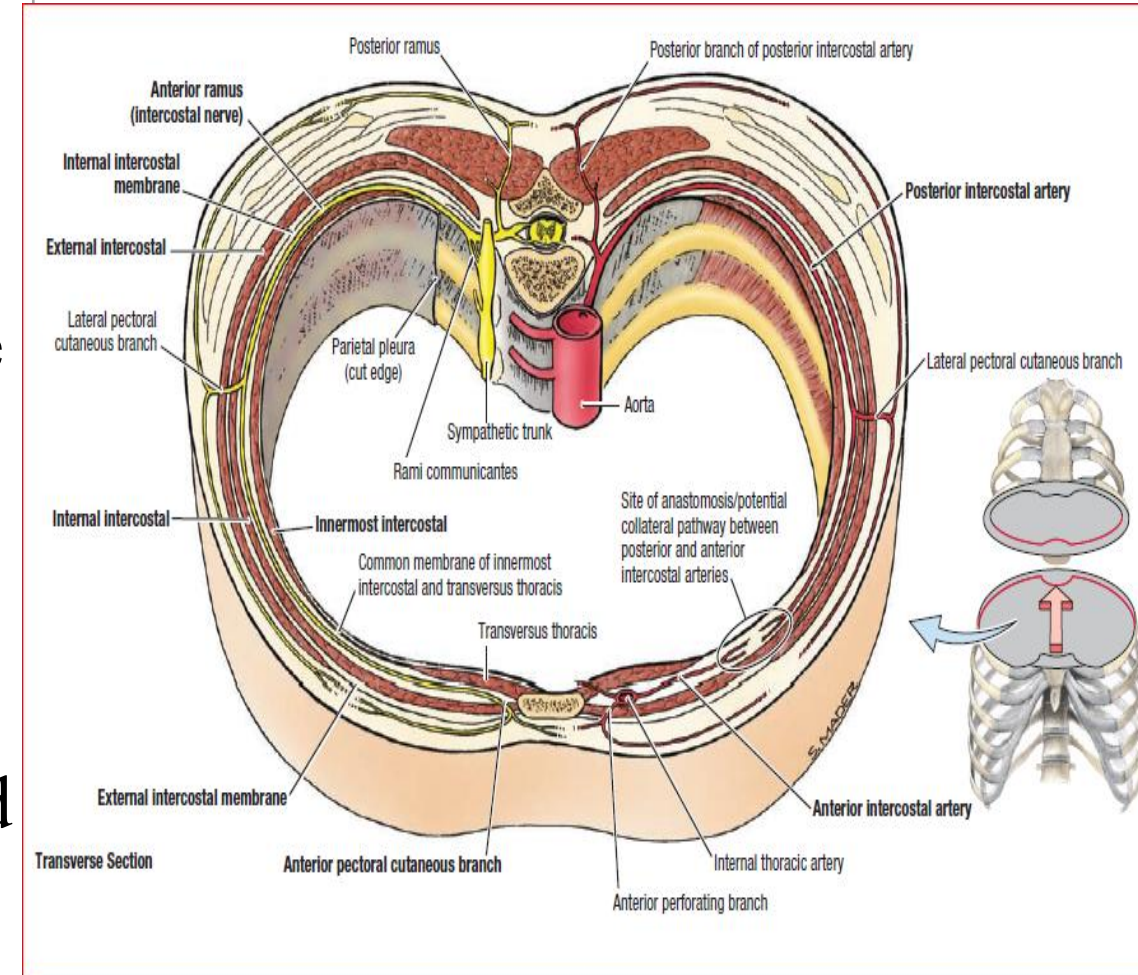
Intercostal muscles,,



Intercostal muscles

Intercostal muscles,,,

- The **internal intercostal muscles** (11 pairs) run deep to and at right angles to the **external intercostals**.
- Their fibers run **inferoposteriorly** from the floors of the costal grooves to the superior borders of the ribs inferior to them.
- Between the ribs posteriorly, medial to the angles, the internal intercostals are replaced by the **internal intercostal membranes**.
- **Are most active during expiration**

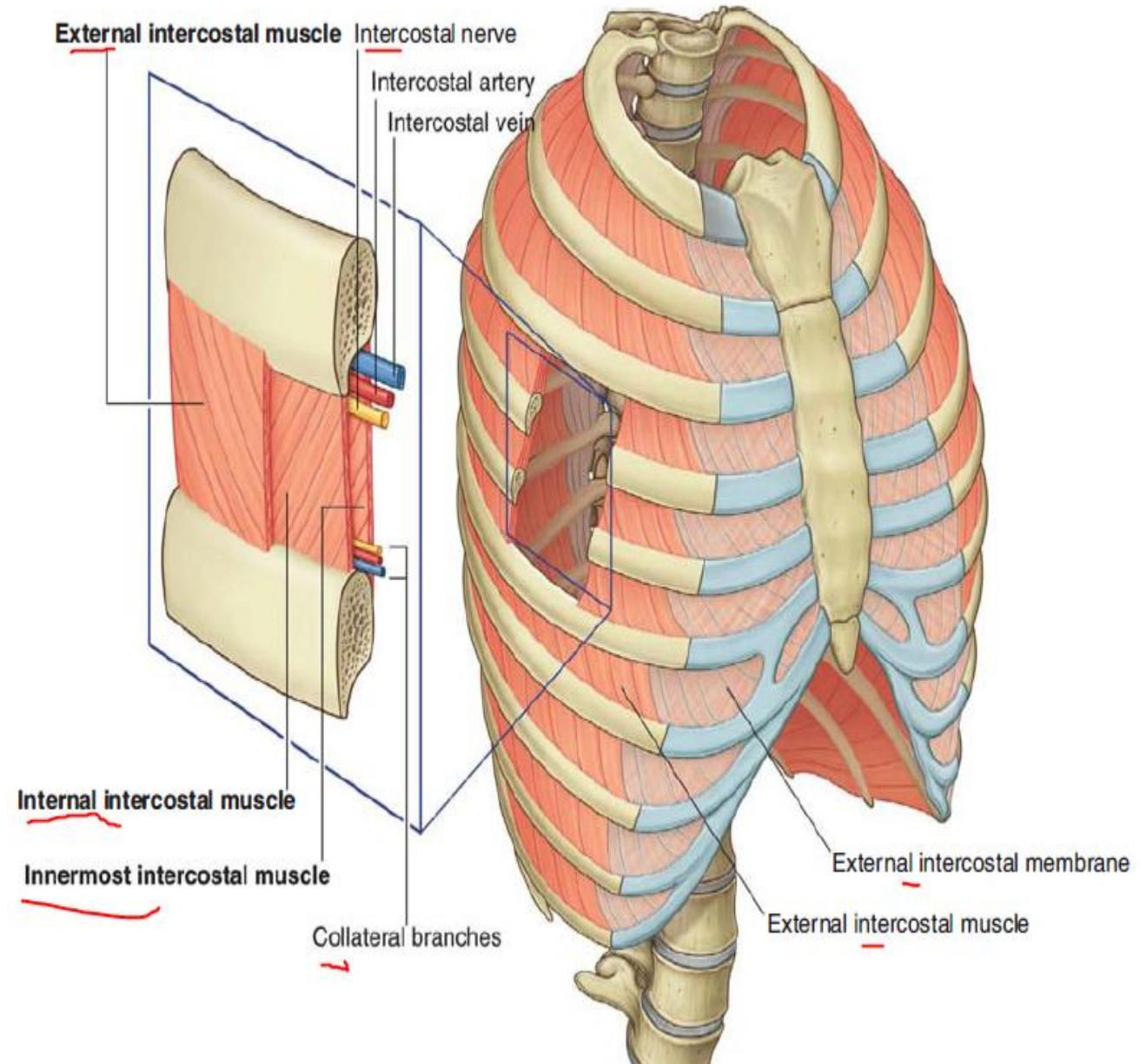


Intercostal muscles

Intercostal muscles,,,

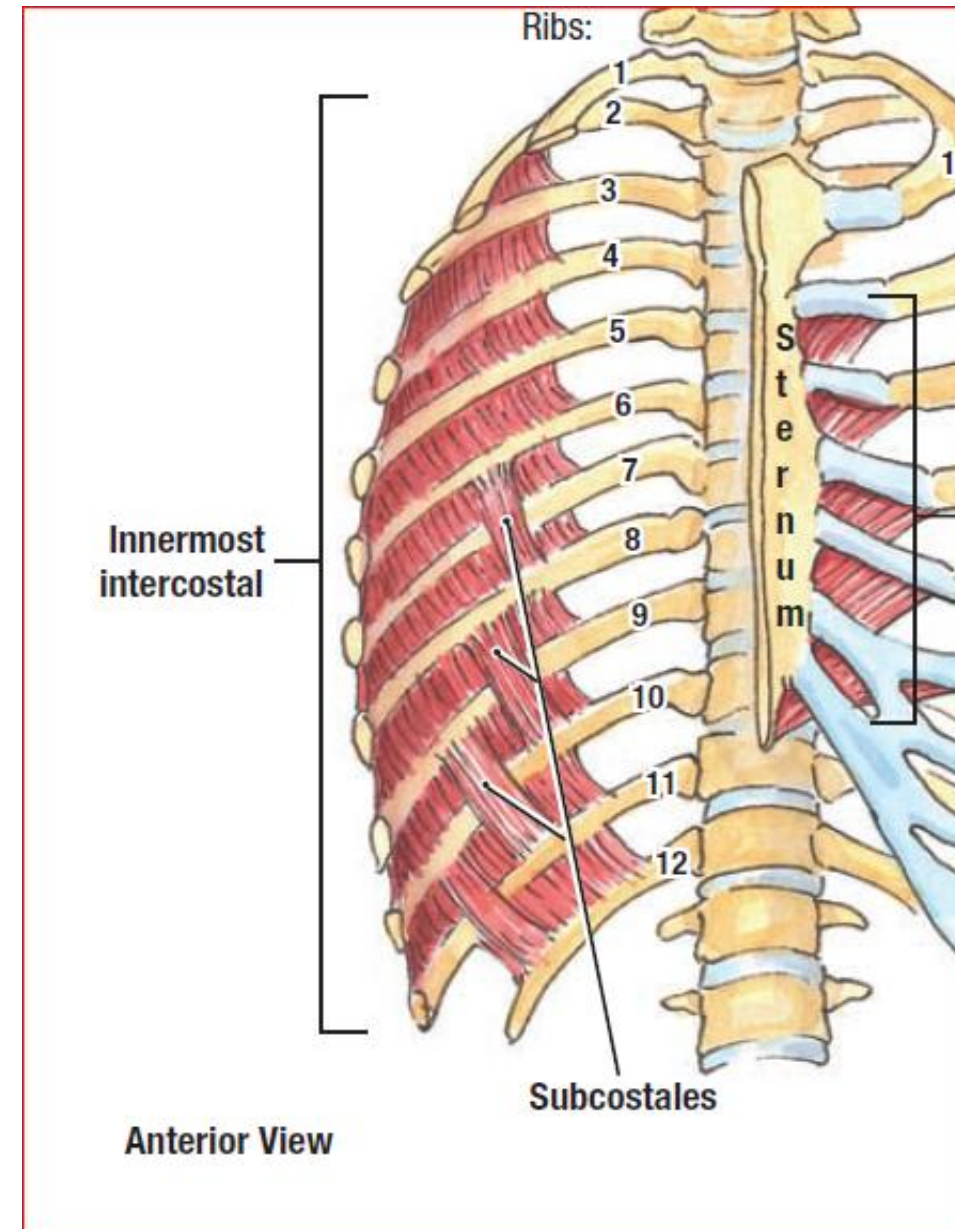
- The **innermost intercostal muscles** are similar to the internal intercostals.
- The innermost intercostal muscle forms the deepest layer
- **The innermost intercostals are separated from the internal intercostals by intercostal nerves and vessels.**

□ **Actions of innermost intercostals** are the same as those of the **internal intercostal muscles.**



Intercostal muscles,,,

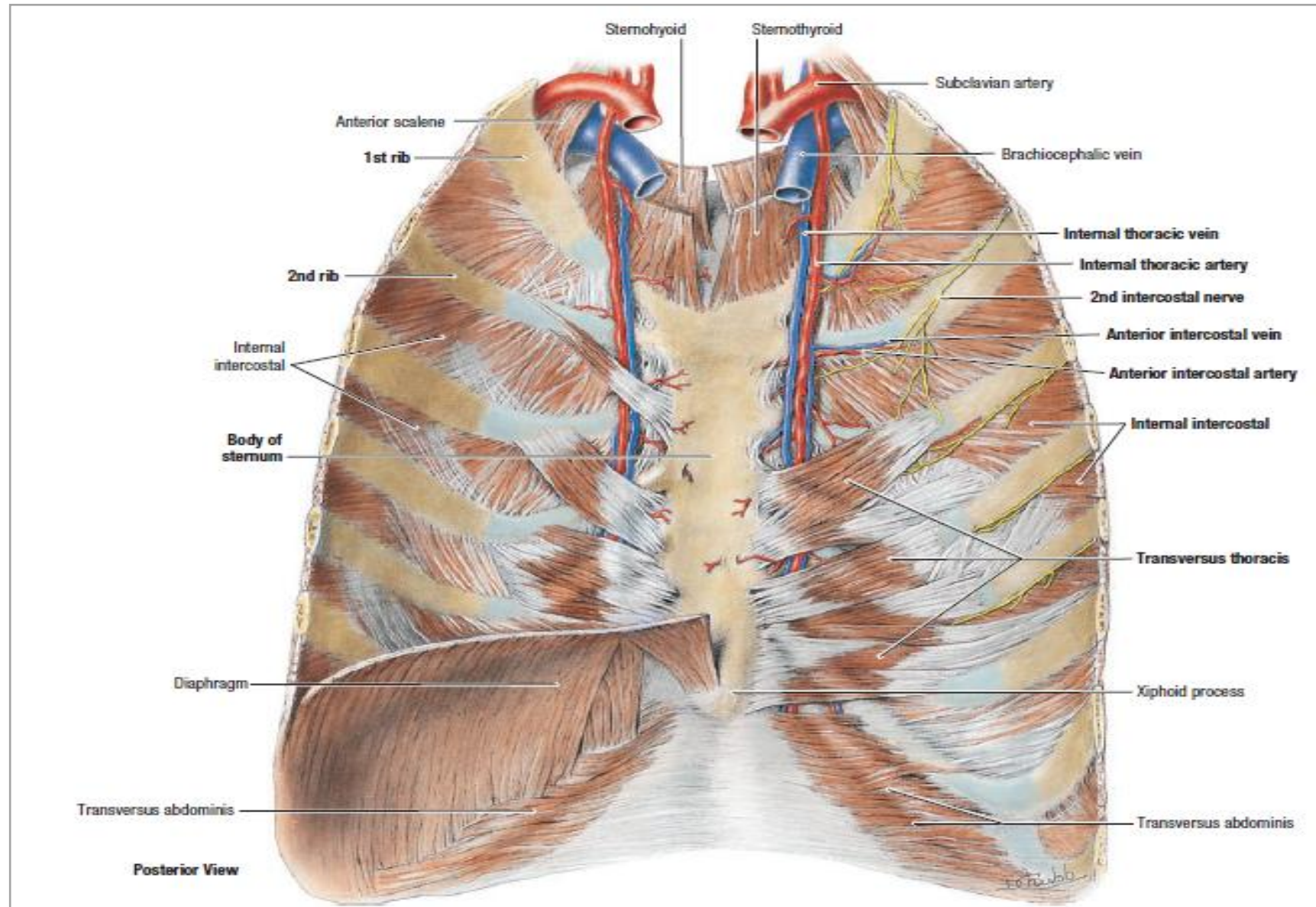
- The **subcostal muscles** are variable in size and shape, usually being well developed only in the lower thoracic wall..
- They extend from the internal surfaces of one rib to the internal surface of the second (next) or third rib below.
- Crossing **one or two** intercostal spaces, the subcostals run in the same direction as the *internal intercostals* and *blend with them*.



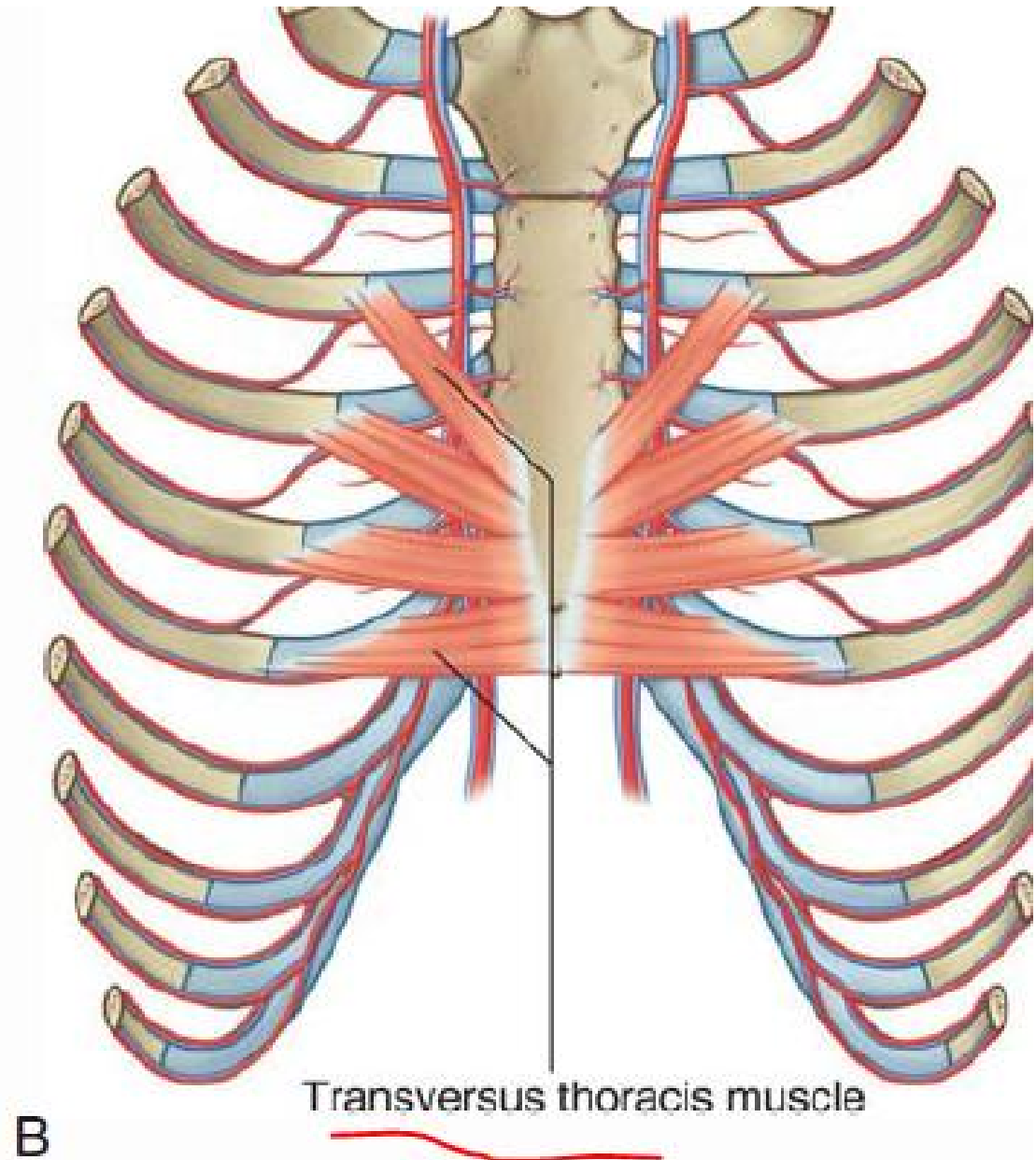
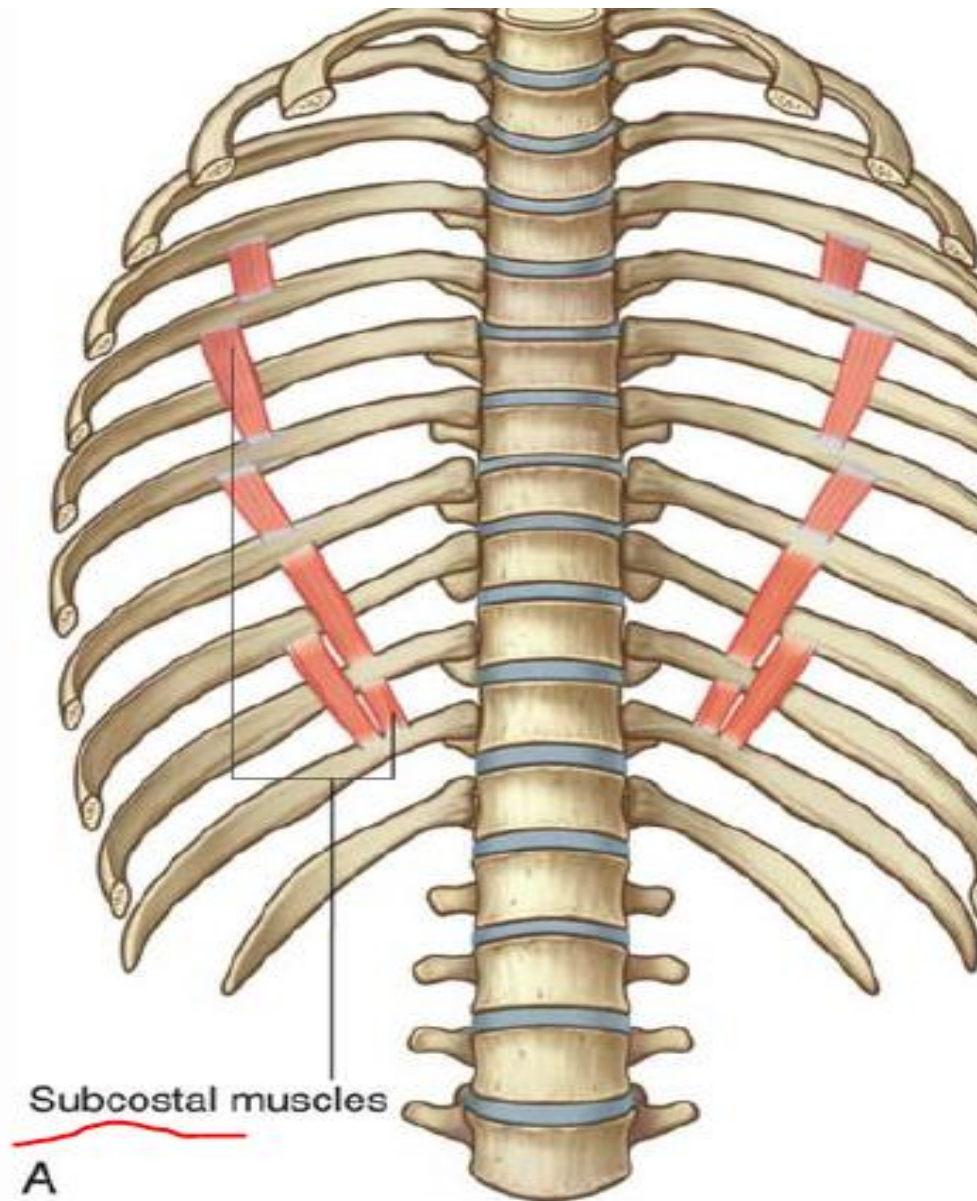
Intercostal muscles,,,

- The **transversus thoracis muscles** consist of 4 or 5 slips
- The transversus thoracis muscles are continuous inferiorly with the **transversus abdominis muscles in the anterolateral body wall.**
- They **originate from the posterior aspect of the xiphoid process, the inferior part of the body of the sternum, and the adjacent costal cartilages of the lower true ribs**
- These muscles appear to have a **weak expiratory function** and may also provide **proprioceptive information.**

Intercostal muscles,,



Intercostal muscles,,

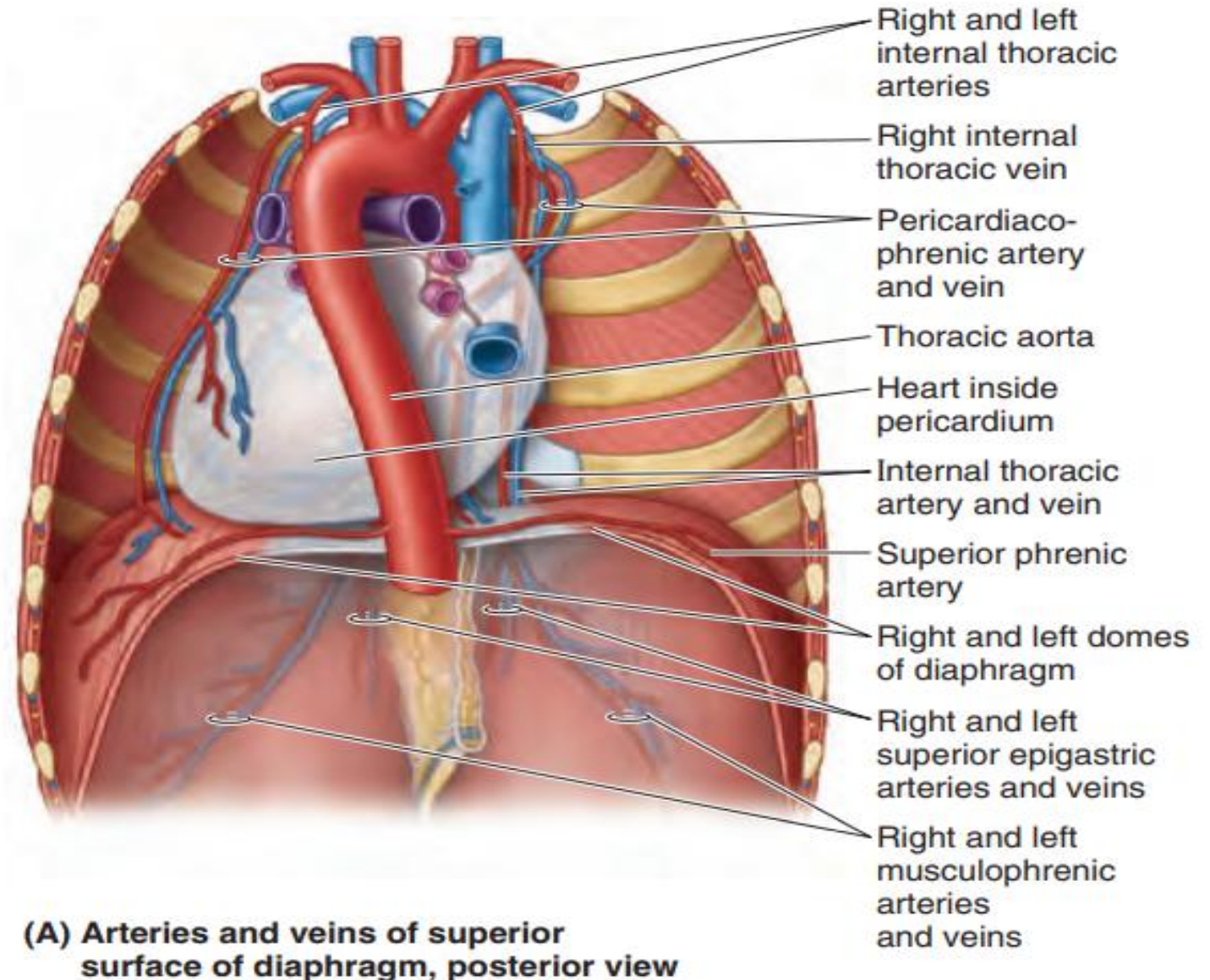


Diaphragm

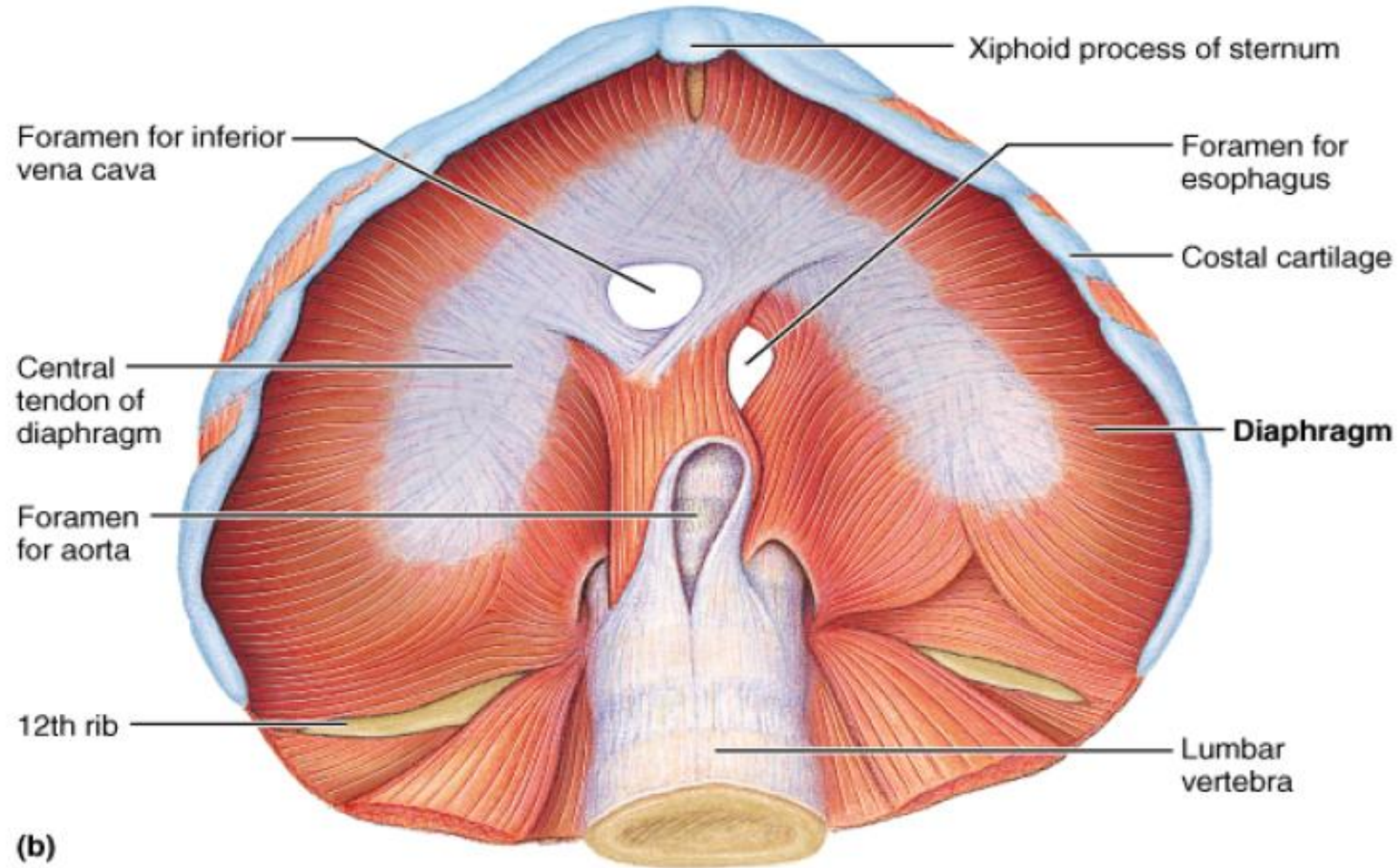
- The **diaphragm** is a double-domed musculotendinous sheet, located at the **inferior-most aspect of the rib cage**.
- ❖ It serves two main functions:
 1. **Separates the thoracic cavity from the abdominal cavity**
 - ❖ **Note:** the word diaphragm is derived from the Greek ‘diáphragma’, meaning partition.
 2. **Undergoes contraction and relaxation**, altering the volume of the thoracic cavity and the lungs, **producing inspiration and expiration**.
- ❖ ***It is the principal muscle of respiration.***
- ❖ The pericardium, containing the heart, lies on the **central part of the diaphragm**.

Diaphragm,,,

- Diaphragm, the main muscle involved in breathing
- It is the chief muscle of respiration
- It consists of a peripheral muscular part, and a centrally placed tendon



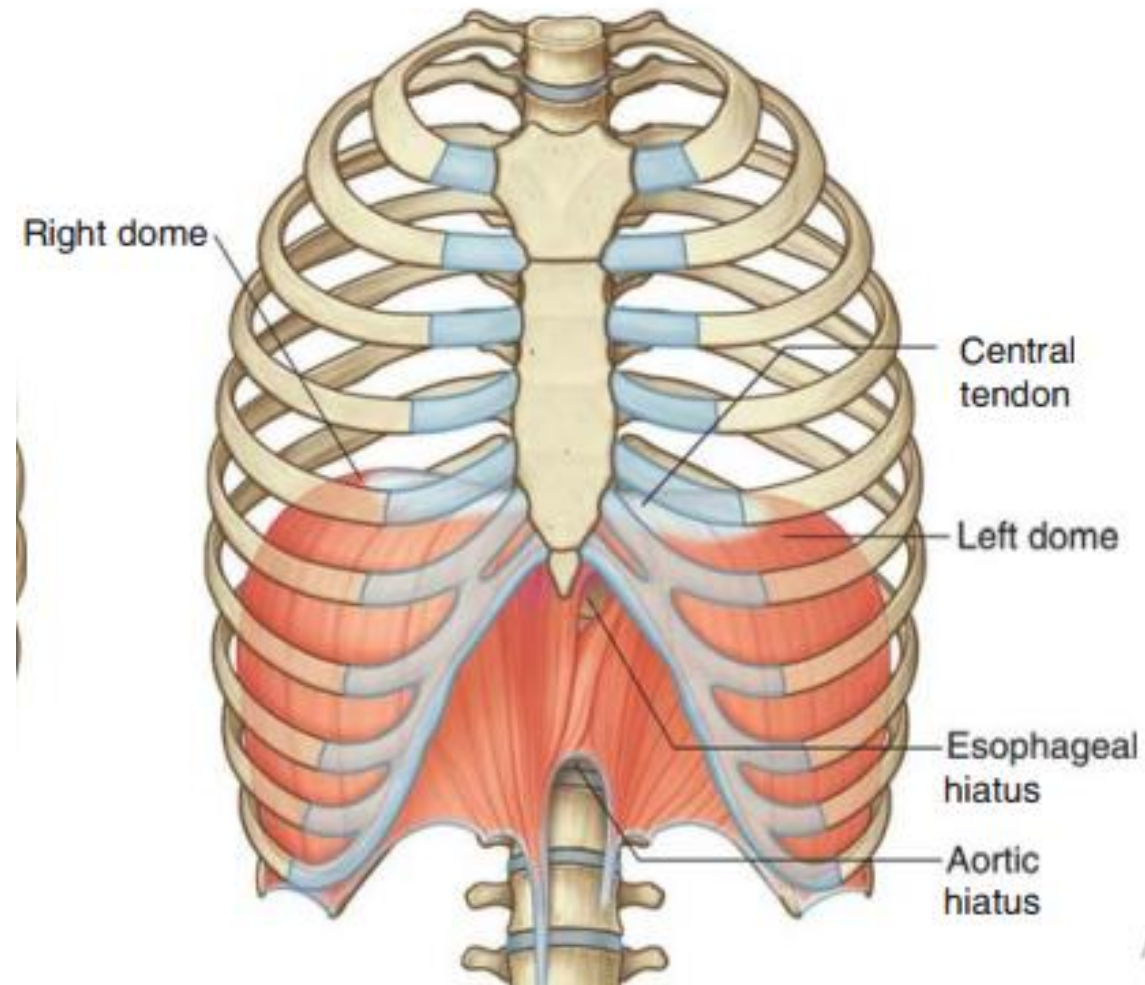
Diaphragm,,,



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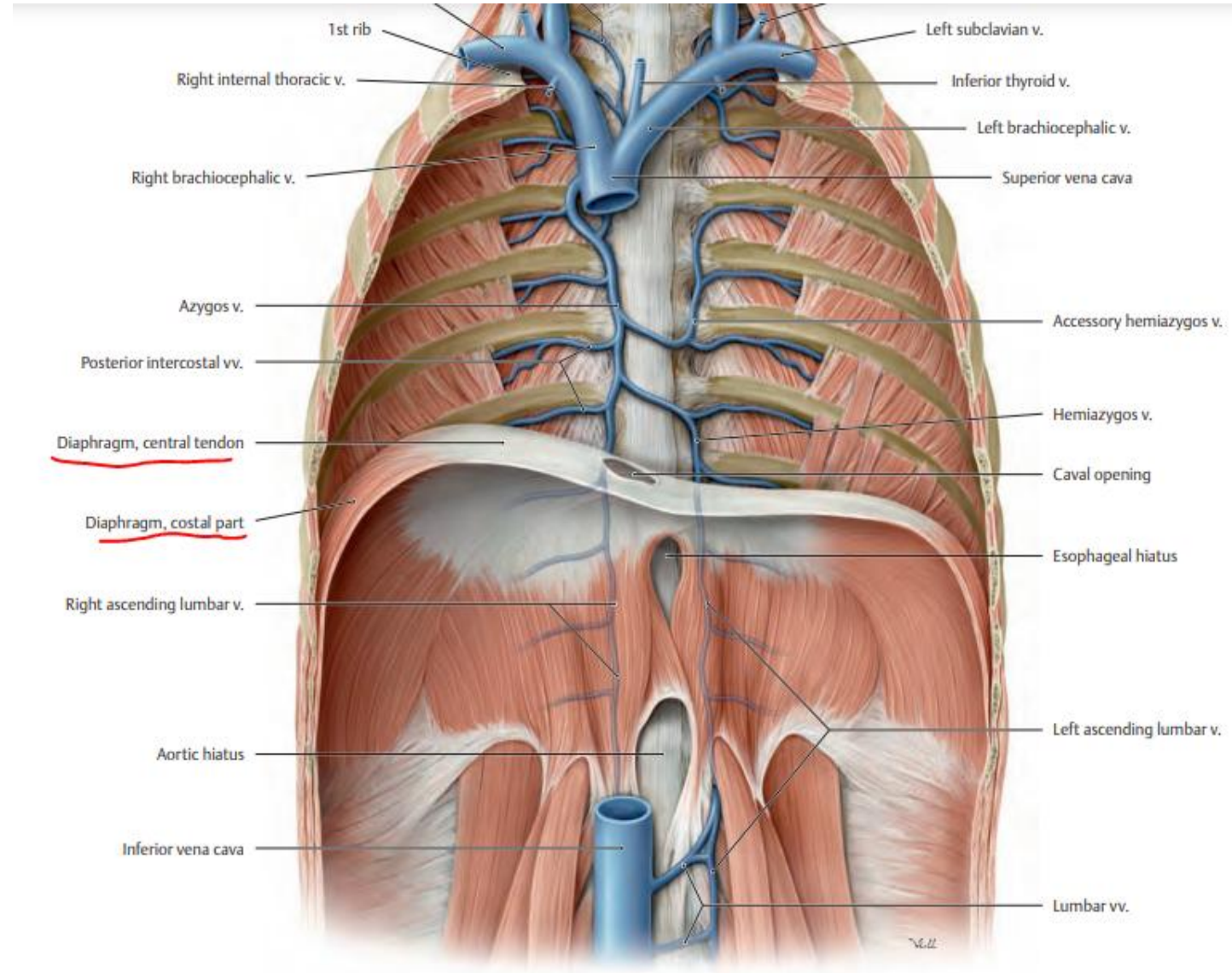
Diaphragm,,

- Because of the oblique angle of the inferior thoracic aperture, the posterior attachment of the diaphragm is inferior to the anterior attachment.
- The diaphragm is not flat; rather, it 'balloons' superiorly, on both the right and left sides, to form domes.
- The right dome is higher than the left, reaching as far as rib V.



Diaphragm,,,

- ❑ Its mainly convex superior surface faces the thoracic cavity, and its concave inferior surface faces the abdominal cavity
- ❑ The diaphragm curves superiorly into right and left domes;
 - normally the right dome is higher than the left dome **owing to the presence of the liver**



Diaphragm,,,

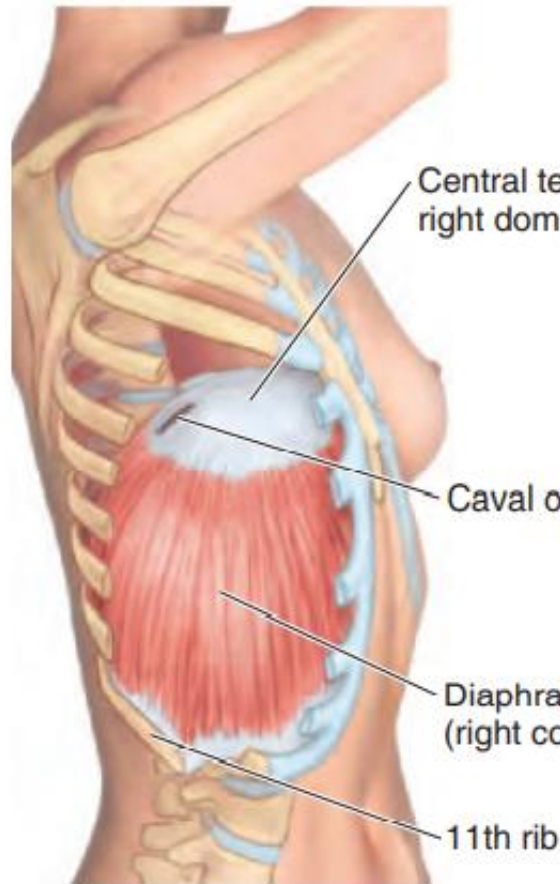
- It descends during inspiration; however, only its central part moves because its periphery, as the fixed origin of the muscle, attaches to the inferior margin of the thoracic cage and the superior lumbar vertebrae.
- ❖ **The level of the domes of the diaphragm varies according to the:**
 - ✓ Phase of respiration (inspiration or expiration).
 - ✓ Posture (e.g., supine or standing).
 - ✓ Size and degree of distension of the abdominal viscera.

Diaphragm,,,

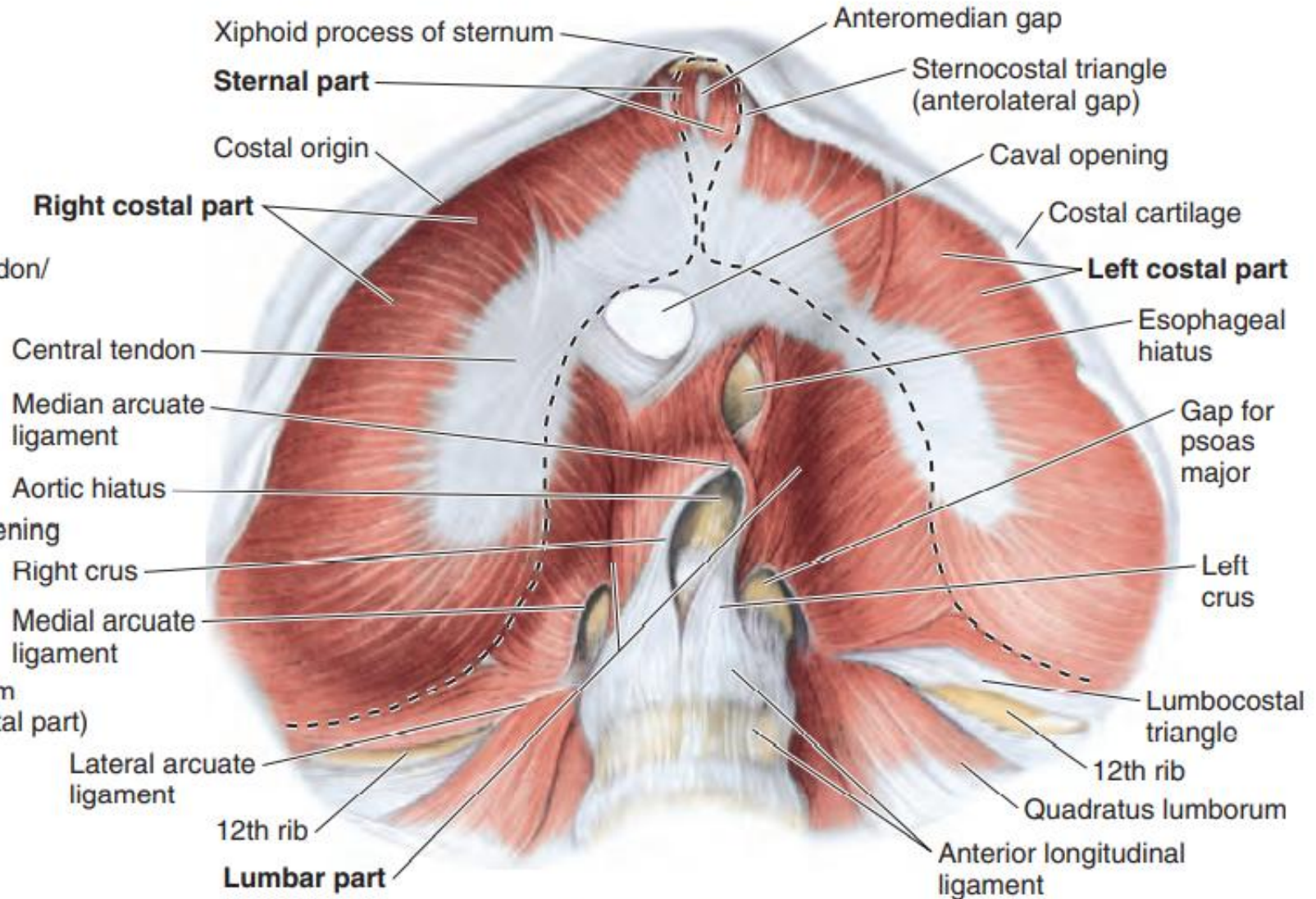
❑ **Based on the peripheral attachments the diaphragm has 3 parts**

- ✓ **Sternal part:** consisting of two muscular slips that attach to the posterior aspect of the xiphoid process; this part is not always present.
- ✓ **Costal part:** consisting of wide muscular slips.
 - the costal parts form **the right and left domes**.
- ✓ **Lumbar part:** arising from two aponeurotic arches, the medial and lateral arcuate ligaments, and the three superior lumbar vertebrae;
 - **the lumbar part forms right and left muscular crura**

Diaphragm,,,



(A)



(B) Inferior view

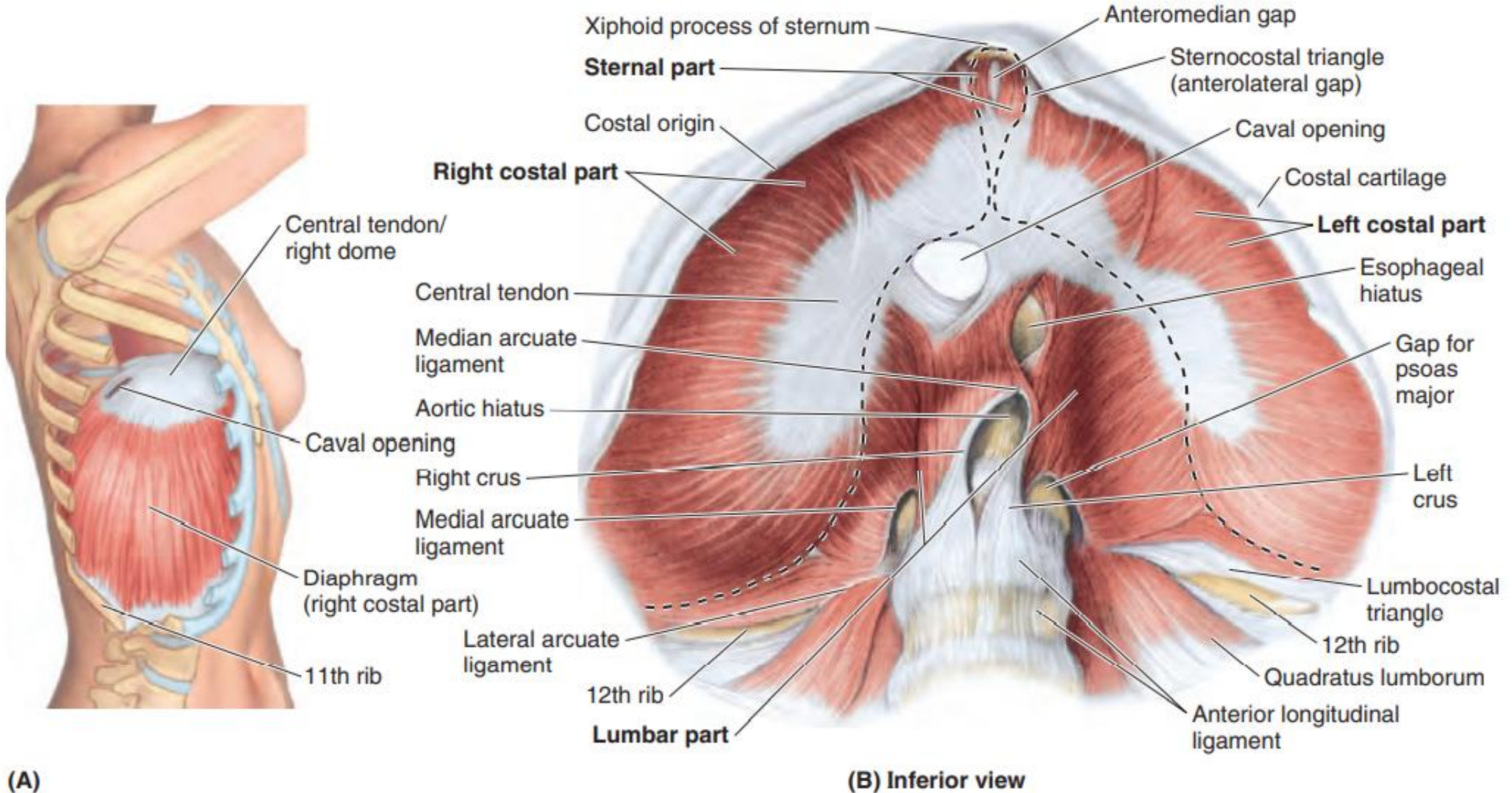
Diaphragmatic Apertures(Opening),,,

- The diaphragmatic apertures (openings, hiatus) permit structures (vessels, nerves, and lymphatics) to pass between the thorax and abdomen.

❖ **There are three large apertures on the diaphragmatic:**

- ✓ Caval opening, for inferior vena cava(IVC)
- ✓ Esophageal, for esophagus, and
- ✓ Aortic hiatus, for aorta and a number of small ones.

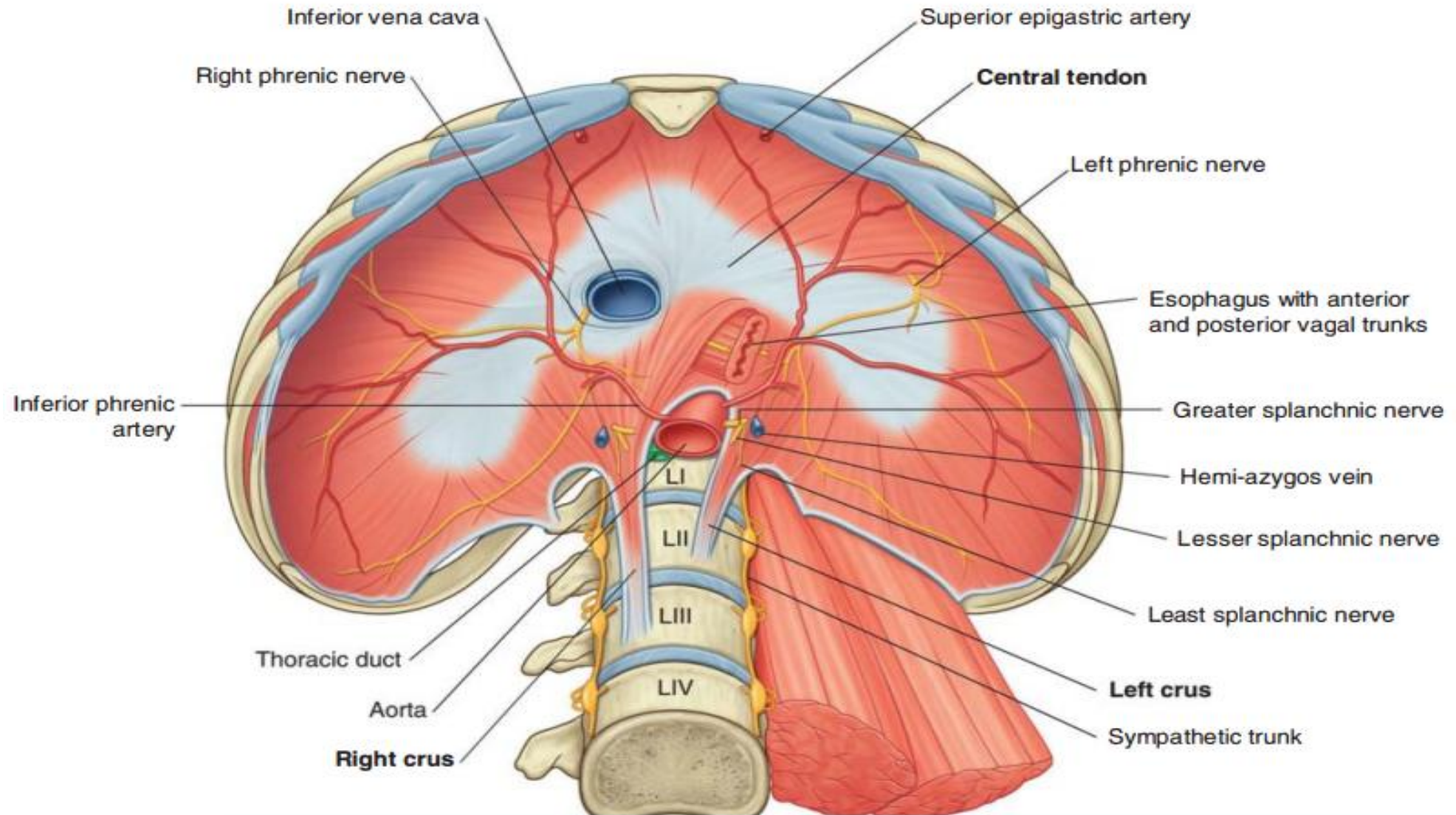
Diaphragmatic Apertures(Opening),,,



Diaphragmatic Apertures(Opening),,,

- ❖ **Caval opening:** The caval opening is primarily for the **IVC** and for terminal branches of **the right phrenic nerve** and a few lymphatic vessels
 - The most superior of the three large diaphragmatic apertures,
 - It lies at the level of the IV disc between the T8 and T9 vertebrae
- ❖ **The esophageal hiatus:** is in the muscle of the right crus of the diaphragm at the level of the T10 vertebra.
 - The esophageal hiatus also transmits the **anterior & posterior vagal trunks**, esophageal branches of the left gastric vessels, and a few lymphatic vessels.

Diaphragmatic Apertures(Opening),,,



Diaphragmatic Apertures(Opening),,,

- The esophageal hiatus is superior to and to the left of the aortic hiatus.
- In most individuals (70%), both margins of the hiatus are formed by muscular bundles of the right crus. In others (30%), a superficial muscular bundle from the left crus contributes to the formation of the right margin of the hiatus.
- ❖ **The aortic hiatus** is the opening posterior in the diaphragm for the **descending aorta**.
 - The aorta passes between the crura of the diaphragm at the level of the inferior border of the T12 vertebra.
 - Aorta does not pierce the diaphragm.
 - The aortic hiatus also transmits the thoracic duct & sometimes the azygos & hemi-azygos veins.

Diaphragm,,,

□ Nerve Supply of the Diaphragm

➤ Motor nerve supply :

- The right and left phrenic nerves (C3 - C5)

➤ Sensory nerve supply :

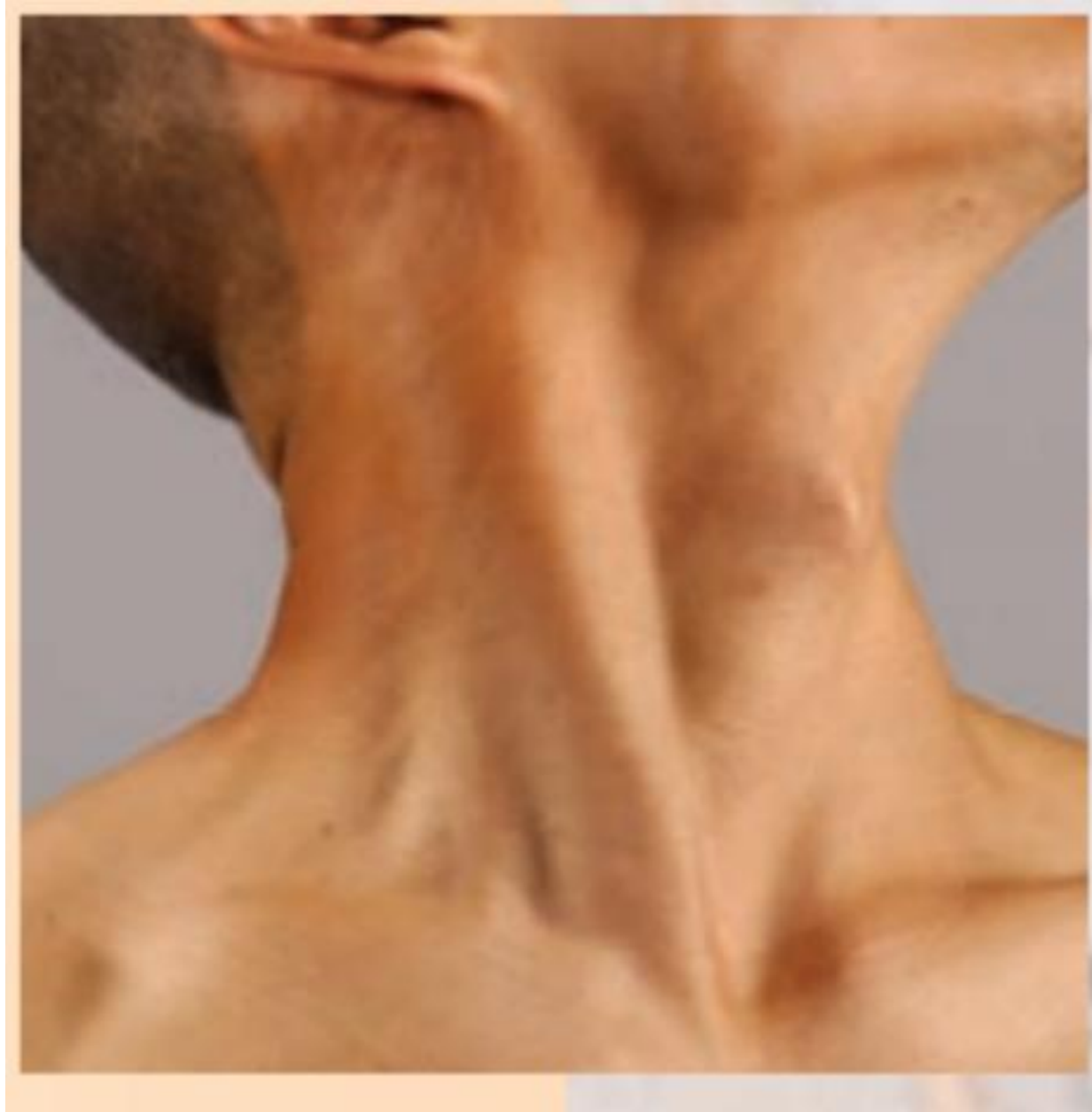
- To the central surfaces of the diaphragm are from the phrenic nerve
- To the periphery of the diaphragm is from the lower six intercostal nerves

Diaphragm,,,

❖ Actions of diaphragm

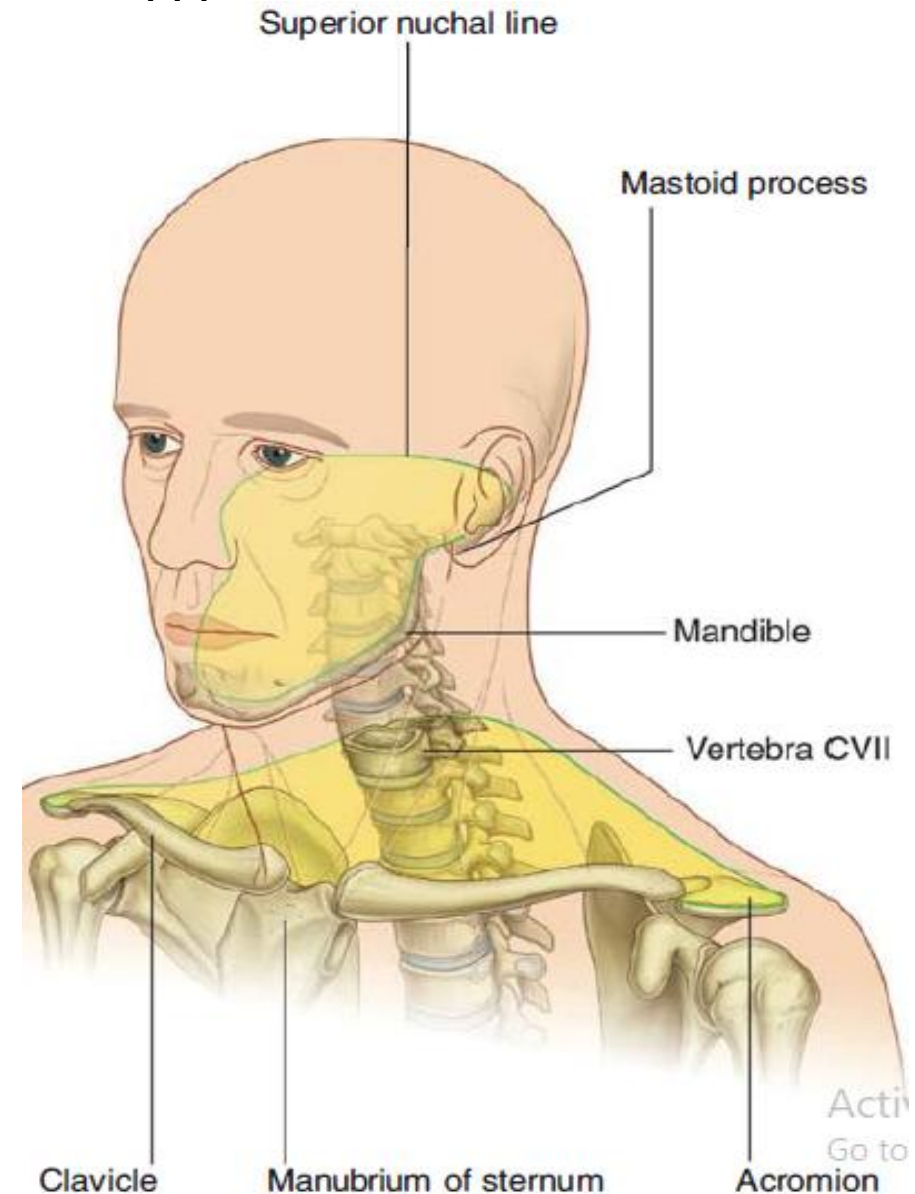
- The diaphragm is the **primary muscle** of respiration.
 - During inspiration, it contracts and **flattens**, increasing the vertical diameter of the thoracic cavity. This produces lung expansion, and air is drawn in.
 - During expiration, the diaphragm passively relaxes and returns to its original dome shape. This reduces the volume of the thoracic cavity.
- ❑ *The diaphragm takes active part in all expulsive acts*

Over view of Anatomy of the neck



Over view of Anatomy of the neck,,

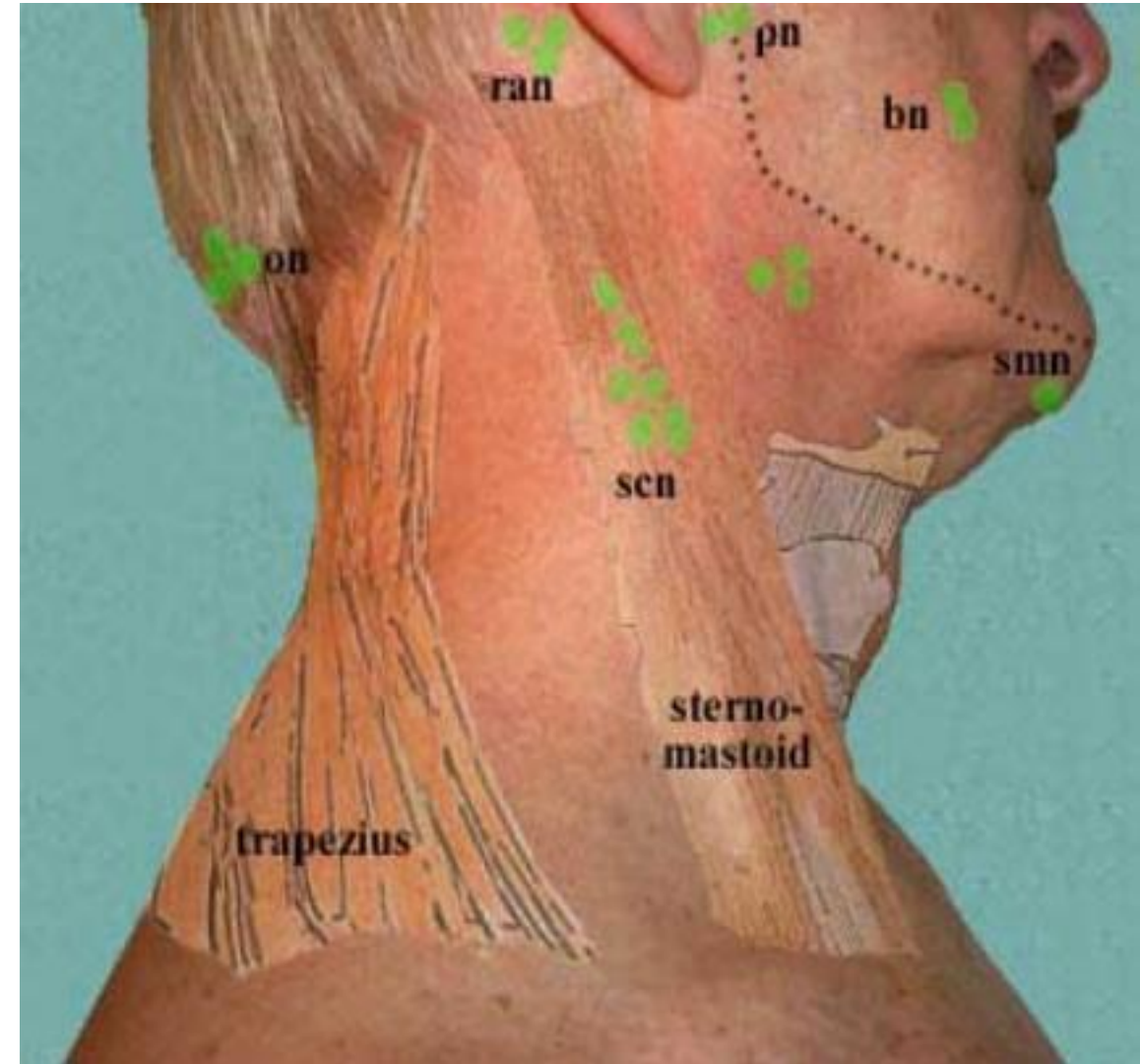
- The term neck is derived from Latin term “**collum or cervix**”
- It is the transitional area b/n the base of the **cranium superiorly & the clavicles inferiorly**.
- The neck joins the head to the trunk and limbs, serving as a major conduit for structures passing between them.



Over view of Anatomy of the neck,,

❖ Boundaries of the neck:

- ✓ **Superior:-** Inferior border of mandible, Mastoid process , External occipital protuberance .
- ✓ **Anterior:-** anterior median line
- ✓ **Posterior:-** anterior border of trapezius
- ✓ **Inferior:-** sternal notch & clavicle



Over view of Anatomy of the neck,,,

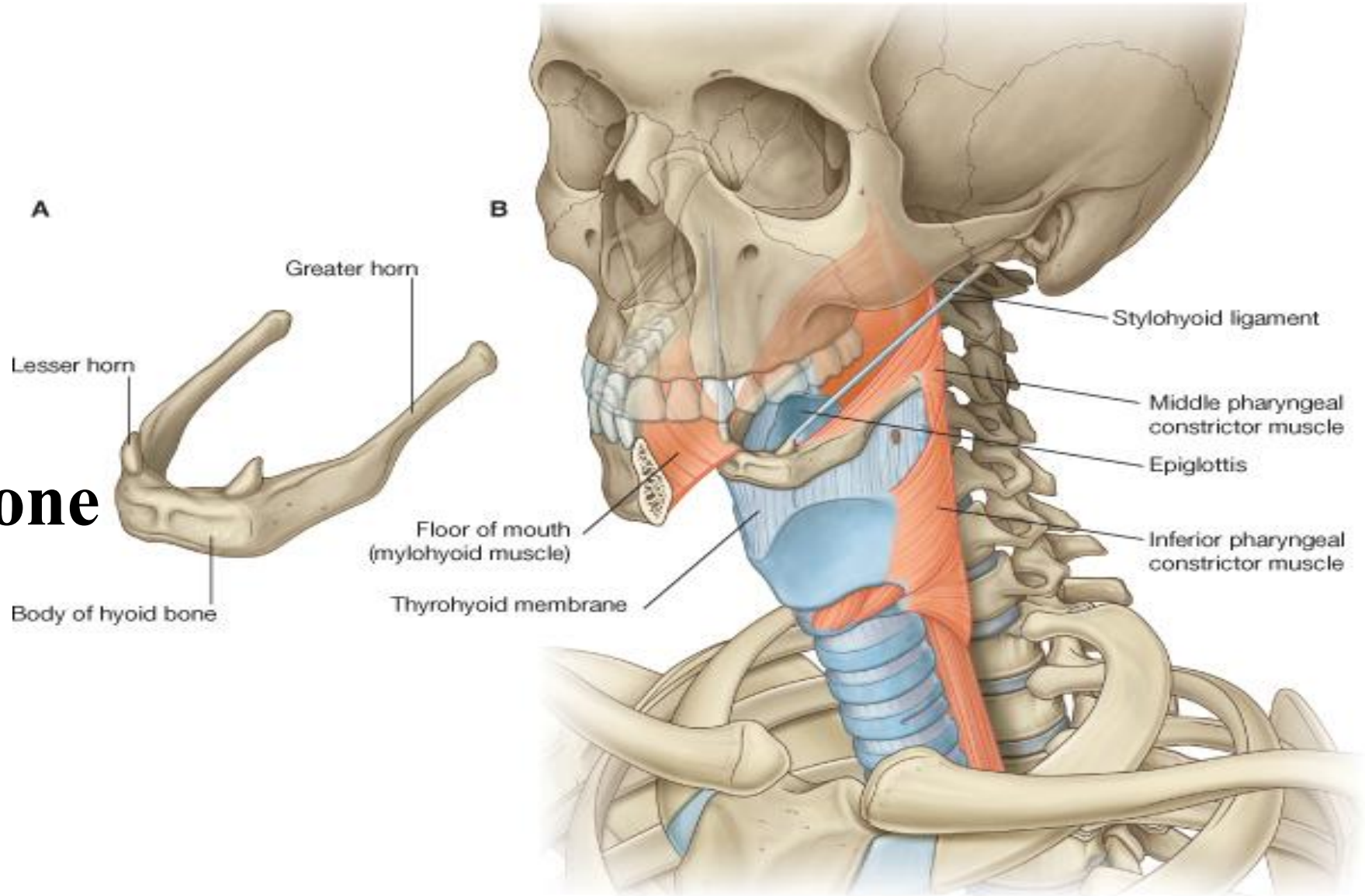
- The neck **serves as a major conduit** for structures passing through it.
- Many important structures are crowded together in the neck, such as muscles, glands, arteries, veins, nerves, lymphatics, trachea, esophagus, and vertebrae.
- **The neck is a well-known region of vulnerability.**
- In the central region of the neck are parts of the respiratory system, namely, the larynx and the trachea, and behind are parts of the alimentary system, the pharynx and the esophagus

Bony Skeletal of the neck

- ❖ The skeleton of the neck is formed by the **cervical vertebrae and their intervertebral disc, hyoid bone, manubrium of the sternum, and clavicles.**
- ❖ These bones are parts of the axial skeleton except the clavicles, which are part of the appendicular skeleton
- Seven cervical vertebrae form the cervical region of the vertebral column, which **encloses the spinal cord and meninges.**
- The vertebral bodies support the head, and the intervertebral (IV) articulations—especially the **craniovertebral** joints.

Bony Skeletal of the neck,,,

❖ Hyoid bone



Cervical vertebrae

- **There are seven cervical vertebrae(C1-C7)**
- **3-6 are typical and 1,2,and 7 are atypical**

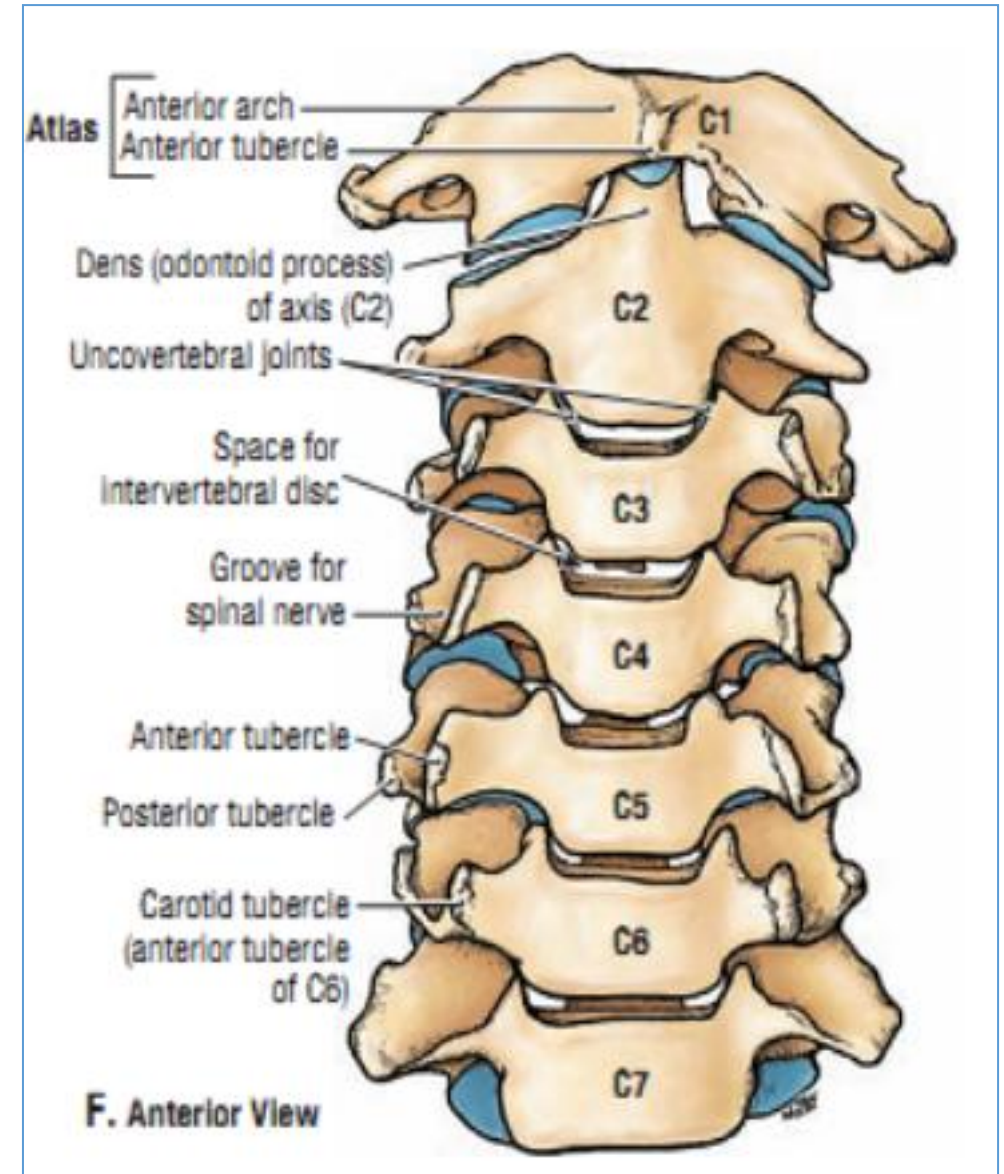
Typical cervical vertebrae have

✓body

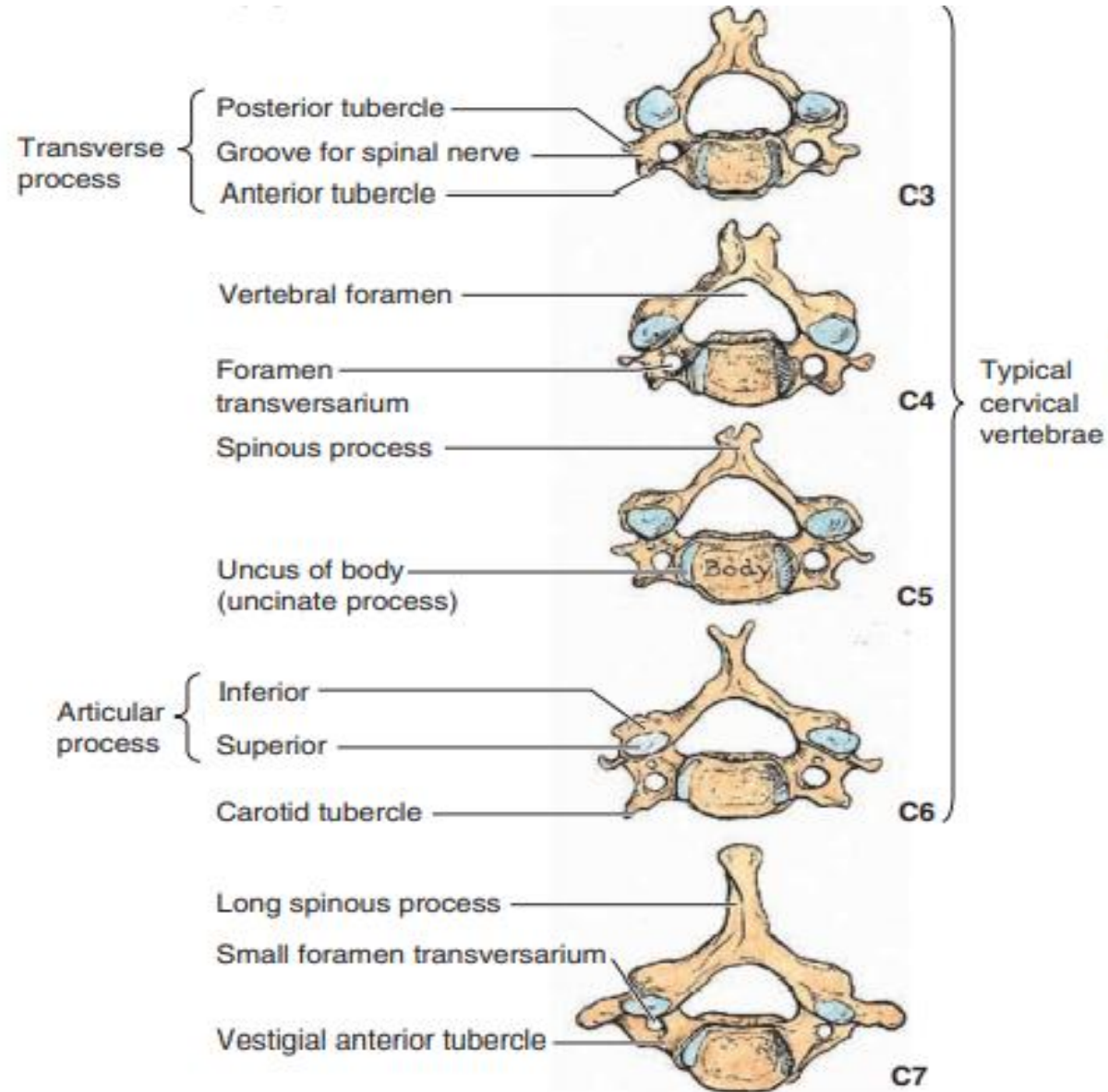
✓vertebral foramen

✓vertebral arch

- Pedicles
- Laminae
- superior facets
- inferior facets
- transverse process
- spine – short and bifid



Cervical vertebrae,,,

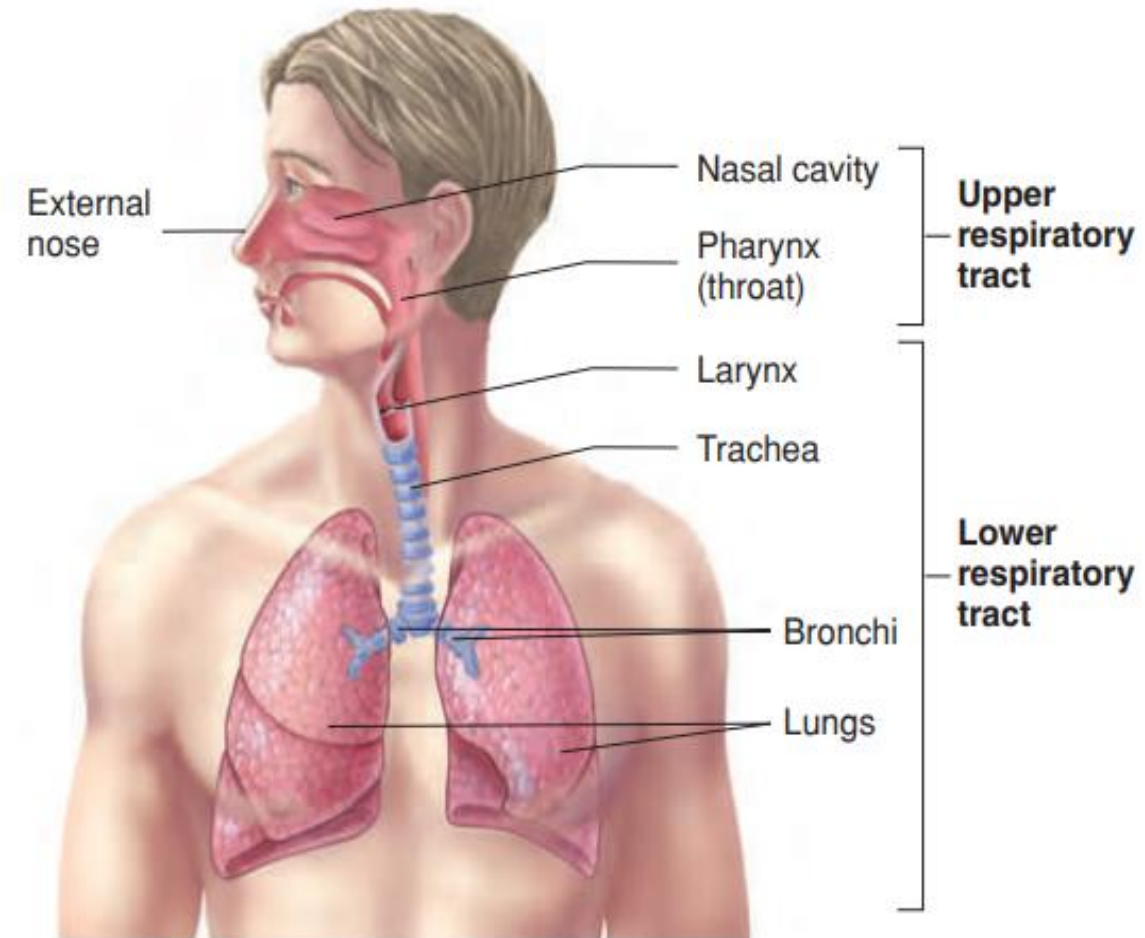


Gross structures and functions of the upper respiratory tract

□ Structurally, the respiratory system

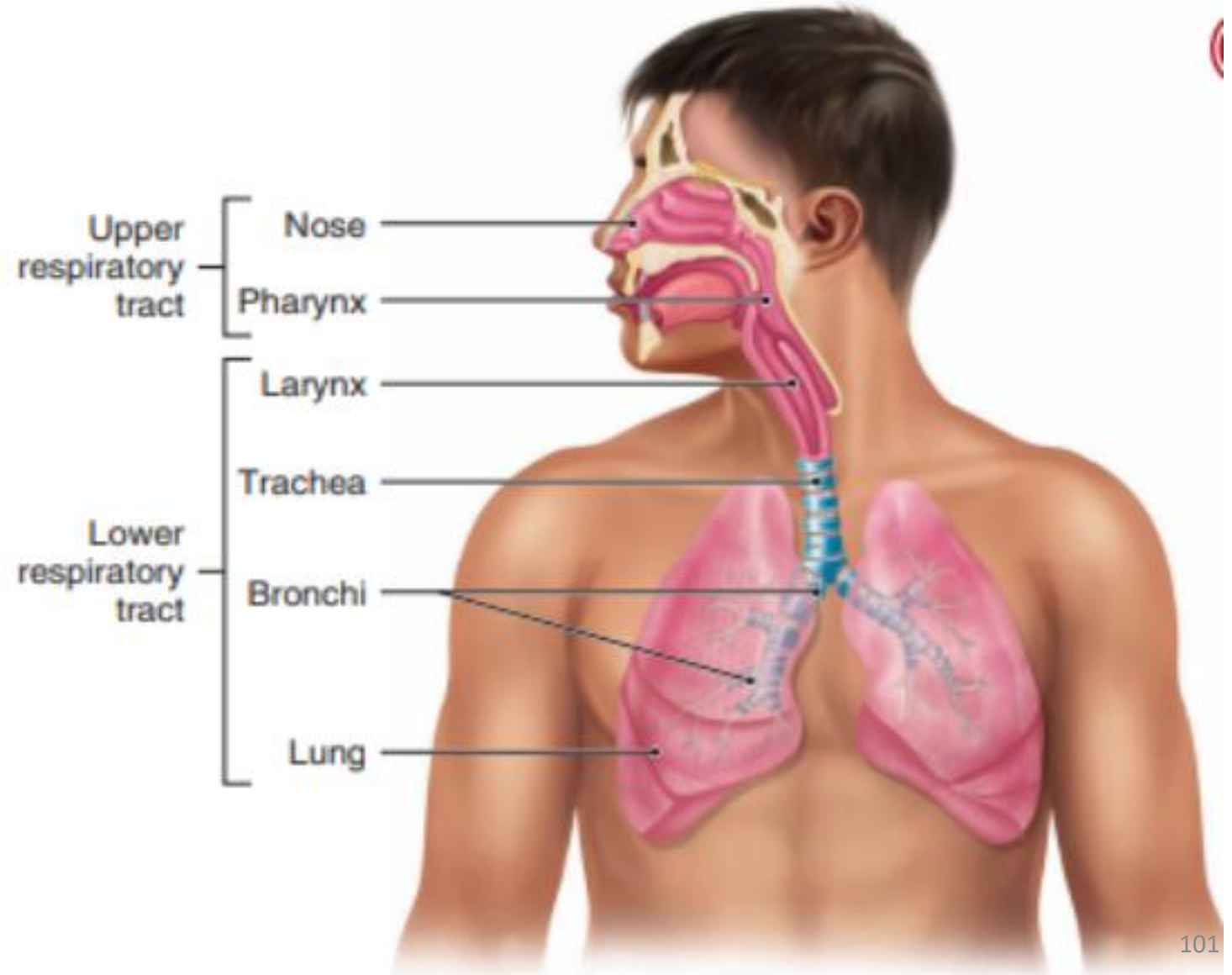
consists of two parts:

- I. the upper respiratory system includes the nose, nasal cavity, pharynx, and associated structures.
- II. the lower respiratory system includes the **larynx**, trachea, bronchi, and lungs



Gross structures and functions of the upper respiratory tract,,

❖ The upper respiratory tract includes, the respiratory organs in **the head and neck.**



Gross structures and functions of the upper respiratory tract,,,

❖ **Functionally, the respiratory system also consists of two parts.**

I. The conducting zone consists of a series of interconnecting cavities and tubes both outside and within the lungs.

- These passageways include the nose, nasal cavity, pharynx, larynx, trachea, bronchi, bronchioles, and terminal bronchioles.
- The structures of the conducting zone also filter, humidify, and warm the incoming air.

II. The respiratory zone consists of tubes and tissues within the lungs where gas exchange occurs.

- These include the respiratory bronchioles, alveolar ducts, alveolar sacs, and alveoli, and are the main sites of gas **exchange between air and blood.**

The extrapulmonary region of the conducting portion

- The extrapulmonary region of the conducting portion consists of the nasal cavities, pharynx, larynx, trachea, and bronchi.
- **Mucosa of the extrapulmonary region** of the conducting portion modifies the inspired air by humidifying, cleansing, and adjusting its temperature.
- This mucosa is composed of
 - ✓ *pseudostratified ciliated columnar epithelium* (respiratory epithelium) with **numerous goblet cells** and
 - ✓ an **underlying connective tissue sheath** that is well endowed with **seromucous glands**.

The extrapulmonary region of the conducting portion,,

- Modulation of the temperature of the inspired air is accomplished mostly in the nasal cavity by the rich vascularity of the CT just deep to respiratory epithelium.
- **In certain areas, the mucosa of the nasal cavity is modified to function in olfaction and is referred to as the olfactory mucosa.**
- The glands in the lamina propria of this region, **known as Bowman's glands,** produce a thin mucous secretion that dissolves **odoriferous substances,** and the **olfactory cells of the pseudostratified columnar olfactory epithelium** perceive these sensory stimuli.

The extrapulmonary region of the conducting portion,,

❖ Odorant binding proteins (**integral membrane proteins that are odorant receptors**)

lying within the plasma membrane of the cilia are sensitive to molecules of specific odor groups, where each of these molecules is **known as an odorant**.

• *The odorant must satisfy at least three requirements:*

I. it must be volatile,

II. water soluble, and

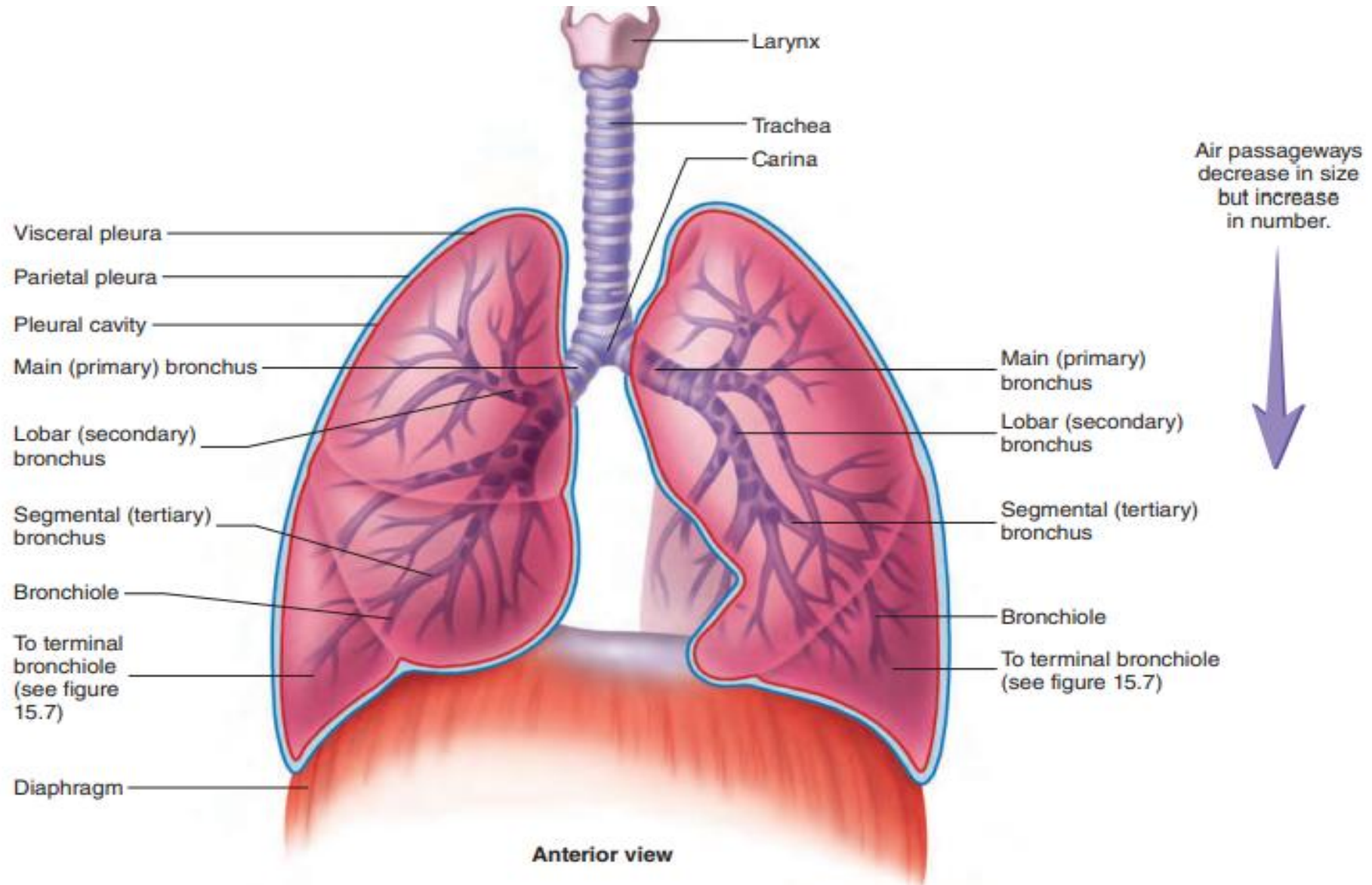
III. lipid soluble, so that it can:

❖ Enter the nasal cavity (volatility), penetrate the mucus (water solubility), and have access to the phospholipid membrane (lipid solubility).

The intrapulmonary region

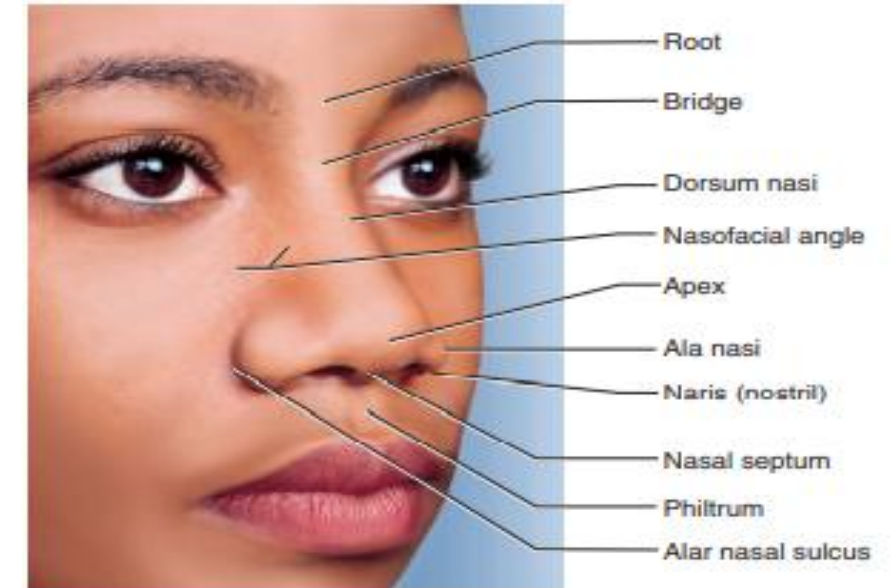
- The intrapulmonary region is composed of intrapulmonary bronchi (secondary bronchi), whose walls are supported by **irregular plates of hyaline cartilage**.
- Each intrapulmonary bronchus gives rise to several bronchioles, tubes of decreasing diameter that do not possess a cartilaginous supporting skeleton.
- **The epithelial lining of the larger bronchioles is ciliated with a few goblet cells, but those of smaller bronchioles are simple columnar, with goblet cells being replaced by Clara cells.**

The intrapulmonary region,,

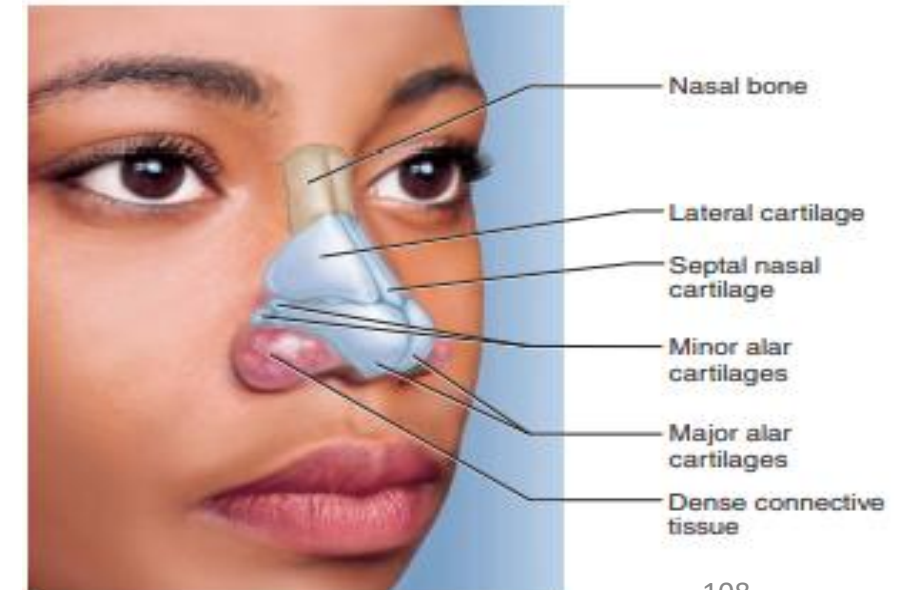


The Nose

- The nose is the only externally visible part of the respiratory system
- The nose consists of the **external nose & the nasal cavity.**
- The external nose is the visible structure that forms a prominent feature of the face.
- Most of the external nose is composed of hyaline cartilage, although the bridge of the external nose consists of bone.



(a)

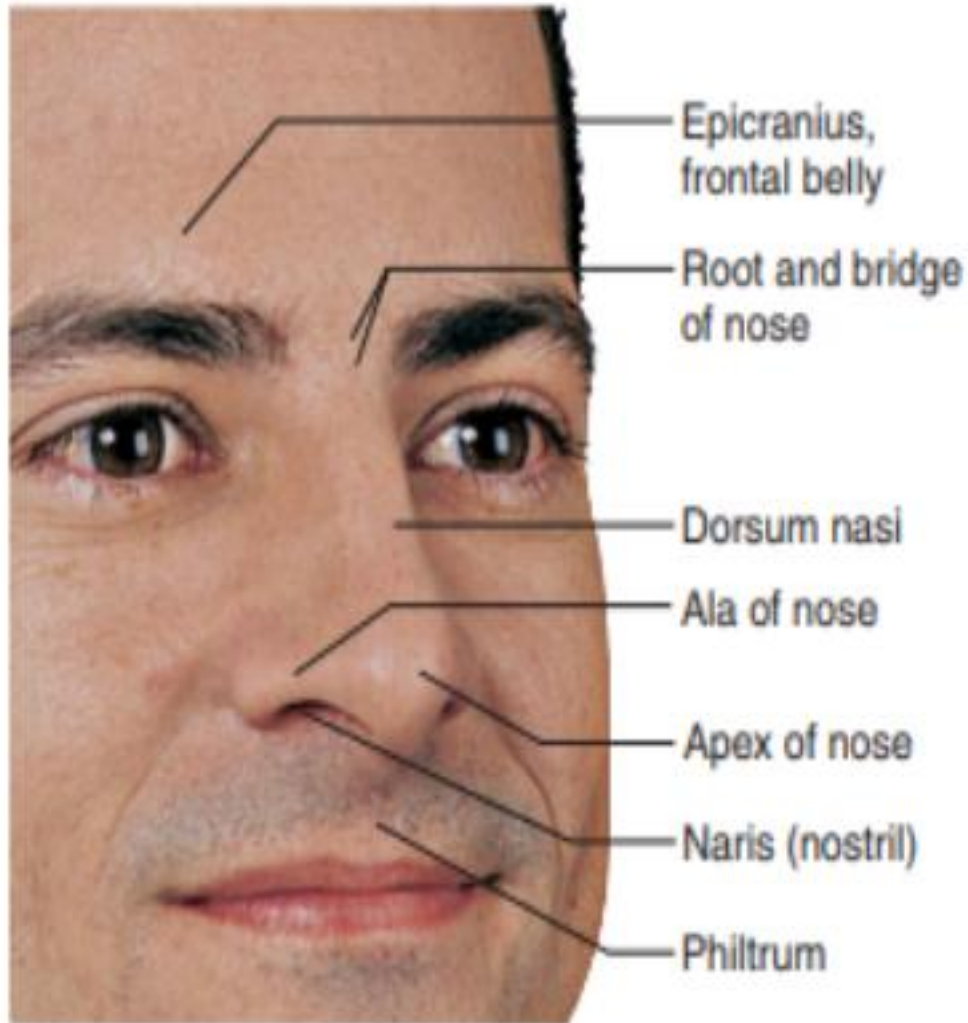


(b)

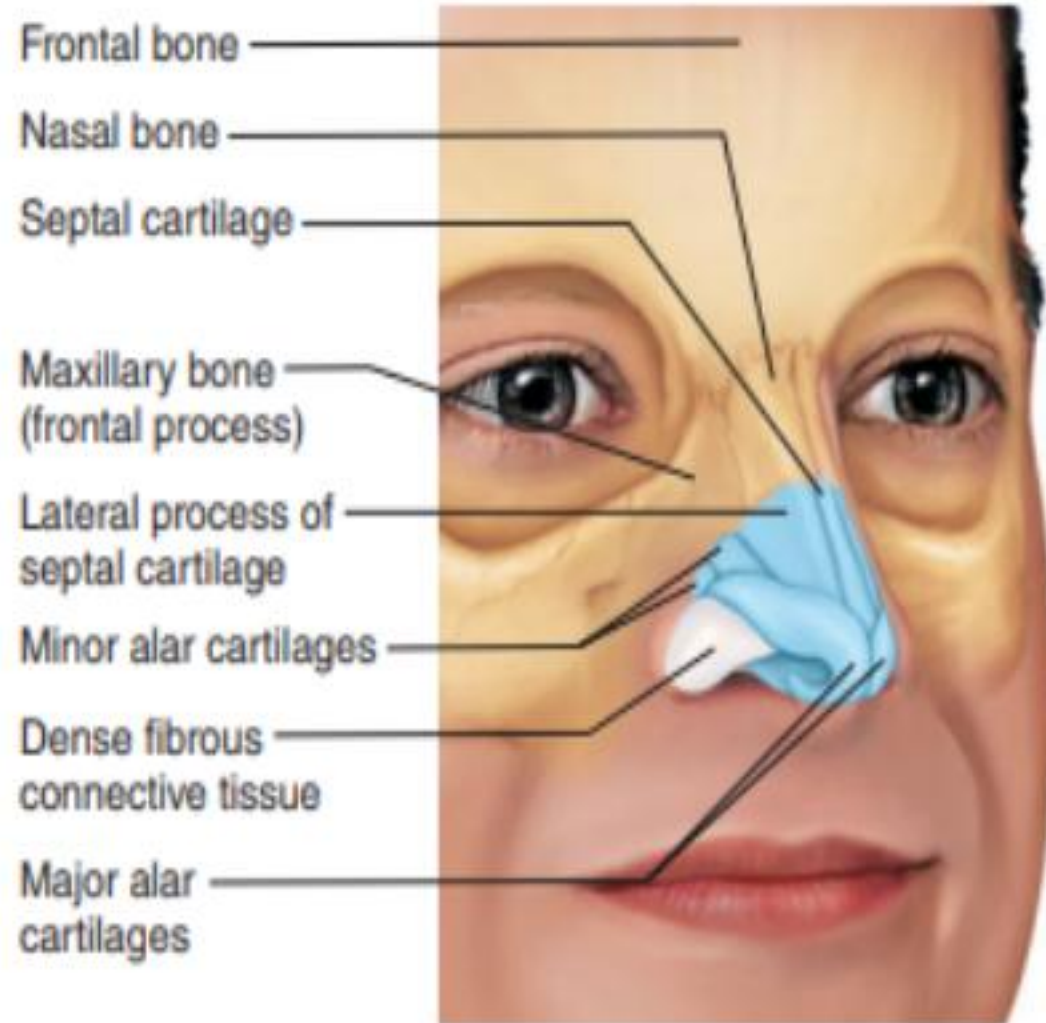
The Nose,,,

- **The skeletal framework of the nose**, consists of the **nasal bones** superiorly, the **maxillary bones laterally**, and flexible plates of hyaline cartilage inferiorly (the lateral, septal, and alar cartilages).
- The tip of the nose, its apex, is formed by the major alar cartilages, and the lateral border of the nostril, the ala, is formed from dense fibrous connective tissue.
- The great variation in nose size and shape is largely due to **differences in the nasal cartilages**.
- The skin covering the nose's anterior and lateral surfaces is thin and contains many sebaceous glands that open into some of the largest skin pores on the face.

The Nose,,,



(a) Surface anatomy



(b) External skeletal framework

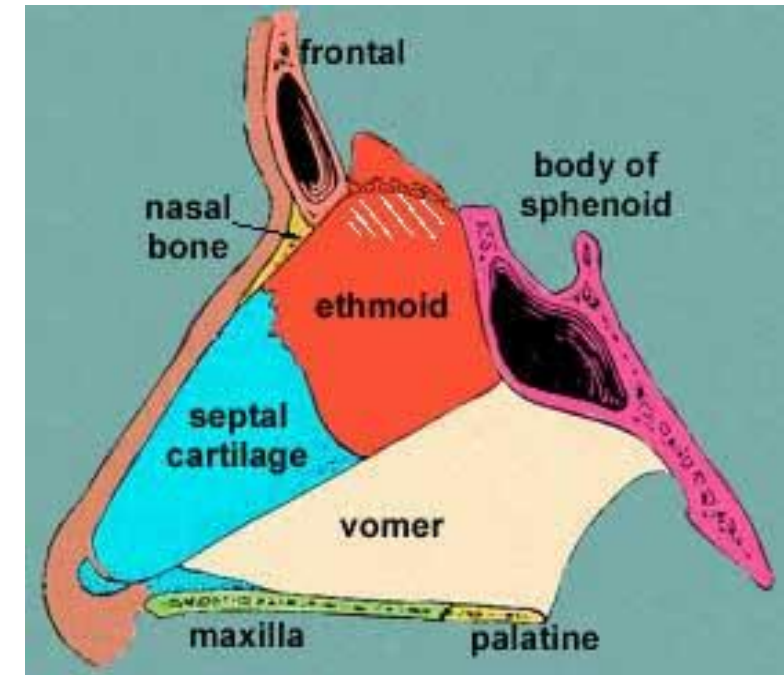
The Nose,,,

- **Important functions of the nose includes.**

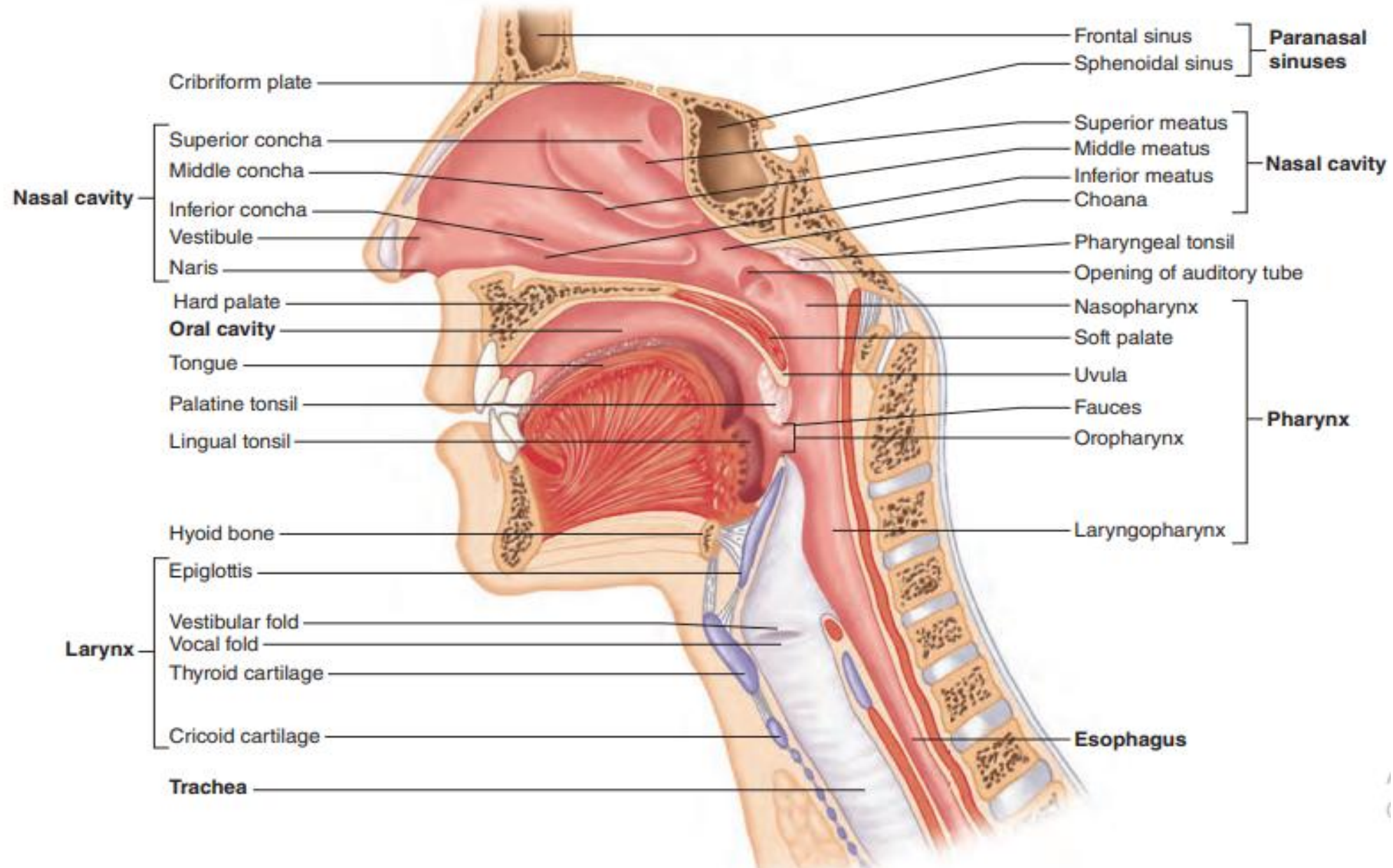
- ✓ provides an airway for respiration,
- ✓ moistens and warms entering air,
- ✓ filters inhaled air to cleanse it of foreign particles,
- ✓ serves as a resonating chamber for speech, and
- ✓ houses the olfactory (smell) receptors

The Nose,,,

- The **nares, or nostrils**, are the external openings of the nose, and the **choanae** are the openings into the pharynx.
- **The nasal cavity extends from nares to the choanae.**
- The nasal cavity is divided into right and left halves by the nasal septum in the midline; formed by the perpendicular plate of the **ethmoid bone, the vomer, & a septal cartilage**
- The hard **palate forms the floor of the nasal cavity**, separating the nasal cavity from the oral cavity.



The Nose,,,



The Nose,,,

- Roof of nasal cavity contains olfactory mucosa that has receptors for **sense of smell**
- Three prominent bony ridges called **conchae** are present on the lateral walls of the nasal cavity.
- The conchae increase the surface area of the nasal cavity.
- **The nasolacrimal ducts**, which carry tears from the eyes, also open into the nasal cavity.
- Sensory receptors for the sense of smell are in the superior part of the nasal cavity
- Sensory receptors detect the foreign substances
- **The sneeze reflex** dislodges foreign substances from the nasal cavity.

The Nose,,,,

- Air enters the nasal cavity through the nares.
- Just inside the nares, the lining of the cavity is composed of **stratified squamous epithelium containing coarse hairs**.
- The hairs **trap some** of the large particles of dust suspended in the air.
- Mucus produced by the goblet cells also **traps debris in the air**.
- The cilia sweep the mucus posteriorly to the pharynx, where it is swallowed.
- The superior nasal concha and superior nasal septum are lined with **olfactory mucosa containing the olfactory epithelium**, the tissue containing the olfactory receptors used in detecting chemicals for the sense of smell.

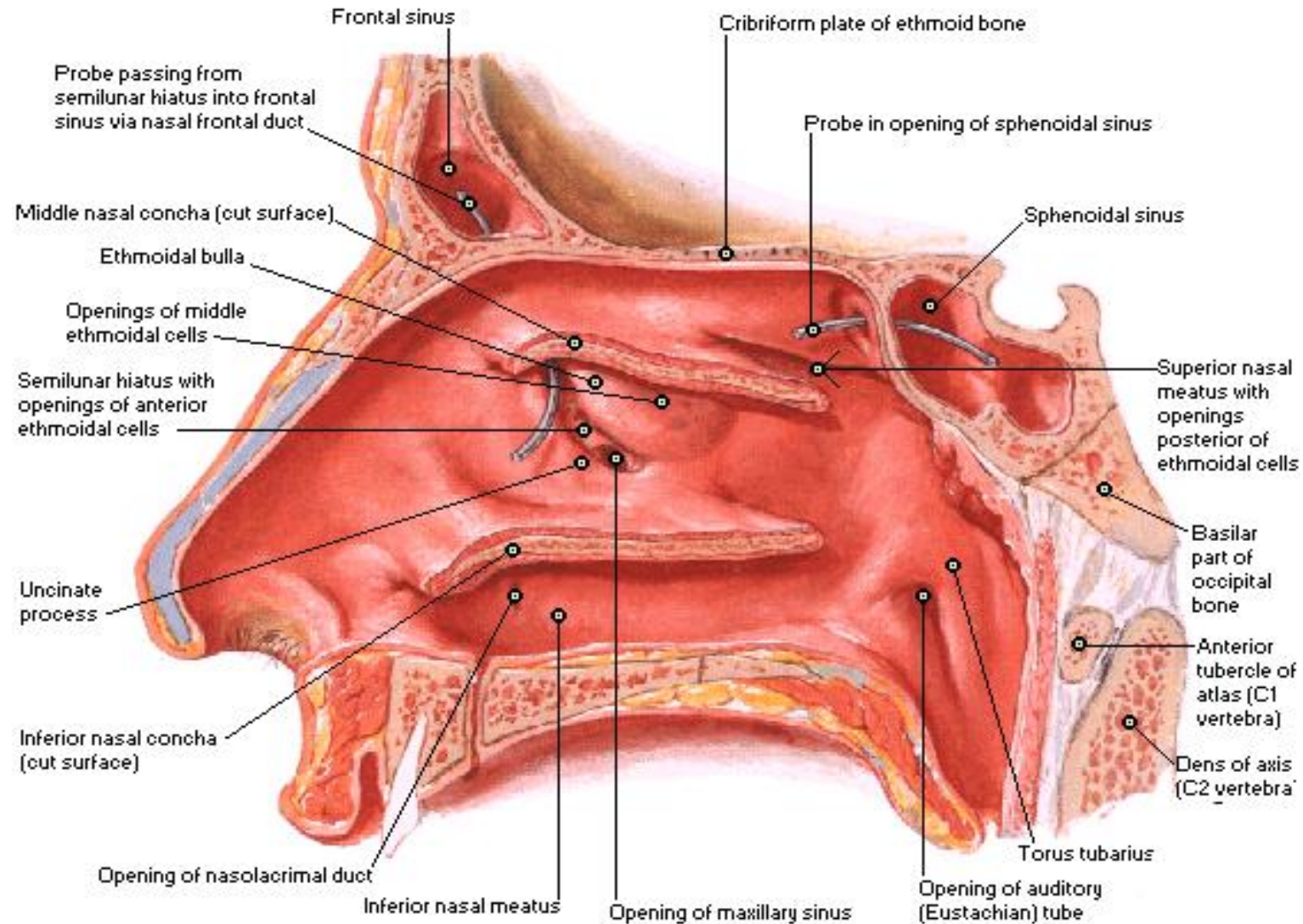
The Nose,,,

- Both of **the olfactory epithelium and respiratory epithelium** are ciliated **pseudostratified columnar epithelia**.
- However, in the olfactory epithelium, **the cilia are immobile** and serve to bind odor molecules but in the respiratory epithelium, they are mobile.
- In the respiratory epithelium the goblet cells secrete mucus, and its ciliated cells propel the mucus posteriorly toward the pharynx.
- Inhaled dust, pollen, bacteria, and other foreign matter stick to the mucus and are swallowed; they are either digested or pass through the digestive tract rather than contaminating the lungs.

The Nose,,,,

- There are several drainage openings into the nasal cavity.
- The paranasal ducts drain mucus from the paranasal sinuses, and the nasolacrimal ducts drain tears from the eyes.
- An excessive secretion of tears causes the nose to run as the tears drain into the nasal cavity.
- The auditory tube from the tympanic cavity enters the upper respiratory tract posterior to the nasal cavity in the nasopharynx.
- With all these accessory connections, it is no wonder that infections can spread so easily from one chamber to another throughout the facial area.

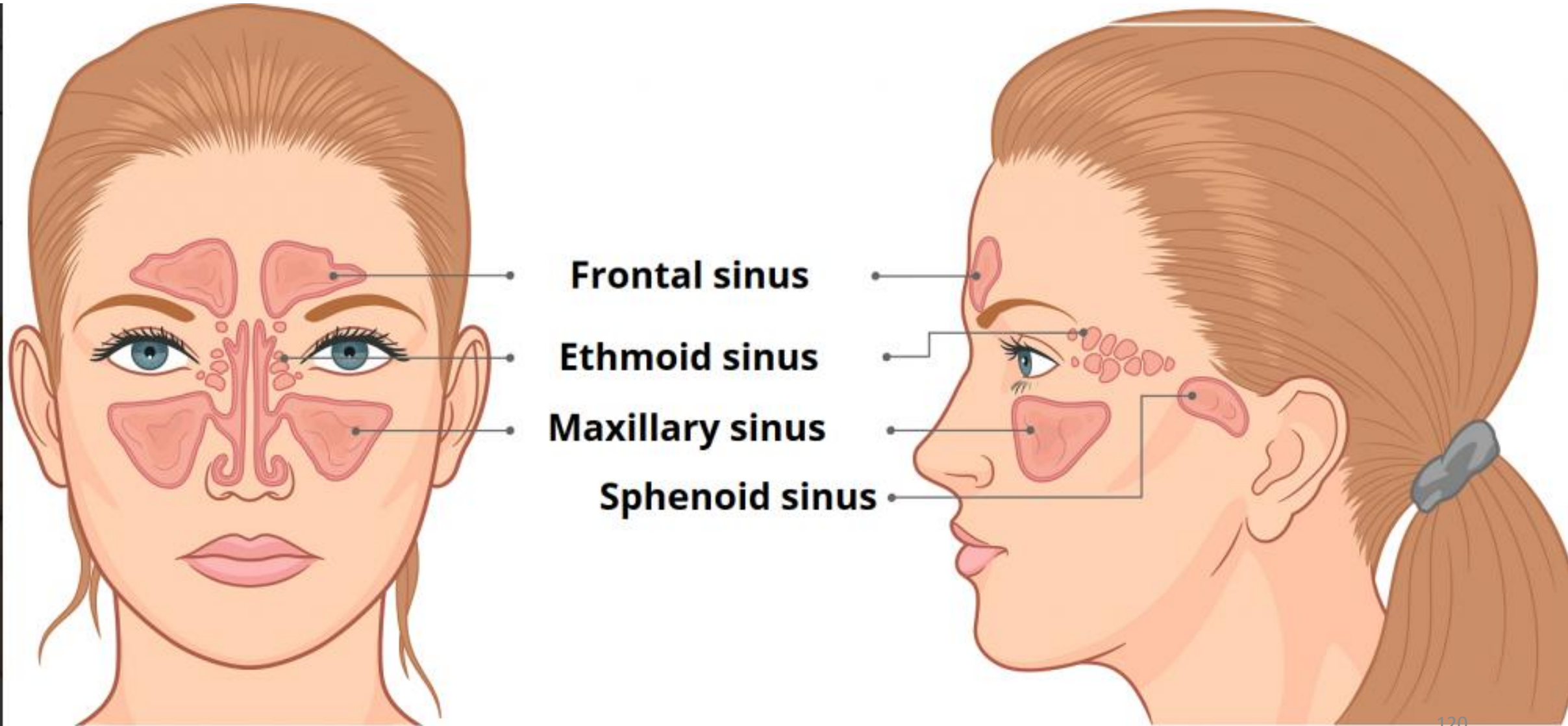
The Nose,,,,



Paranasal sinuses

- The paranasal sinuses are **air-filled spaces within bone**.
- They are **extensions of the nasal cavity**.
- There are four paired sinuses – named according to the bone in which they are located – maxillary, frontal, sphenoid and ethmoid.
- The paranasal **sinuses open into the nasal cavity**.
- Paranasal sinuses are lined by the same mucosa, and perform the same air-processing functions as does that nasal cavity.

Paranasal sinuses,,,



Paranasal sinuses,,,

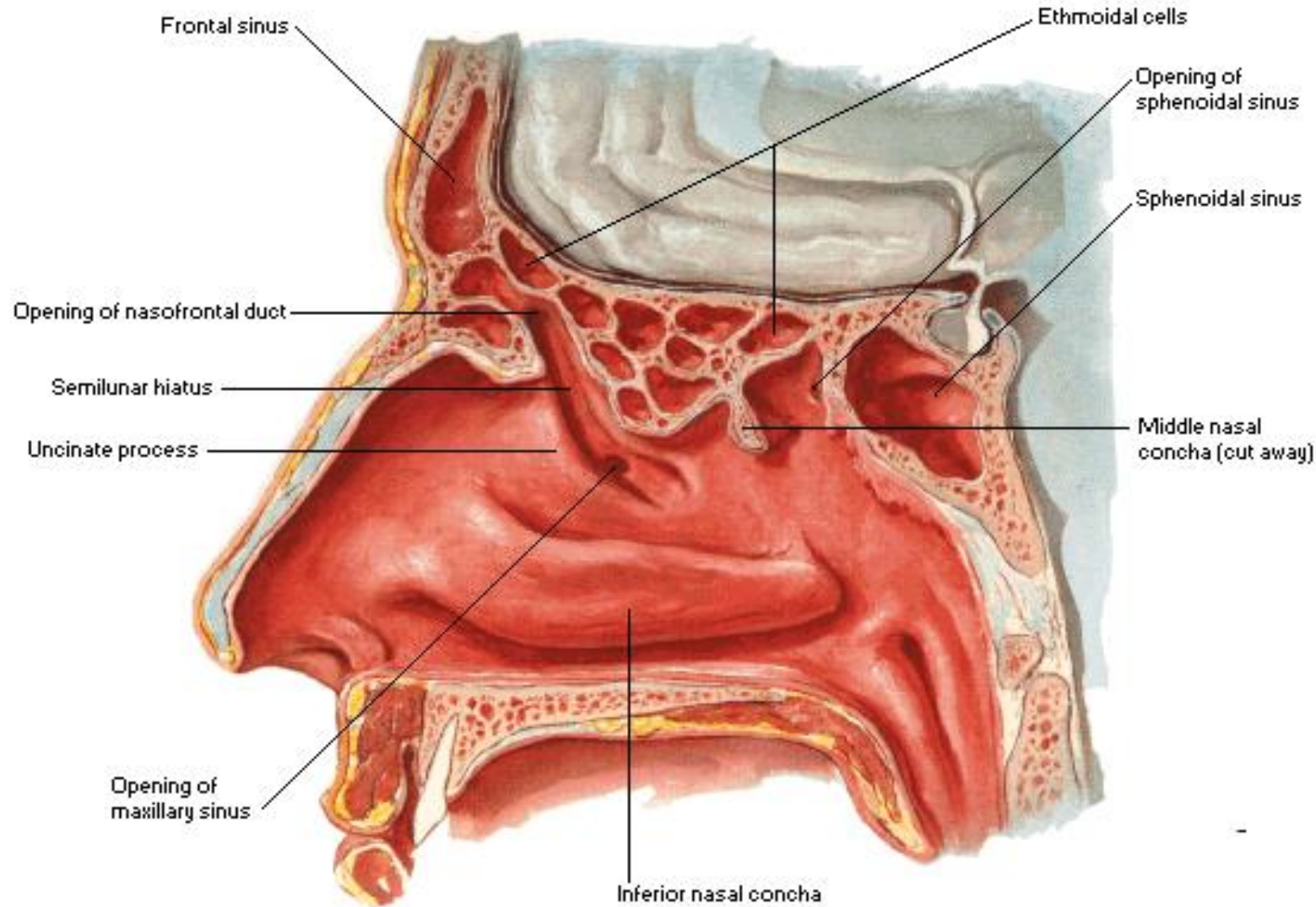
❖ **Paranasal sinuses** reduce the weight of the skull, produce mucus, and influence the quality of the voice by acting as resonating chambers.

▪ **Roles of the Paranasal sinuses can be summarised as:**

- ✓ Lightening the weight of the head
- ✓ Supporting immune defence of the nasal cavity
- ✓ Humidifying inspired air
- ✓ Increasing resonance of the voice

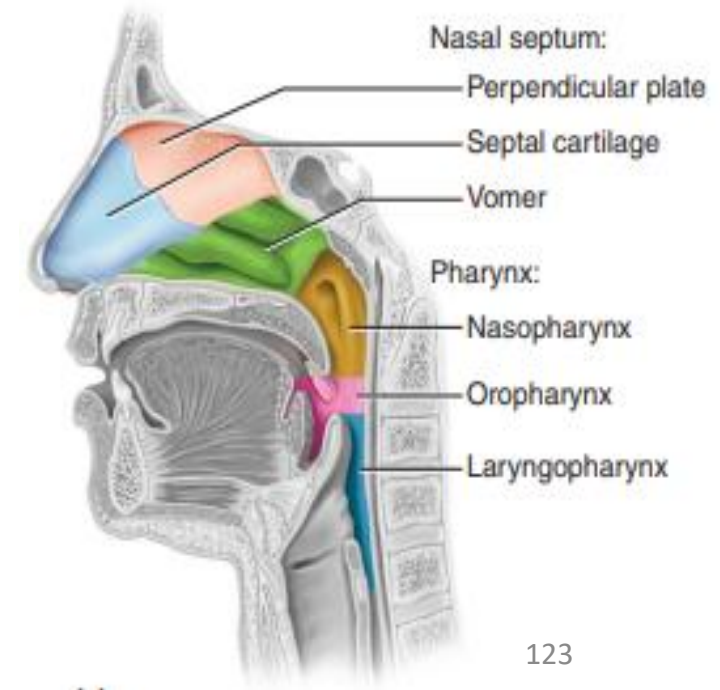
Paranasal sinuses,,,

Sagittal Section

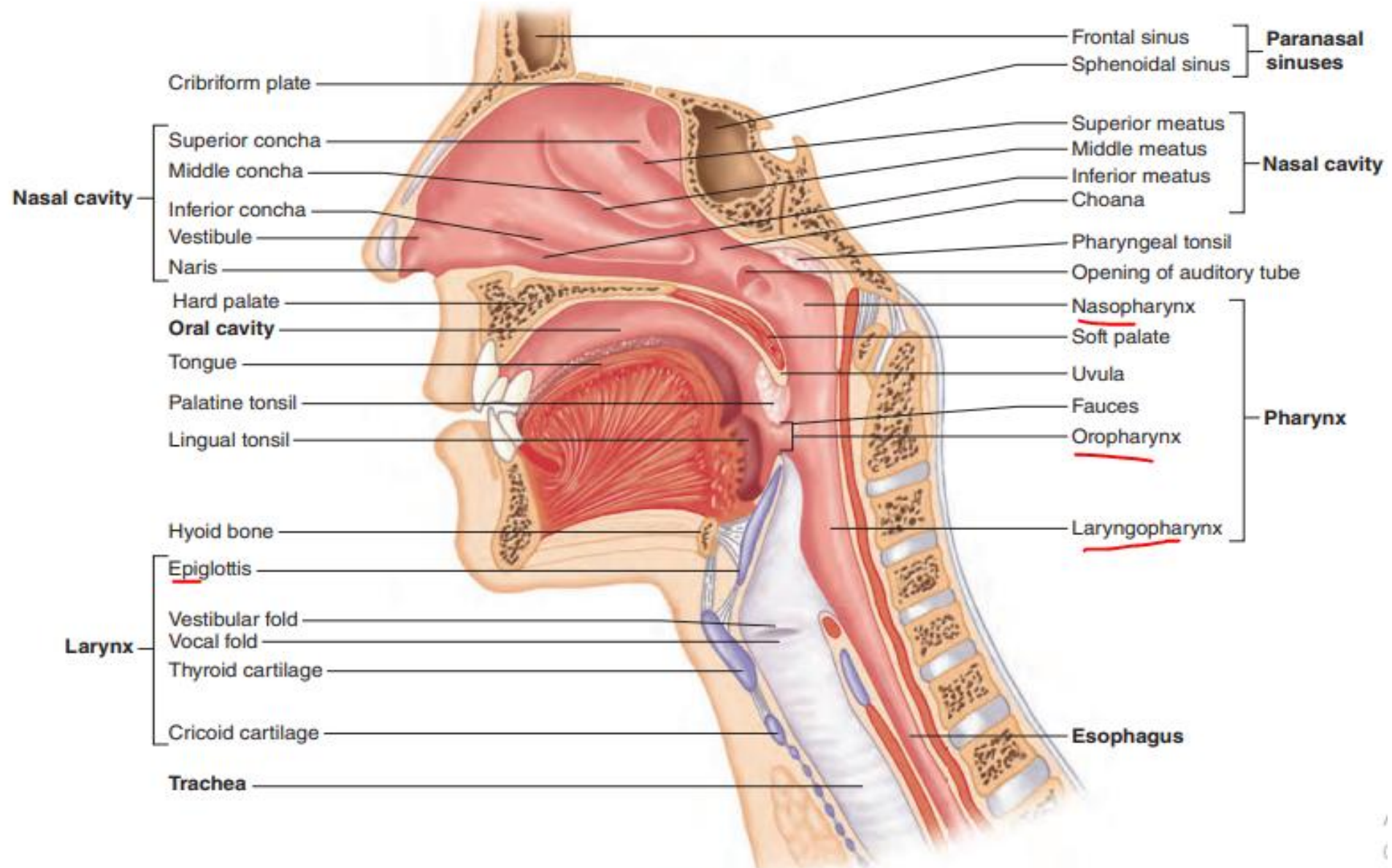


Anatomy of Pharynx

- ❖ The pharynx is the common passageway for **both the respiratory and the digestive systems**.
- ❖ Air from the nasal cavity and air, food, and water from the mouth pass through the pharynx.
- Inferiorly, the **pharynx leads** to the rest of the respiratory system through the opening into **the larynx** and to the digestive system through the **esophagus**.
- The pharynx can be divided into three regions: the **nasopharynx, the oropharynx, and the laryngopharynx**



Anatomy of Pharynx,,,



(a)

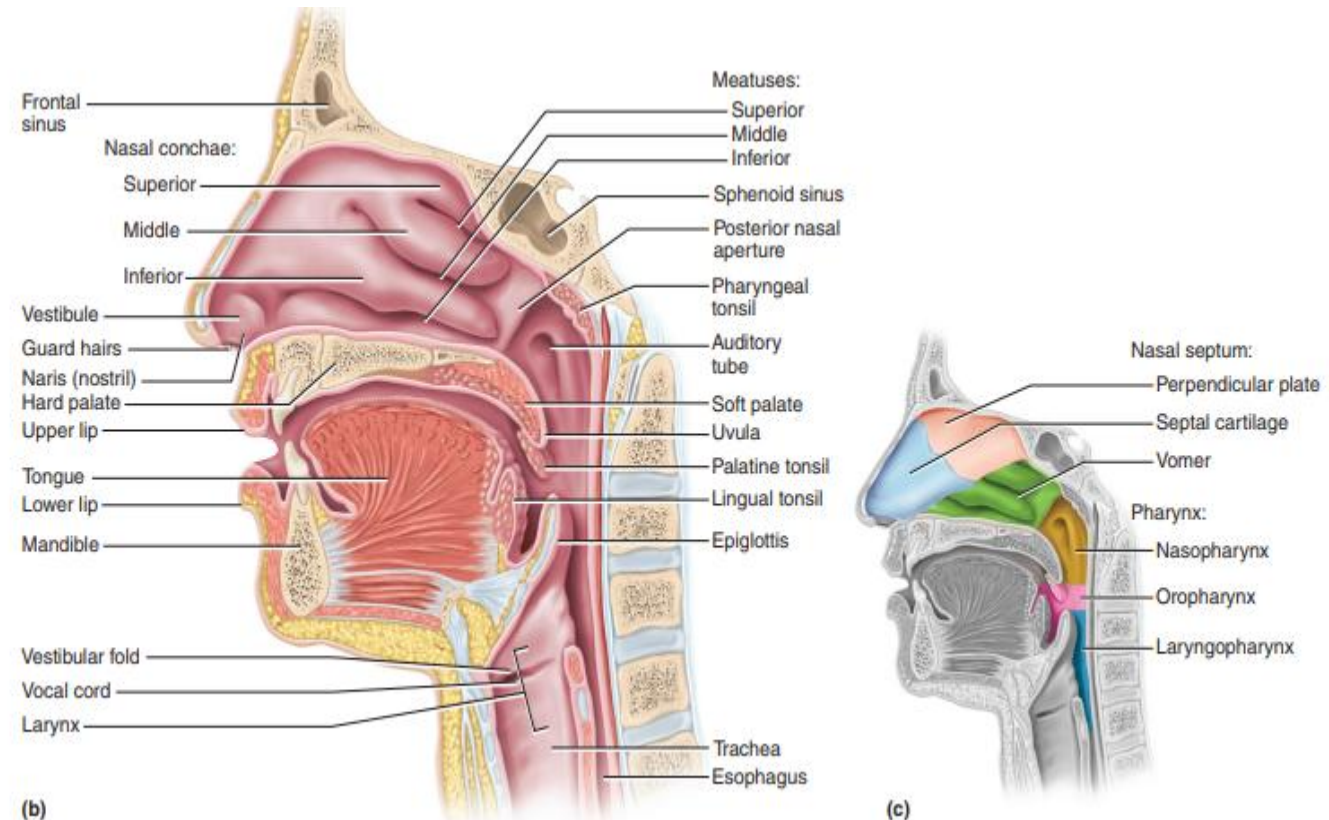
Medial view

Anatomy of Pharynx,,,

- The nasopharynx is the **superior part of the pharynx**.
- It is located posterior to the **choanae & superior to the soft palate**, which is an incomplete muscle and connective tissue partition separating the nasopharynx from the oropharynx.
- The soft palate forms the floor of the nasopharynx.
- The nasopharynx is lined with **pseudostratified ciliated columnar epithelium**
- The auditory tubes extend from the middle ears and open into the nasopharynx.
- The soft palate is elevated during swallowing to closes the nasopharynx

Anatomy of Pharynx,,,

- The posterior part of the nasopharynx **contains the pharyngeal tonsil**, which helps defend the body against infection.
- **A pharyngotympanic (auditory) tube**, which drains the middle ear, opens into each lateral wall of the nasopharynx.



Anatomy of Pharynx,,

- The oropharynx extends from the uvula to the epiglottis, and the oral cavity opens into the oropharynx. *Thus, food, drink, and air all pass through the oropharynx.*
- As the nasopharynx blends into the oropharynx, the lining epithelium changes from **pseudostratified columnar** to a thick, protective **stratified squamous epithelium..**
- Two sets of tonsils, the **palatine tonsils and the lingual tonsil**, are located near the opening between the **mouth and the oropharynx.**

Anatomy of Pharynx,,

□The Laryngopharynx:

- Like the oropharynx superior to it, the laryngopharynx serves as a **common passageway for food and air**
- Food and drink pass through the laryngopharynx to the esophagus
- A small amount of air is usually swallowed with the food and drink.
- Swallowing too much air can cause excess gas in the stomach and may result in belching
- The laryngopharynx is lined with **stratified squamous epithelium and ciliated columnar epithelium**

The end!!!

Thank you for your attention!!!